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بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

دعاء

وَقُلْ اَعْمَلُوا فَسَيَرَى اللَّهُ عَمَلَكُمْ وَرَسُولُهُ وَالْمُؤْمِنُونَ

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الحمد لله الذي من علينا فأفضل و أعطانا فأجزل

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"أَعْمَلَ صَالِحًا تَرْضَاهُ وَأَدْخِلْنِي بِرَحْمَتِكَ فِي عِبَادِكَ الصَّالِحِينَ

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نستغفرك و نتوب إليك

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المخلص:

تتناول هذه الأطروحة دراسة وتحليل سلوك تشغيل البرونز في عمليات الخراطة تحت حالتان مختلفة من التزيت: التشغيل الجاف، والتشغيل باستخدام زيت صناعي تقليدي. ويتمثل الهدف الرئيسي في تقييم تأثير معاملات القطع واستراتيجيات التزيت المختلفة على خشونة السطح (Ra)، والاهتزازات الناتجة عن القطع، وتآكل الحد القاطع للأداة (Vb)، مع مراعاة الجوانب البيئية والاقتصادية المرتبطة بكل أسلوب.

يقدم الفصل الأول مراجعة شاملة للأسس النظرية لعملية الخراطة، وخصائص مواد أدوات القطع، ومعايير تقييم الأداء، إضافةً إلى تقنيات التزيت المختلفة. كما يستعرض الدراسات السابقة التي تناولت تأثير التزيت على كفاءة التشغيل، وسلوك الاهتزازات، والجوانب المتعلقة بالاستدامة في عمليات إزالة المادة.

أما الفصل الثاني فيعرض المنهجية التجريبية المعتمدة في هذه الدراسة، حيث يوضح تجهيزات التشغيل المستخدمة، وخصائص قطعة الشغل المصنوعة من البرونز، ومواصفات أدوات القطع، وأجهزة القياس المستعملة، بما في ذلك جهاز قياس الخشونة ($Profilometer$)، والمجهر لتحليل التآكل، ومقياس التسارع لرصد الاهتزازات. وقد تم إعداد خطة تجريبية منظمة تسمح بإجراء مقارنة منهجية بين حالات التزيت الثلاث.

ويخصص الفصل الثالث للتحليل الإحصائي للنتائج التجريبية، حيث تم استخدام تحليل التباين ($ANOVA$) ومنهجية سطح الاستجابة (RSM) لبناء نماذج تنبؤية تربط بين معاملات القطع ومؤشرات الأداء. كما تمت مقارنة تأثيرات التشغيل الجاف، والزيت الصناعي، والزيت الصديق للبيئة على جودة السطح، ومستويات الاهتزاز، ومعدل تآكل الأداة، مع مناقشة النتائج بشكل مفصل.

ويتناول الفصل الرابع تحسين ظروف القطع، من خلال تطبيق طريقة تاكوشي ($Taguchi$) ودالة الرغبة ($Desirability Function$) لتحديد أفضل توليفة من معاملات التشغيل التي تحقق تقليلًا متزامنًا لخشونة السطح والاهتزازات وتآكل الأداة. كما تم اقتراح استخدام الشبكات العصبية الاصطناعية (ANN) والشبكات العصبية العميقة (DNN) كاتجاه مستقبلي لتعزيز نمذجة الأداء والتنبؤ به على المدى الطويل.

وتخلص الدراسة إلى أن استخدام الزيوت الصديقة للبيئة يمثل خيارًا واعدًا ومستدامًا، إذ يحقق توازنًا مناسبًا بين كفاءة الأداء الصناعي والتكلفة والمسؤولية البيئية في عمليات خراطة البرونز.

الكلمات المفتاحية: الخراطة، البرونز، استراتيجيات التزيت، خشونة السطح، اهتزازات القطع، تآكل الأداة، التحليل الإحصائي، تحسين العمليات، الزيوت الصديقة للبيئة.

Résumé

Cette thèse porte sur l'étude et l'analyse du comportement d'usinage du **bronze** lors des opérations de tournage sous 2 conditions de lubrification différentes : tournage à sec, tournage avec une huile industrielle conventionnelle. L'objectif principal est d'évaluer l'influence des paramètres de coupe et des stratégies de lubrification sur la rugosité de surface (R_a), les vibrations de coupe et l'usure en dépouille de l'outil (V_b), tout en tenant compte des aspects environnementaux et économiques associés à chaque méthode de lubrification.

Le premier chapitre présente une revue approfondie des fondements théoriques du procédé de tournage, des caractéristiques des matériaux des outils de coupe, des critères d'évaluation des performances ainsi que des différentes techniques de lubrification. Il examine également les travaux antérieurs portant sur l'influence de la lubrification sur l'efficacité de l'usinage, le comportement vibratoire et les enjeux liés à la durabilité dans les opérations d'enlèvement de matière.

Le deuxième chapitre décrit la méthodologie expérimentale adoptée dans cette recherche. Il détaille l'équipement d'usinage utilisé, les propriétés de la pièce en bronze, les spécifications des outils de coupe ainsi que les instruments de mesure employés, notamment un rugosimètre pour l'évaluation de l'état de surface, un microscope pour l'analyse de l'usure et un accéléromètre pour la mesure des vibrations. Un plan expérimental structuré a été élaboré afin de permettre une comparaison systématique des trois conditions de lubrification.

Le troisième chapitre est consacré à l'analyse statistique des résultats expérimentaux. L'Analyse de la Variance (ANOVA) et la Méthodologie des Surfaces de Réponse (RSM) ont été utilisées pour développer des modèles prédictifs reliant les paramètres de coupe aux indicateurs de performance. Les effets comparatifs du tournage à sec, de l'huile industrielle et de l'huile écologique sur la qualité de surface, le niveau des vibrations et l'usure de l'outil sont examinés et discutés en détail.

Le quatrième chapitre traite de l'optimisation des conditions de coupe. La méthode de Taguchi et la fonction de désirabilité ont été appliquées afin de déterminer la combinaison optimale des paramètres permettant de minimiser simultanément la rugosité de surface, les vibrations et l'usure de l'outil. L'intégration des Réseaux de Neurones Artificiels (ANN) et des Réseaux de Neurones Profonds (DNN) est également proposée comme perspective future pour améliorer la modélisation et la prédiction à long terme des performances d'usinage.

Cette étude conclut que l'utilisation d'huiles écologiques constitue une alternative prometteuse et durable, offrant un équilibre satisfaisant entre performance industrielle, maîtrise des coûts et responsabilité environnementale dans les applications de tournage du bronze.

Mots-clés : Tournage, Bronze, Stratégies de lubrification, Rugosité de surface, Vibrations de coupe, Usure de l'outil, Analyse statistique, Optimisation des procédés, Huiles écologiques.

Abstract:

This dissertation investigates the machining behavior of **bronze** during turning operations under three distinct lubrication conditions: dry cutting, machining with conventional industrial oil. The primary aim is to evaluate how different cutting parameters and lubrication approaches affect surface roughness (R_a), cutting-induced vibrations, and flank tool wear (V_b), while also considering the environmental and economic implications of each lubrication strategy.

The first chapter provides a comprehensive review of the theoretical foundations of turning processes, tool material characteristics, performance evaluation criteria, and lubrication techniques. It also examines prior studies addressing the influence of lubrication on machining efficiency, vibration behavior, and sustainability concerns in metal cutting operations.

The second chapter outlines the experimental procedure adopted in this research. It details the machining setup, the properties of the bronze workpiece, the specifications of the cutting tools, and the measurement devices employed, including a profilometer for surface assessment, a microscope for wear analysis, and an accelerometer for vibration monitoring. A structured experimental plan was implemented to enable a systematic comparison of the three lubrication conditions.

The third chapter presents a statistical interpretation of the experimental findings. Techniques such as Analysis of Variance (ANOVA) and Response Surface Methodology (RSM) were applied to establish predictive models linking cutting variables to performance indicators. The comparative effects of dry, industrial oil, and eco-friendly lubrication strategies on surface finish quality, vibration levels, and tool degradation are thoroughly examined.

The fourth chapter focuses on optimizing the machining parameters. The Taguchi method and desirability function approach were utilized to determine the optimal combination of cutting conditions that minimize surface roughness, vibration amplitude, and tool wear simultaneously. Additionally, the potential integration of Artificial Neural Networks (ANN) and Deep Neural Networks (DNN) is proposed for future research to enhance long-term predictive modeling of machining performance.

The study concludes that eco-friendly lubrication presents a viable and sustainable alternative, offering a balanced compromise between machining efficiency, cost considerations, and environmental responsibility in bronze turning applications.

Keywords: Turning, Bronze, Lubrication Strategies, Surface Roughness, Cutting Vibrations, Tool Wear, Statistical Analysis, Process Optimization, Eco-Friendly Oils.

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Nomenclature

Nomenclature

- a** – Depth of cut (mm)
a_l – Width of cut (mm)
a_t – Total depth of cut (mm)
D – Milling cutter diameter (mm)
F_n – Tangential cutting force (N)
f – Feed per revolution (mm/rev)
f_z – Feed per tooth (mm/tooth)
K_s – Specific cutting pressure (N/mm²)
k-b – Allowable bending stress of the shaft (MPa)
k-t – Allowable torsional stress of the shaft (MPa)
L – Workpiece length (mm)
L_a – Shaft length between supports (mm)
N-min – Minimum spindle speed (rev/min)
N_p – Batch size (number of parts per batch)
n – Spindle speed (rev/min)
P – Required cutting power (W)
P_m – Machine rated power (kW)
R_t – Total surface roughness (mm)
r_e – Tool nose radius (mm)
T – Machining time (min)
T_a – Setup and rapid traverse time (min/part)
T_d – Tool change time (min)
T_L – Loading and unloading time per part (min)
T_m – Effective machining time (min)
T-pr – Total production time per part (min/part)
V-c – Cutting speed (m/min)
Z – Number of cutting edges (teeth)
η – Machine efficiency
λ_s – Cutting edge inclination angle (°)

General Introduction

In today's industrial environment, improving machining processes has become essential for enhancing productivity, ensuring high-quality manufactured components, and promoting environmental sustainability. Turning, as one of the principal material removal operations, is extensively employed in shaping a wide range of engineering materials, including **bronze**, which is widely used in mechanical and industrial applications due to its favorable mechanical and tribological properties. Nevertheless, machining performance is strongly influenced by cutting parameters and lubrication strategies, as these factors directly affect tool life, surface integrity, and overall energy consumption.

Conventional machining practices commonly rely on abundant use of industrial cutting fluids to minimize friction and dissipate heat generated in the cutting zone. While effective in improving machining conditions, these traditional fluids raise significant environmental and health concerns because of their chemical composition and disposal requirements. In response to these challenges, modern manufacturing trends increasingly emphasize sustainable solutions, such as dry machining and the application of biodegradable vegetable-based oils. These alternatives aim to reduce the ecological footprint of machining operations while maintaining acceptable levels of technical performance.

Accordingly, this research work is dedicated to investigating and modeling the turning performance of **bronze** under three different machining conditions: dry cutting, machining with conventional industrial oil, and machining with environmentally friendly oil. The primary objective is to evaluate how these lubrication strategies influence critical process indicators, including surface roughness, tool wear, and cutting forces, in order to determine the optimal conditions that balance technical efficiency, economic feasibility, and environmental responsibility.

This thesis is organized as follows: it begins with a comprehensive review of previous studies related to sustainable machining and lubrication in turning processes. The experimental methodology adopted in this study is then described in detail, followed by the presentation and discussion of the obtained results. Finally, conclusions and recommendations are provided to guide future research efforts in this field.

I.Chapiter I: Study Bibliographic

I.1. Introduction:

Material removal machining remains one of the most widely applied manufacturing processes in the production of mechanical components. By progressively removing excess material from a raw workpiece, this process enables the achievement of precise geometries, tight dimensional tolerances, and high surface quality through the use of advanced machine tools. Over time, machining technologies have undergone continuous development to meet the growing industrial demands for higher productivity, improved accuracy, and reduced environmental impact. Today, manufacturers are expected not only to deliver components with superior technical performance, but also to adopt sustainable and economically viable production strategies.

Among engineering materials, **bronze** occupies an important position due to its excellent mechanical strength, corrosion resistance, and favorable tribological behavior. These properties make it particularly suitable for applications such as bearings, bushings, gears, and marine components. However, despite its good machinability compared to many ferrous alloys, machining bronze still presents specific challenges related to heat generation, tool wear, and surface finish quality. Consequently, the careful selection of cutting parameters and lubrication strategies plays a crucial role in ensuring stable machining performance and extending tool life.

In this context, the present study focuses specifically on the turning of bronze, with a comparative analysis of three lubrication conditions: dry machining, machining with conventional industrial oil, and machining using environmentally friendly oil. The objective is to determine the optimal cutting conditions that ensure high surface integrity, reduced tool wear, and controlled cutting forces, while simultaneously minimizing environmental impact and improving overall process efficiency.

I.2. Fundamental Concepts of Turning:

I.2.1. Turning:

Turning is a conventional machining process in which material is removed from a rotating workpiece by means of a single-point cutting tool to produce cylindrical or conical shapes with high dimensional accuracy and controlled surface finish. The operation is typically performed on a lathe, where the workpiece rotates about its longitudinal axis while the cutting tool advances linearly along or across the axis of rotation. Material removal occurs through shear deformation, generating chips as the cutting tool penetrates the surface under specific cutting parameters such as cutting speed, feed rate, and depth of cut. Turning can be applied to produce external and internal surfaces, including straight, tapered, grooved, and threaded profiles. The quality of the final product depends on factors such as tool geometry, cutting conditions, workpiece material properties, and lubrication strategy, all of which influence surface roughness, tool wear, cutting forces, and thermal effects during machining.[1]

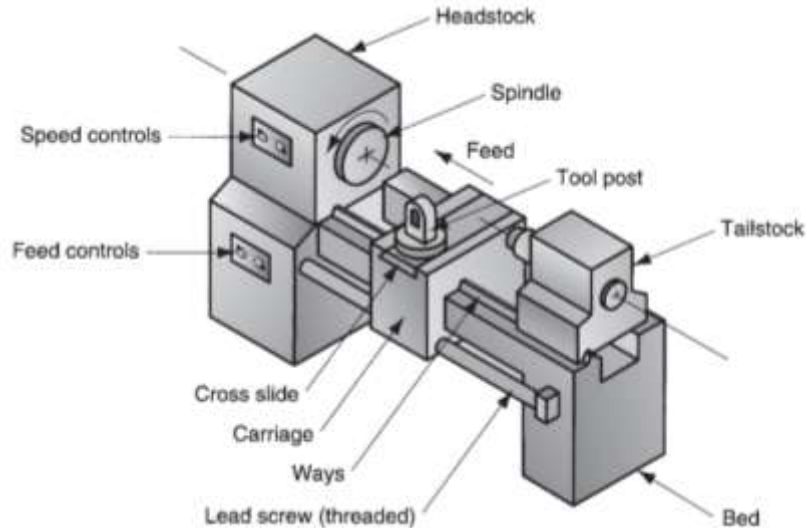


Figure I-1 Diagram of an engine lathe, indicating its principal components[1]

I.2.2. Turning Operations:**a) Straight Turning:**

Straight turning is a fundamental turning operation that consists of machining an external cylindrical or conical surface of a rotating workpiece. This operation is carried out using a cutting tool that moves parallel to, or at a certain angle relative to, the axis of the workpiece. The main objective of straight turning is to obtain uniform surfaces with precise dimensions and surface finishes that meet the required specifications. Straight turning can be performed in successive passes, ranging from roughing operations, which remove large amounts of material, to finishing operations, which ensure geometric accuracy and the final surface quality of the part. This operation plays a crucial role in the manufacturing of many mechanical components such as shafts, spindles, and bushings, where dimensional accuracy and surface quality are essential criteria.[2]

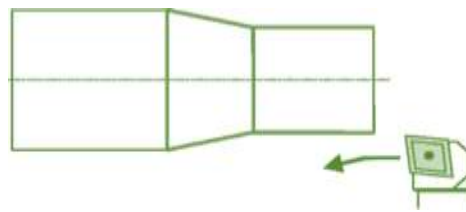


Figure I-2 Straight Turning [2]

b) Boring:

Boring is a turning operation that consists of machining the interior of an existing hole in order to improve its dimensional accuracy and surface finish. This operation is generally performed after drilling, using an appropriate cutting tool to obtain a precise diameter and tight tolerances.[2]

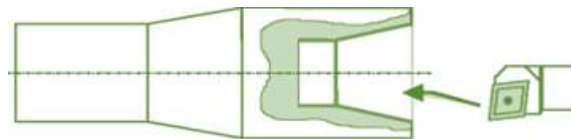


Figure I-3 Boring [2].

c) Facing:

Facing is a turning operation that consists of machining a flat surface, generally on the end face of a cylindrical workpiece. The cutting tool moves perpendicular to the axis of rotation of the workpiece in order to obtain a flat surface that is perpendicular to this axis [2].

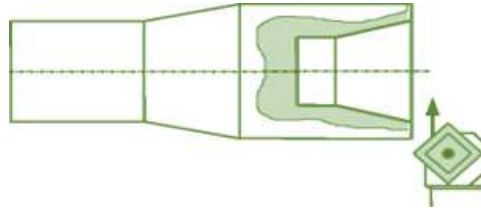


Figure I-4 Facing [2].

d) Drilling:

Drilling is a machining operation that consists of creating a hole in a solid workpiece using a drill bit. This operation is often the first step before other internal machining processes such as boring, and it allows the creation of pilot holes with a specified depth and diameter [2].

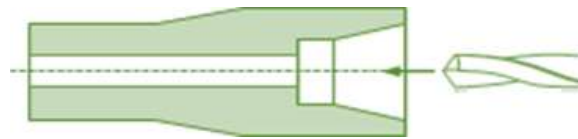


Figure I-5 Drilling [2].

e) Grooving:

Grooving is a turning operation that consists of machining a groove (or channel) on the external or internal surface of a workpiece. This operation is performed using a specific cutting tool and allows the creation of spaces for assembly elements such as circlips, seals, or for specific functional purposes [2].

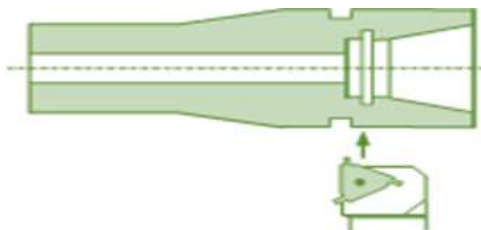


Figure I-6 Grooving [2].

f) Chamfering:

Chamfering is a turning operation that consists of machining a bevel on the edge of a workpiece, generally at a 45° angle. This operation helps facilitate assembly, remove sharp edges, and improve both the aesthetics and safety of mechanical parts [2].

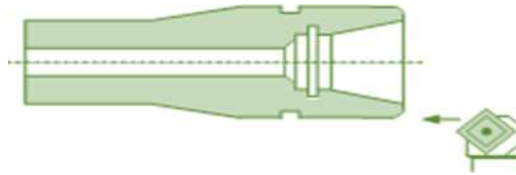


Figure I-7 Chamfering [2].

g) Parting:

Parting is a turning operation that consists of separating a workpiece from the raw material or cutting a specific section of an already machined part. This operation uses a thin cutting tool to make a clean cut perpendicular to the axis of rotation of the workpiece [2].

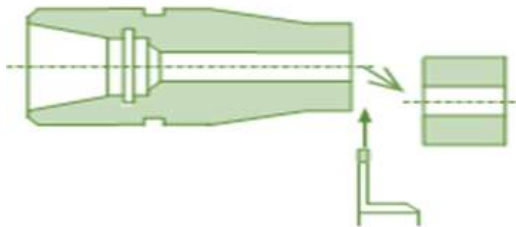


Figure I-8 Parting [2].

h) Threading:

Threading is a turning operation that consists of machining a helical thread on the external or internal surface of a workpiece to enable assembly by screwing. This operation requires a specific cutting tool and can be performed as external threading (such as for a bolt) or internal threading (such as for a nut) [2].

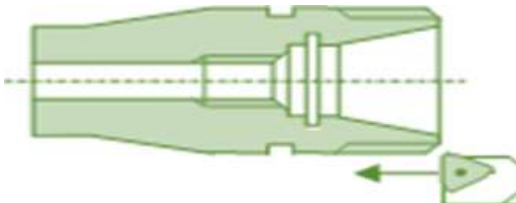


Figure I-9 Threading [2].

i) Knurling:

Knurling is the process of embossing a diamond shaped pattern on the surface of the workpiece. The knurling tool holder has one or two hardened steel rollers with edges of required pattern. The tool holder is pressed against the rotating work. The rollers emboss the required pattern. The tool holder is fed automatically to the required length. Knurls are available in coarse, medium and fine pitches. The patterns may be straight, inclined or diamond shaped.[4]

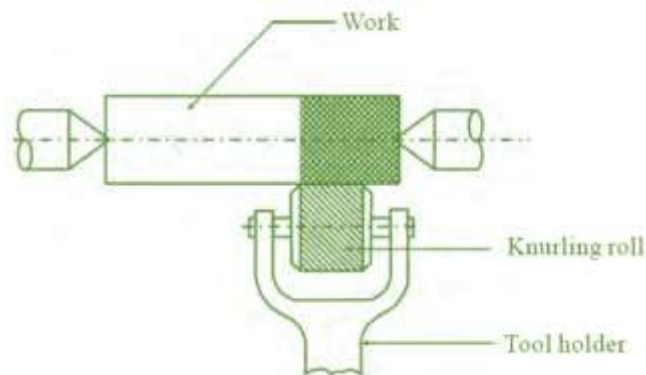


Figure I-10 Knurling [4].

j) Undercutting:

Undercutting is done (i) at the end of a hole (ii) near the shoulder of stepped cylindrical surfaces (iii) at the end of the threaded portion in bolts. It is a process of enlarging the diameter if done internally and reducing the diameter if done externally over a short length. It is useful mainly to make fits perfect. Boring tools and parting tools are used for this operation.[4]

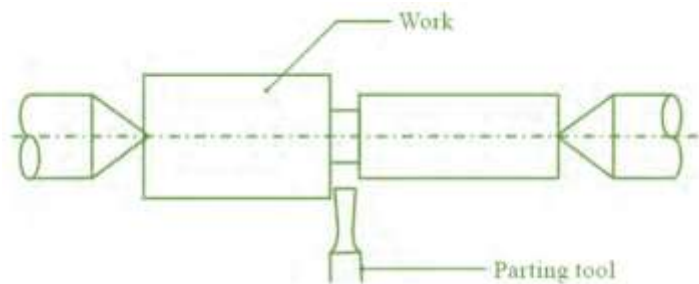


Figure I-11 Undercutting [4].

I.2.3. Lathes :

The lathe is a very versatile and important machine to know how to operate. This machine rotates a cylindrical object against a tool that the individual controls. The lathe is the forerunner of all machine tools. The work is held and rotated on its axis while the cutting tool is advanced along the line of a desired cut. The lathe is one of the most versatile machine tools used in industry. With suitable attachments, the lather may be used for turning, tapering, form turning, screw cutting, facing, dulling, boring, spinning, grinding, polishing operation. Cutting operations are performed with a cutting tool fed either parallel or at right angles to the axis of the work. The cutting tool may also be fed at an angle, relative to the axis of the work, for machining taper and angles. On a lathe, the tailstock does not rotate. Instead, the spindle that holds the stock rotates. Collets, centers, three jaw chucks, and other work-holding attachments can all be held in spindle. The tailstock can hold tools for drilling, threading, reaming, or cutting tapers. Additionally, it can support the end of the workpiece using a center and can be adjusted to adapt to different workpiece lengths.[3]

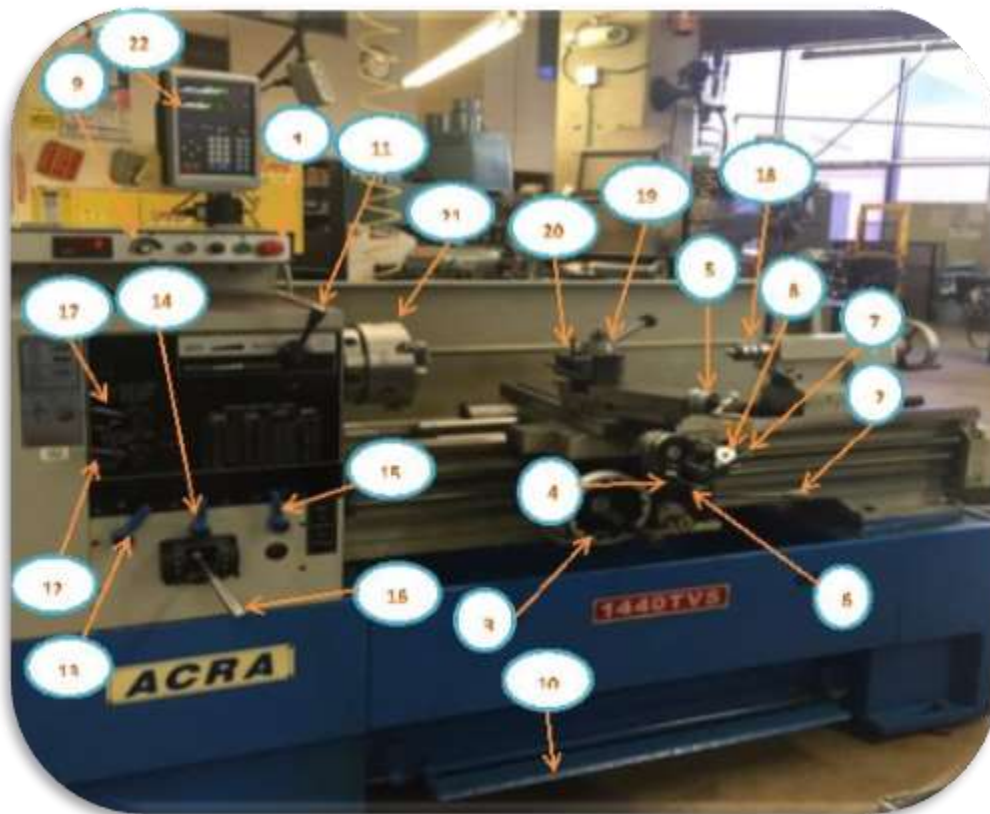


Figure I-12 Parts of a lathe [3]

1- Power On/Off	9- Brake	16- Gear Box Low/High
2- Spindle Forward/Reverse (flip handle up or down)	10- Spindle High/Low Range	17- Tailstock
3- Carriage Handwheel	11- Thread/Feed Reverse (push in/pull out)	18- Tool Post
4- Compound Feed Handwheel	12- Feed Ranges (A, B, C)	19- Toolholder
5- Carriage/Cross Feed Engage	13- Feed Ranges (R, S, T)	20- Three – Jaw Chuck
6- Threading Half Nut	14- Feed Ranges (V, W, X, Y, Z) – V and Z are settings for threading	21- DRO (Digital Read Out) Threading/Feed Selector (see item15)
7- Threading Dial		
8- Spindle Speed	15- Gear Box	

I.2.4. Tools used in a lathe:

Tools used in a lathe are classified as follows:

A. According to the construction, the lathe tools are classified into three types:[4]

1. Solid tool
2. Brazed tipped tool
3. Tool bit and tool holders

B. According to the operation to be performed, the cutting tools are classified as:[4]

- | | |
|------------------------|---------------------------------|
| 1. Turning tool | 6. Grooving tool |
| 2. Thread cutting tool | 7. Boring tool |
| 3. Facing tool | 8. Internal thread cutting tool |
| 4. Forming tool | 9. Knurling tool |
| 5. Parting tool | |

C. According to the direction of feed movement, the following tools are used:[4]

1. Right hand tool
2. Left hand tool
3. Round nose tool

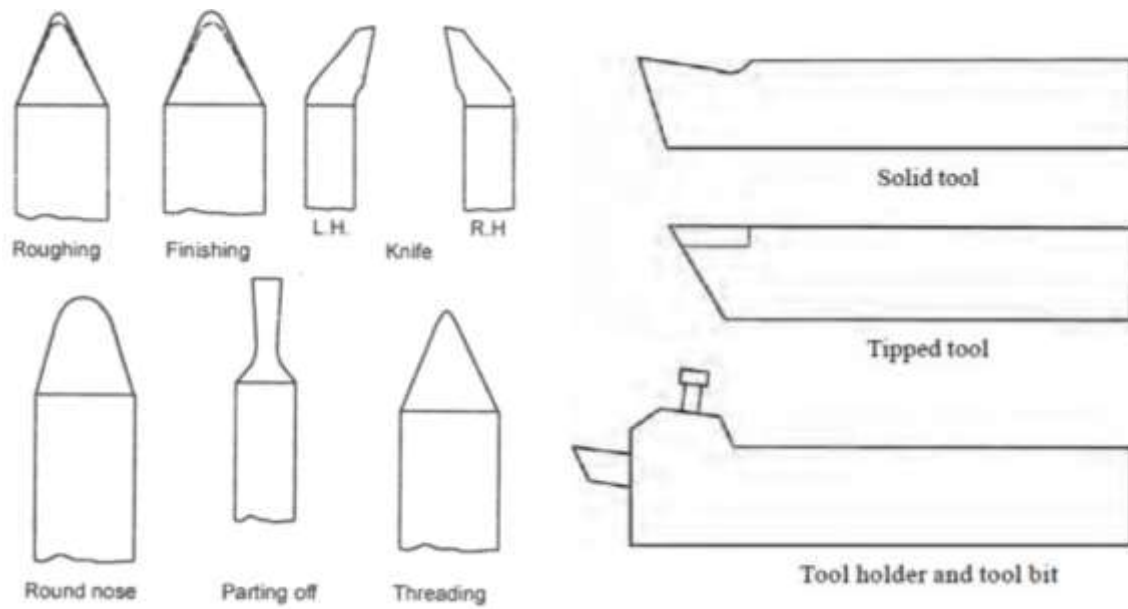


Figure I-13 Types of tools [4]

I.2.5. Cutting Tool Terminology:

There are many different tools that can be used for turning, facing, and parting operations on the lathe. Each tool is usually composed of carbide as a base material, but can include other compounds. This section covers the different appearances and uses of lathe cutting tools.[3]

Figure A: depicts a standard turning tool to create a semi-square shoulder. If there is enough material behind the cutting edge, the tool can also be used for roughing.[3]

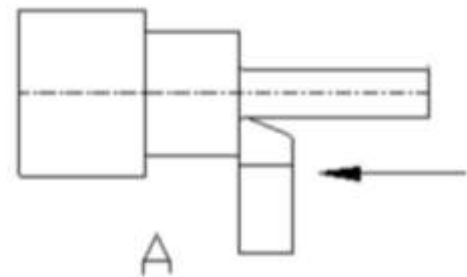


Figure B: depicts a standard turning tool with a lead angle. This angle enables for heavy roughing cuts. It is also possible to turn the tool to create a semi-square shoulder.[3]

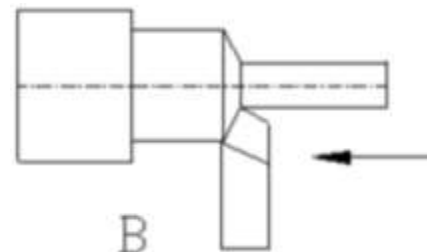


Figure C: nose has a very large radius, which helps with fine finishes on both light and heavy cuts. The tool can also be used to form a corner radius.[3]

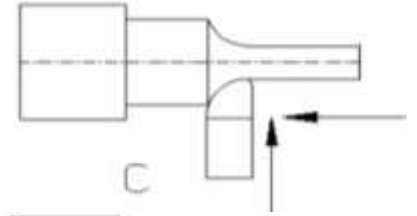


Figure D: depicts a rotated standard turning tool. Its nose leads the cutting edge to create light finishing cuts on the outside diameter and face of the shoulder.[3]

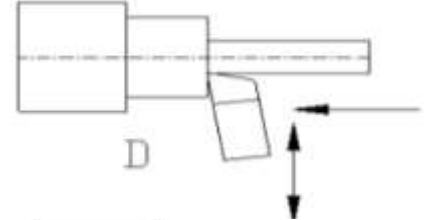


Figure E: depicts a form tool. Different forms can be ground into the tool, which will be reproduced onto the part.[3]

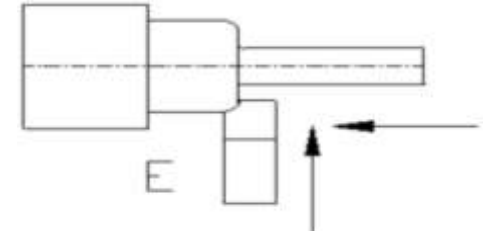


Figure F: depicts a facing tool. This cutter is used to face the end of a workpiece to provide for a smooth, flat finish. If the stock has a hole in the center, utilize a half-center to stabilize and support the workpiece.[3]

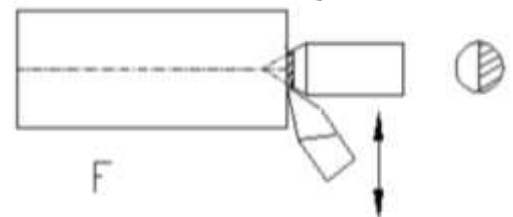


Figure G: depicts a grooving or under-cutting tool. As shown, it is used to cut grooves into the workpiece. When there are proper clearances, the tool can cut deeply, or cut to the left or right.[3]

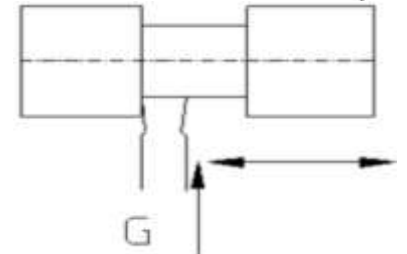


Figure H: depicts a parting tool. Parting tools cut off the stock at a certain length. This tool requires a preformed blade and holder.[3]

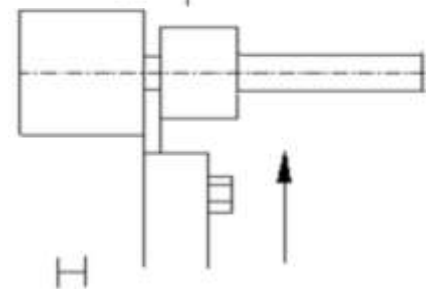
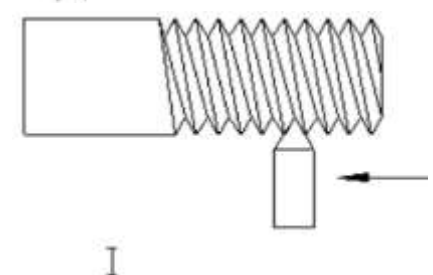


Figure I: depicts a 60° threading tool used to thread stock.[3]



I.3. Fundamental Concepts of Milling:

I.3.1. Milling:

Milling is a cutting process that involves, in coordination, the rotational motion of a multi-edge cutting tool (cutting motion M_c) and the linear feed motion of the workpiece (called feed motion M_f). There is also a relative displacement of the tool with respect to the workpiece, which can be carried out in practically any direction.[8]

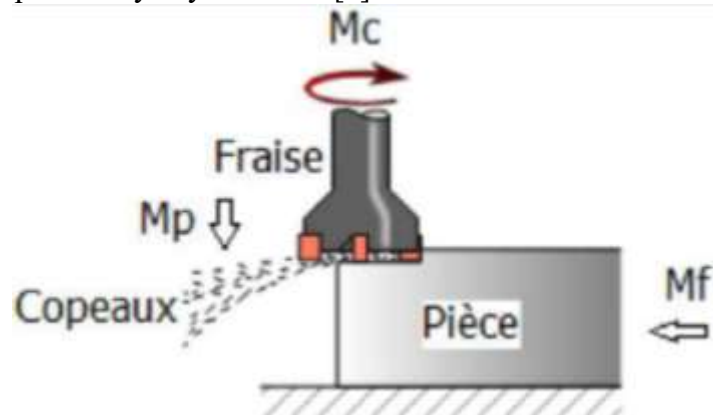


Figure I-14 An Image of Milling [8]

I.3.2. Types of Milling:

A- Face Milling (or End Milling):

- The axis of the milling cutter is perpendicular to the machined surface.
- The axis of the milling cutter is parallel to the machined surface.

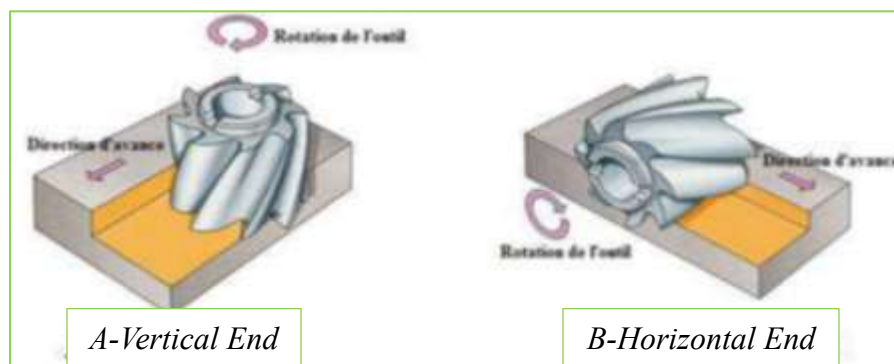


Figure I-15 Vertical End Mill and Horizontal End Mill.[8]

B- Conventional Milling and Climb Milling:

- In **conventional milling (up milling)**, the feed direction is opposite to the direction of rotation of the cutter.
 - ❖ The chip thickness starts at zero and increases toward the end of the cut. The cutting forces tend to push the cutter away from the workpiece.
 - ❖ The high tensile stresses exerted on the cutting edge as it leaves the material often lead to edge failure or breakage.
- In **climb milling (down milling)**, the tool advances in the same direction as its rotation.
 - ❖ Climb milling should always be preferred if the machine tool, clamping system, and workpiece rigidity allow it [7].
 - ❖ In climb contour milling, the chips are thicker at the entry than at the exit. This prevents the cutting edge from rubbing against and burnishing the surface before engaging in the cut.

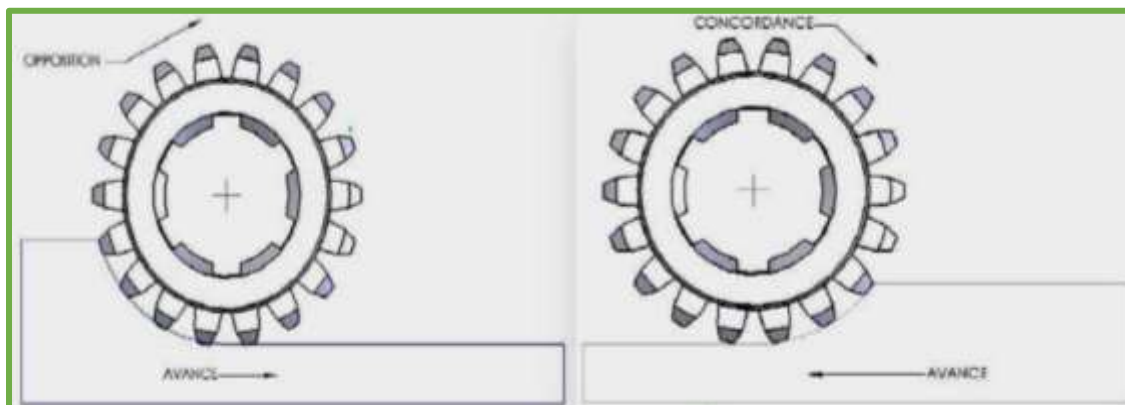


Figure I-16 Conventional (Up) Milling and Climb (Down) Milling [8].

I.3.3. Characteristics of Milling Cutters (Tools):

A- Type of Cutting Edges: according to the number of faces carrying cutting edges, the following are distinguished (Figure I-17) [7]:

- ✓ Single-edge milling cutters.
- ✓ Double-edge milling cutters.
- ✓ Triple-edge milling cutters.



Figure I-17 The Milling Cutters Edges [6].

B- Shape: according to the profile of the generatrices relative to the tool axis, the following types are distinguished (Figure I-18) [6]:

- ❖ Cylindrical milling cutters.
- ❖ Conical and double-conical milling cutters.
- ❖ Inserted-blade (or inserted-tooth) milling cutters.
- ❖ Form milling cutters.
- ❖ Shell milling cutters.

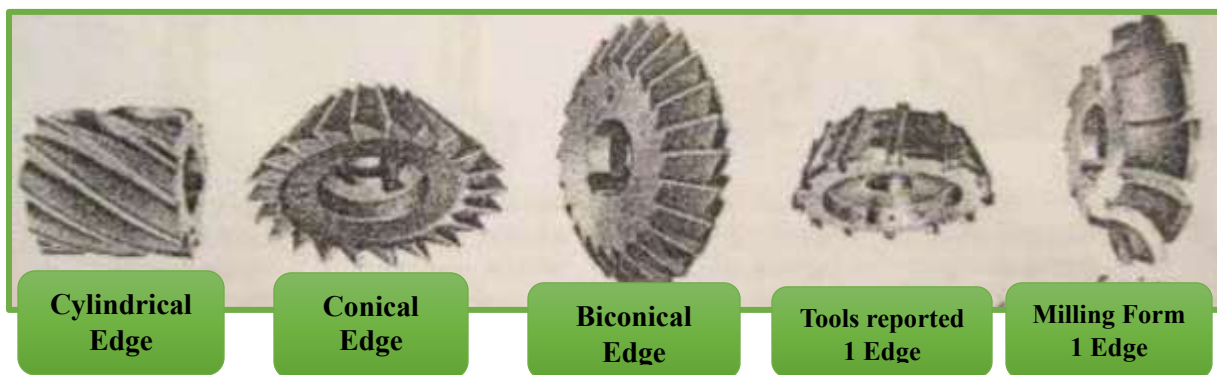


Figure I-18 Shapes of Milling Cutters [6].

C- Tooth Geometry (Helix/Teeth):

- If the cutting edge of the milling cutter is parallel to the cutter axis, it is called a **straight-tooth cutter**.
- If the cutting edge is inclined relative to the cutter axis, there are:
 - **Right-hand or left-hand helical teeth.**
 - **Alternating double-helix teeth.**

A milling cutter is also characterized by its **number of teeth** [7].



Figure I-19 Tooth Geometries of Milling Cutters [6].

D- Mounting Method:

Two distinct mounting methods are distinguished:

- **Hole mounting:** plain, plain with keyway, or threaded.
- **Shank mounting:** cylindrical or tapered [7].

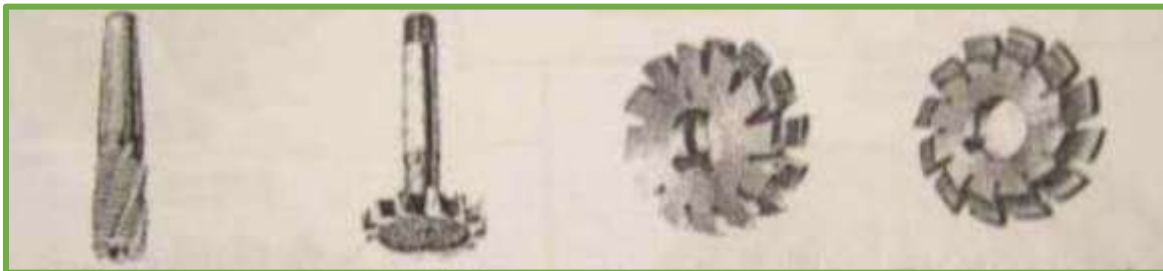


Figure I-20 Modes de fixation des fraises [6].

E-Tool Materials:

Cutting tools are used to remove material from the workpiece by cutting. Therefore, cutting tools must be made from a material harder than the workpiece material and must be capable of withstanding the heat generated during the metal cutting process.[8]

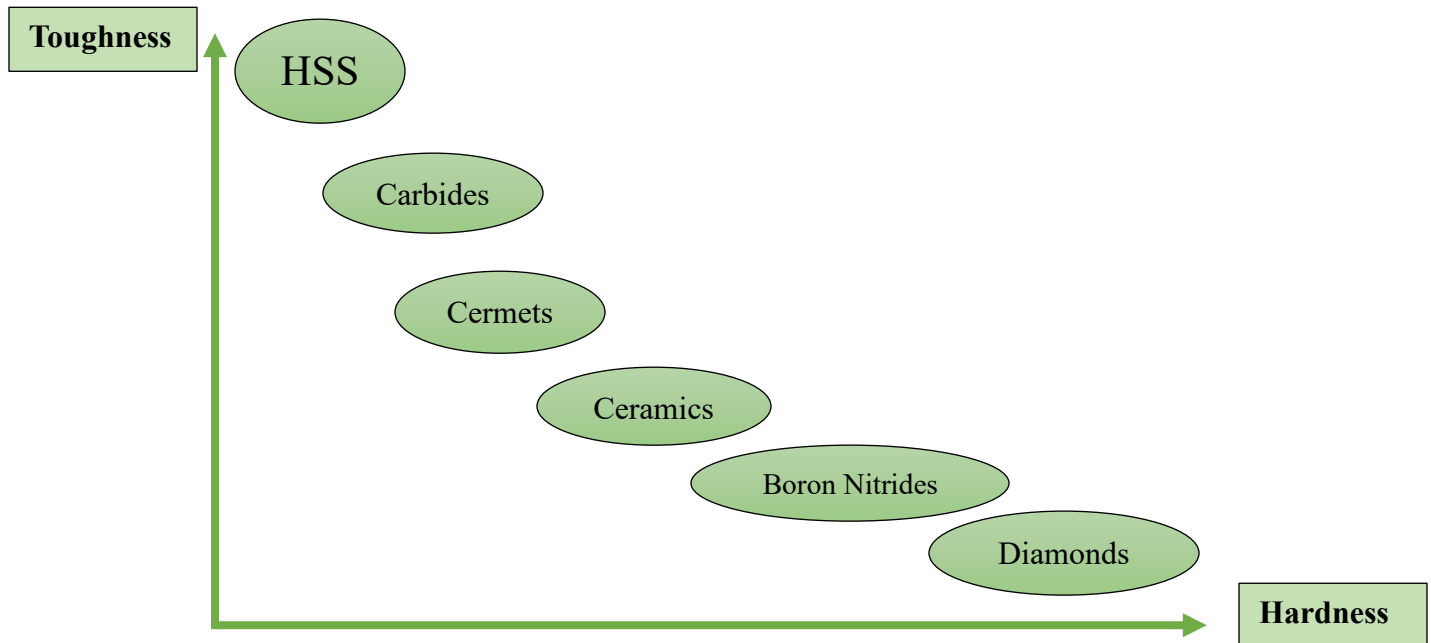


Figure I-21 Chart Showing Tool Materials in Terms of Hardness and Toughness.[5]

a) High-Speed Steel (HSS):

HSS tools (High-Speed Steel) are made from a low-alloy steel that undergoes heat treatment. They are commonly used for certain types of tools such as drills or tools requiring a very small cutting-edge angle. HSS tools do not allow high cutting speeds because excessive heating can remove the hardness of the tool, leading to rapid failure of the cutting edge [5].

Manufacturing: shell casting or powder metallurgy.

Composition:

- Minimum 0.7% Carbon
- Approximately 4% Chromium
- Tungsten, Molybdenum, Vanadium, and Cobalt for the hardest grades.

Hardness: 63–66 HRC

b) Carbides:

Carbide tools are currently the most widely used. They are available in various shapes for each type of material and machining operation. They are typically supplied as inserts that are mounted on a tool holder, allowing very quick replacement. These inserts are often coated with a harder carbide, resulting in a tool with a tough core and an extremely hard outer surface [5].

Manufacturing: powder sintering followed by coating.

Composition:

- Core: Tungsten carbide (melting point 2600 °C), or titanium carbide (3100 °C), tantalum carbide (3780 °C), or molybdenum carbide (3500 °C)
- Binder: cobalt (most common) or nickel
- Coating: aluminum oxide (ceramic called corundum: Al_2O_3)

c) Cermets:

The name comes from ceramic–metal because these are carbides containing particles of titanium, titanium carbonitride, or titanium nitride. These tools are usually combined with molybdenum carbide to increase their toughness. They are used for high cutting speeds with low feed rates, making them suitable for finishing operations. Since the material is brittle, cut interruptions (such as joint planes) should be avoided [5].

d) Ceramics:

For cutting tools, ceramics include oxides and nitrides, such as aluminum oxide and silicon nitride. Ceramic tools have very high hardness (but low toughness), excellent thermal stability, and no chemical reaction with the workpiece material. Ceramics allow high material removal rates, but require high machine stability, strict adherence to cutting conditions, and an adapted machining method (tool approach) [5].

e) Cubic Boron Nitride (CBN):

CBN offers very high hardness, making it the hardest material after diamond. Compared to diamond, it has the advantage of **not oxidizing at high temperatures**. It is also used for making grinding wheels and for machining hard workpieces.

Its use requires:

- A stable machine.
- High rigidity of the workpiece and the work holding device.
- Coolant or lubrication [5].

f) Diamond:

Diamond is widely used as a component of grinding wheels or as grains for re-sharpening wheels. It has a **low coefficient of friction**, which limits built-up edge formation and reduces clogging.

However, its major drawback is **instability at high temperatures**. A diamond exposed to temperatures above 650 °C transforms into ordinary graphite, so it **cannot be used for ferrous materials**.

On the other hand, it is suitable for **non-ferrous materials** machined at low temperatures, such as aluminum, copper, magnesium alloys, and thermosetting resins [5].

I.3.4. Milling Machines:

Milling machines must provide a rotating spindle for the cutter and a table for fastening, positioning, and feeding the workpart. Various machine tool designs satisfy these requirements. To begin with, milling machines can be classified as horizontal or vertical. A horizontal milling machine has a horizontal spindle, and this design is well suited for performing peripheral milling (e.g., slab milling, slotting, side and straddle milling) on workparts that are roughly cube shaped. A vertical milling machine has a vertical spindle, and this orientation is appropriate for face milling, end milling, surface contouring, and diesinking on relatively flat workparts. Other than spindle orientation, milling machines can be classified into the following types: [1]

- | | |
|---------------------|---------------------------|
| (1) knee-and-column | (4) tracer mills |
| (2) bed type | (5) CNC milling machines. |
| (3) planer type | |

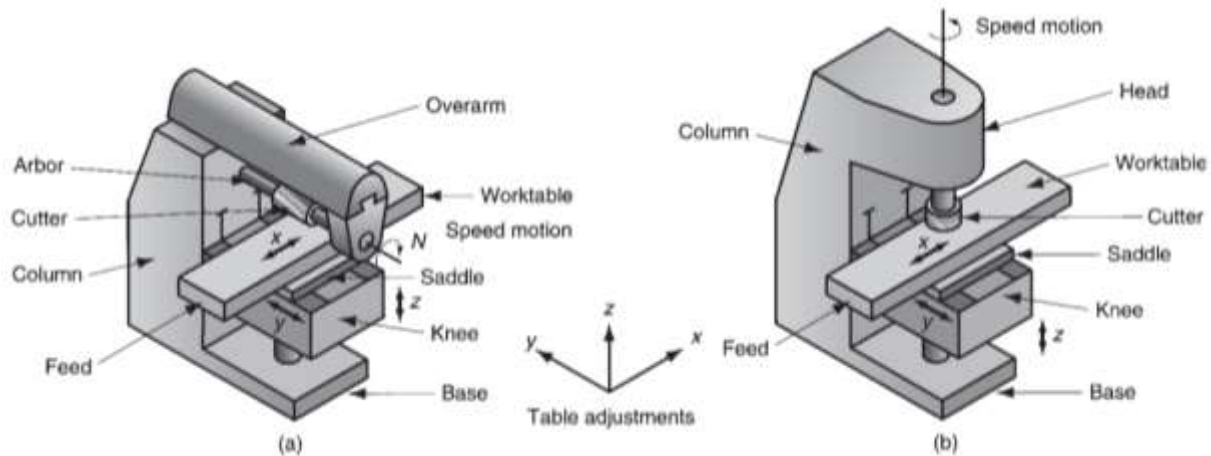


Figure I-22 Two basic types of knee-and-column milling machine: (a) horizontal and (b) vertical.[1]

The knee-and-column milling machine is the basic machine tool for milling. It derives its name from the fact that its two main components are a column that supports the spindle, and a knee (roughly resembling a human knee) that supports the worktable. It is available as either a horizontal or a vertical machine, as illustrated in **Figure 22**. In the horizontal version, an arbor usually supports the cutter. The arbor is basically a shaft that holds the milling cutter and is driven by the spindle. An overarm is provided on [1]

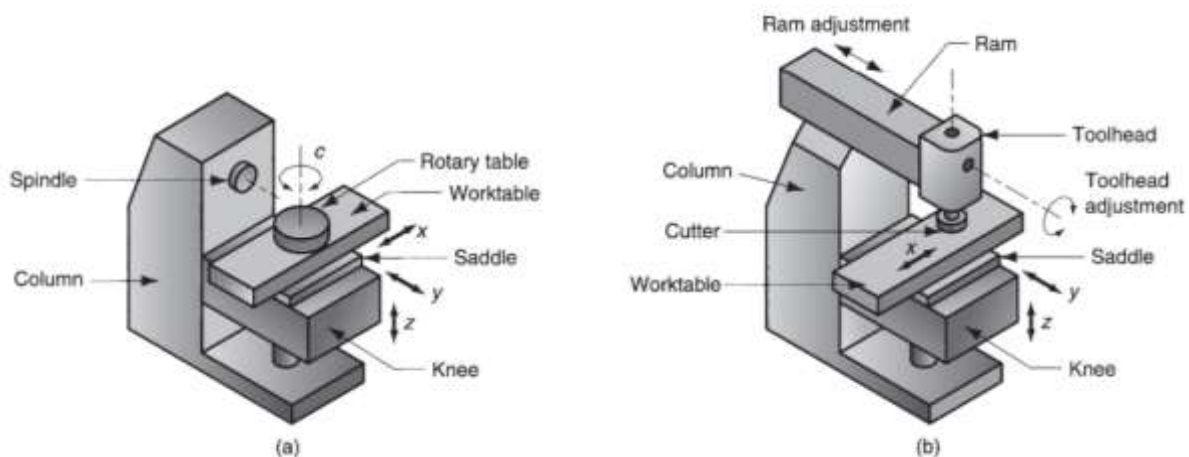


Figure I-23 Special types of knee-and-column milling machine: (a) universal-overarm, arbor, and cutter omitted for clarity: and (b) ram type.[1]

Bed-type milling machines are designed for high production. They are constructed with greater rigidity than knee-and-column machines, thus permitting them to achieve heavier feed rates and depths of cut needed for high material removal rates. The characteristic construction of the bed-type milling machine is shown in Figure 24.

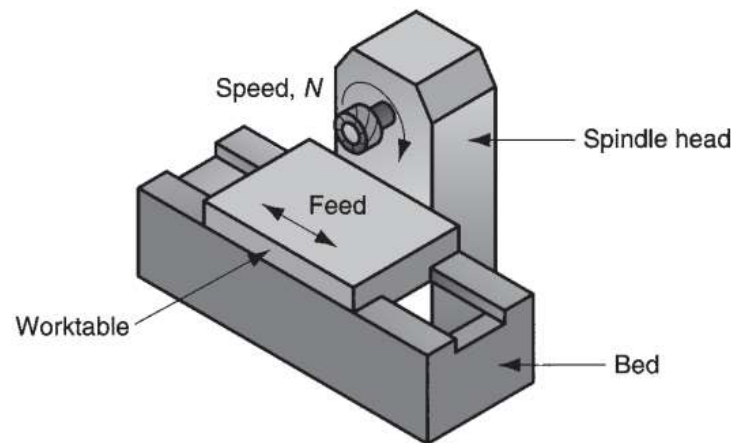


Figure I-24 Simplex bedtype milling machine horizontal spindle.[1]

Planer type mills are the largest milling machines. Their general appearance and construction are those of a large planer (see Figure 25); the difference is that milling is performed instead of planing. Accordingly, one or more milling heads are substituted for the single-point cutting tools used on planers, and the motion of the work past the tool is a feed rate motion rather than a cutting speed motion. Planer mills are built to machine very large parts. The worktable and bed of the machine are heavy and relatively low to the ground, and the milling heads are supported by a bridge structure that spans across the table.[1]

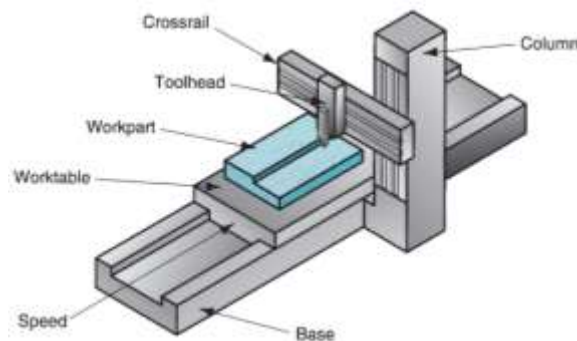


Figure I-25 Planer type mills.[1]

CNC turning center, Figure 26, is capable of performing various turning and related operations, contour turning, and automatic tool indexing, all under computer control. In addition, the most sophisticated turning centers can accomplish (1) workpart gaging (checking key dimensions after machining), (2) tool monitoring (sensors to indicate when the tools are worn), (3) automatic tool changing when tools become worn, and even (4) automatic workpart changing at the completion of the work cycle.[1]



Figure I-26 Computer numerical control, four-axis turning center.
(Courtesy of Cincinnati Milacron, Batavia, Ohio.).[1]

I.4. Cutting Parameters and Performance Criteria:

Cutting parameters are the fundamental input variables governing turning operations. The principal parameters are cutting speed (V_c), feed rate (f), and depth of cut (a_p). These variables significantly influence cutting forces, temperature generation, tool wear, and surface integrity [9].

I.4.1. Input Parameters:

A) Cutting Speed (V_c)

Cutting speed is defined as the tangential velocity between the tool and the rotating workpiece and is expressed as:

$$V_c = \frac{\pi DN}{1000} \quad (1)$$

where:

V_c = cutting speed (m/min)

D = workpiece diameter (mm)

N = spindle speed (rev/min) [11]

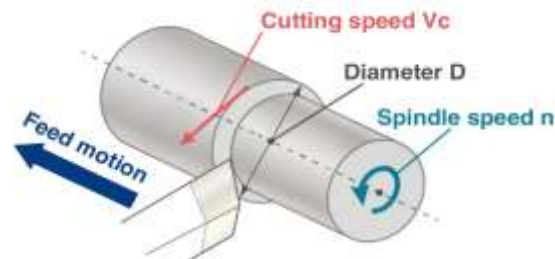


Figure I-27 Cutting speed

Tool life is commonly estimated using Taylor's tool life equation:

$$V_c T^n = C \quad (2)$$

where:

T = tool life (min)

n = tool life exponent

C = constant depending on tool–workpiece combination

This equation and its derivation are presented in [10],[9],

B) Feed Rate (f):

In turning, feed rate is expressed in mm/rev. Surface roughness in ideal turning conditions can be approximated by:

$$Ra = \frac{f^2}{32r} \quad (3)$$

where:

Ra = arithmetic average roughness (μm)

f = feed (mm/rev)

r = tool nose radius (mm)[10].

The lower the Ra value, the less variation between the peaks and troughs on a surface, making the surface smoother. For example, a lego block will have a low Ra value, as will a laptop's

touchpad. With a higher R_a value, these products would be highly textured and coarser and therefore potentially unsuitable for their desired use. This contrast of R_a values illustrates the importance of determining the desired surface roughness of a product before the manufacturing process begins. Without such determinations, machining finishes of a product may vary significantly from what was initially intended.[16]

The illustrations below demonstrate the variations between the values of R_a (the numerical average of all the peaks and valleys across the length of the test) and R_z (the average of consecutive highest peaks and lowest valleys).[16]

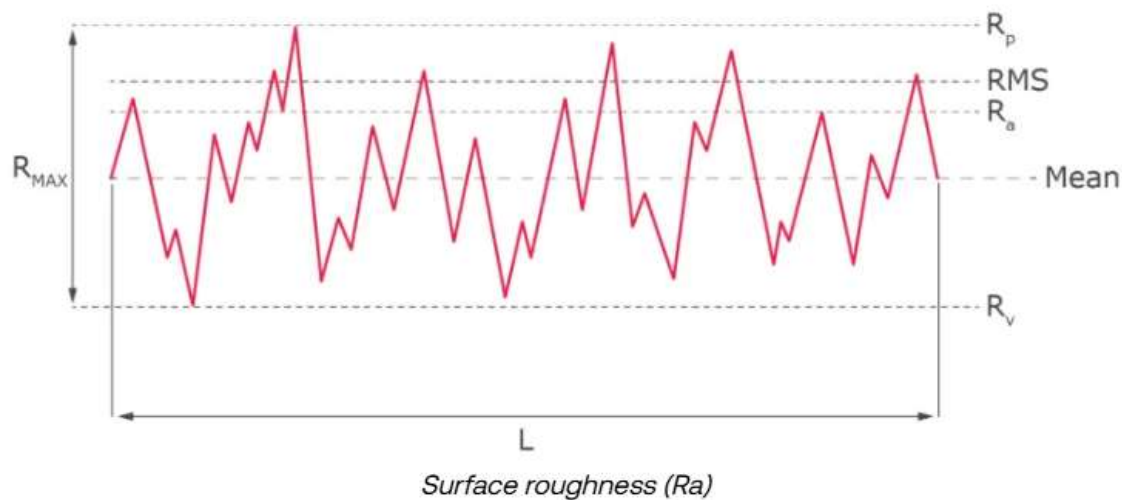


Figure I-28 Surface Roughness (R_a).[16]

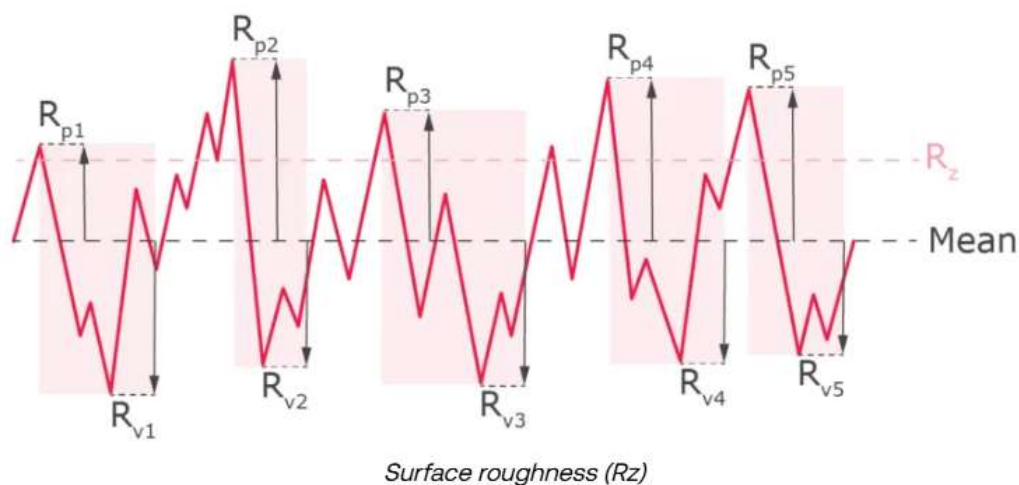


Figure I-29 Surface Roughness (R_z).[16]

Surface roughness terminology:

Ra - The numerical average of all the peaks and valleys across the length of the test. It's also called the Center Line Average (CLA).

Rz - The average of consecutive highest peaks and lowest valleys. Vertical distance between the highest peak and lowest valley, the distance of the second highest peak and the second-lowest valley, etc. This is usually done for the five biggest deviations, and then an average is calculated.

Rp - The calculated distance between the profile's tallest peak and the mean line within the evaluation length.

Rv - The calculated distance between the profile's lowest valley and the mean line within the evaluation length.

Rmax - The biggest successive deviation between the highest peak and the lowest valley, calculated within the evaluation length.

RMS - Calculated within the evaluation length, this is the root mean square average of profile height variation from the mean line.[16]

C) Depth of Cut (a_p):

Depth of cut is defined as:

$$a_p = \frac{D_0 - D_f}{2} \quad (4)$$

where:

D_0 = initial diameter

D_f = final diameter [11],

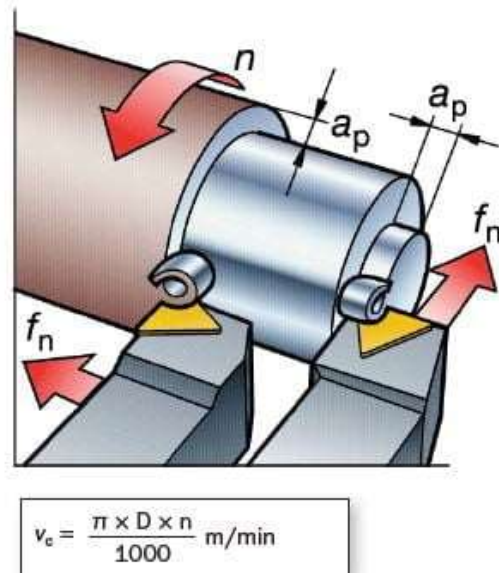


Figure I-30 Depth of cut

D) Cutting Temperature

The heat distribution in metal cutting can be summarized as follows [9], [15]:

60–80% carried by the chip

10–20% absorbed by the workpiece

~10% conducted into the tool

The cutting temperature rise affects tool life and dimensional accuracy. According to Shaw [15], temperature depends strongly on cutting speed and friction conditions.

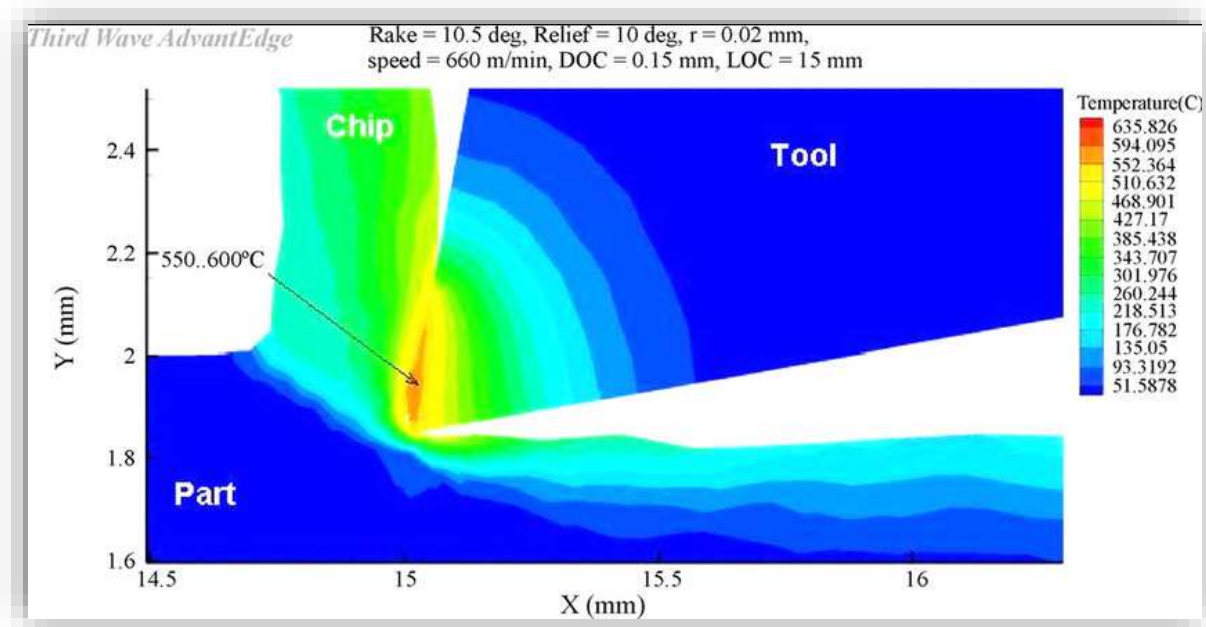


Figure I-31 Cutting Temperature

E) Lubrication Effects:

Lubrication is the process or technique used to reduce friction and wear between one or two surfaces in close proximity and moving relative to each other by introducing a substance called a lubricant between them. There are various types of lubricants, including oils, oil–water emulsions, pastes, gels, aerosols (mists), and air or other gases. They can be produced from petroleum distillates, animal fats, vegetable oils, water and air, or other raw materials.[2]

Depending on the context and the type of cutting fluid considered, it may be referred to as a cutting fluid, cutting oil, cutting agent, coolant, or lubricant. Lubrication in hard machining presents specific challenges. The selection of an appropriate lubricant, the determination of optimal lubrication parameters (pressure, flow rate, droplet size, etc.), and the optimization of overall performance are critical factors in achieving effective machining conditions.[2]

Cutting fluids reduce friction and heat generation. The coefficient of friction (μ) at the tool–chip interface influences the shear angle (ϕ), which is given by Merchant's equation:

$$\phi = 45^\circ + \frac{\alpha}{2} - \frac{\beta}{2} \quad (5)$$

where:

α = rake angle

β = friction angle ($\tan\beta = \mu$)

This relationship is discussed in [15],

Lubrication mechanisms are detailed in [12],



Figure I-32 Lubrication Effects

I.4.2. Output Performance Criteria

A) Surface Roughness Parameters

According to ISO 4287 [13], R_a is defined as:

$$R_a = \frac{1}{L} \int_0^L |y(x)| dx \quad (6)$$

RMS roughness:

$$R_q = \sqrt{\frac{1}{L} \int_0^L y^2(x) dx} \quad (7)$$

Definitions and illustrations can be found in [11].

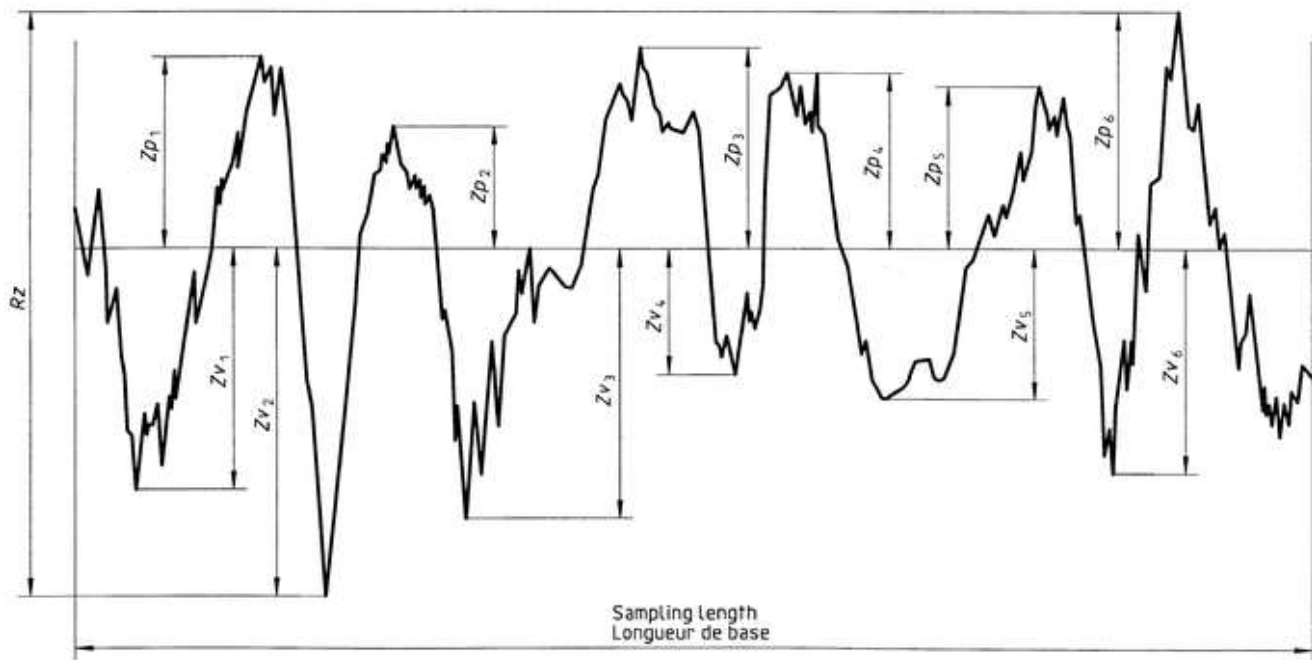


Figure I-33 ISO 4287:1997 (Fr), Geometrical Product Specifications (GPS) — Surface texture: Profile method — Terms, definitions and surface texture parameters

Methods of Roughness Measurement:

In industry, the most commonly used method is a contact-type surface roughness measuring device, where a probe contacts the surface to be characterized (2D mechanical profilometer). Depending on its level of sophistication, this type of equipment allows the measurement of all or part of the surface texture parameters. In fact, the evaluation of primary profile and waviness parameters requires an external reference for guidance.[2]



Figure I-34 Digital Surface Roughness Tester

B) Tool Wear:

Tool wear is a critical aspect in the field of machining. Over time, cutting tools undergo a wear process that can significantly affect their performance and the quality of the machined parts. Understanding the mechanisms and factors influencing tool wear is essential for optimizing machining processes, extending tool life, and improving part quality [9][11].

- **Crater Wear:** Crater wear occurs when the friction between the chip and the rake face of the tool leads to a decrease in tool hardness, either by diffusion or by adhesion of particles onto the chip due to increased cutting temperature. This type of wear is common, especially when machining materials with low thermal conductivity. The maximum depth of the crater, defined by K_T , characterizes the crater shape [9].
- **Flank Wear:** Flank wear results from friction between the machined surface and the flank face of the tool, appearing as a shiny, striated band parallel to the cutting edge. This wear

is measurable by the average flank wear width V_B and directly affects surface finish and dimensional accuracy [9][11].

- **Chipping and Edge Cracking:** Small fragments of the tool break due to mechanical impact or thermal shock from cyclic heating and cooling. This type of wear is common in interrupted cuts or unstable machining conditions (vibration or chatter) [11].
- **Notch Wear:** Notch wear develops at the intersection of crater and flank wear, where the main cutting edge meets the workpiece surface along the depth-of-cut line. It often occurs during turning of materials prone to work hardening and is characterized by V_N [11].
- **Plastic Deformation:** Excessive temperature and high pressures in the cutting zone, particularly at the tool tip, can cause permanent deformation of the tool, manifesting as displacement on the flank faces [9][11].
- **Sudden Tool Breakage:** This affects cutting tools with low toughness or fracture resistance compared to the workpiece material, as is common with ceramic tools [11].

Flank wear width (V_B) and crater wear depth (K_T) are standard tool wear indicators [9].

Tool life criterion often defined as:

$$VB_{max} = 0.3 \text{ mm (carbide tools) (8)}$$

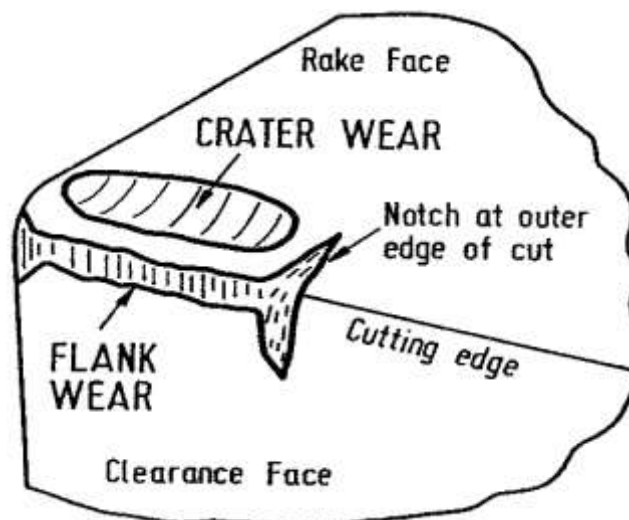


Figure I-35 Crater and flank wear.[10]

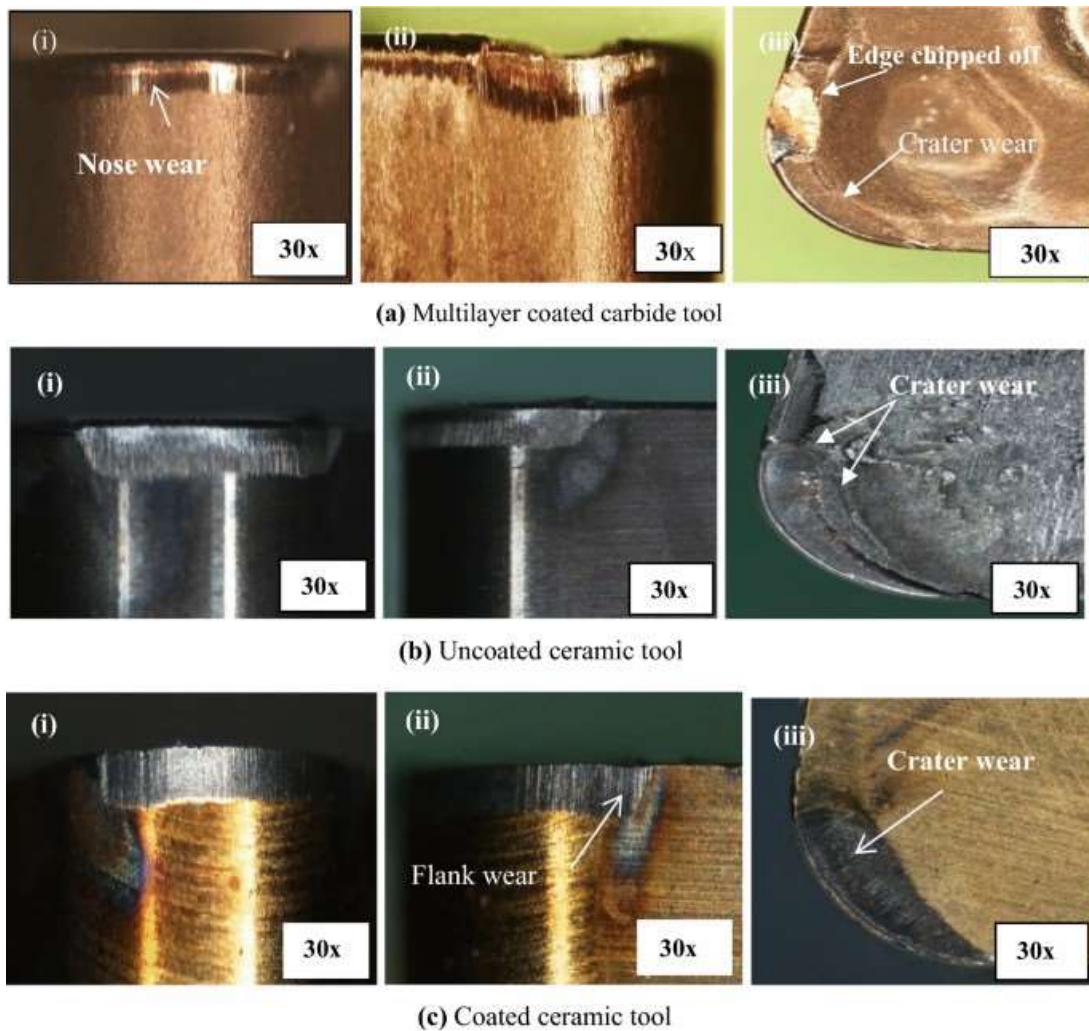


Figure I-36 Experimental Investigation on Tool Wear in AISI H13 Die Steel Turning Using RSM and ANN Methods

C) Vibrations in Turning:

Vibration is a fundamental dynamic phenomenon encountered in turning operations. During the interaction between the cutting tool and the rotating workpiece, dynamic cutting forces are generated, which may induce oscillatory motion in the machine–tool–workpiece system. These vibrations can significantly influence surface quality, dimensional accuracy, tool wear, and process stability [9][14].

According to Altintas [14], machining vibrations are generally classified into:

- Free vibrations (due to initial disturbances),
- Forced vibrations (caused by periodic external forces such as spindle imbalance),
- Self-excited vibrations (chatter), which are the most critical in machining.

Self-excited vibrations arise from the regenerative effect, where the waviness left from a previous tool pass modulates chip thickness in the subsequent pass, amplifying oscillations [14][15].

Dynamic Model of Vibration in Turning

The machining system can be approximated as a single-degree-of-freedom (SDOF) system:

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

Where:

- m = equivalent mass
- c = damping coefficient
- k = stiffness of the system
- x = displacement
- $F(t)$ = cutting force excitation

The natural frequency of the system is:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

When the excitation frequency approaches the natural frequency, resonance may occur, leading to excessive vibration amplitudes [14].

Time-Domain Vibration Parameters:

When a mechanical system vibrates, it oscillates around an equilibrium position. The motion can be expressed as:

$$v(t) = \frac{dx(t)}{dt}$$
$$a(t) = \frac{d^2x(t)}{dt^2}$$

These parameters are commonly used in vibration monitoring in machining environments [11].

Main Indicators:

- Peak-to-Peak Value: Difference between maximum and minimum amplitude; useful for evaluating total displacement and mechanical clearance [11].
- Peak Value: Maximum instantaneous amplitude; important for detecting shocks and impacts.
- RMS (Root Mean Square):

$$RMS = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}$$

RMS represents the effective vibration energy and is widely used in condition monitoring and predictive maintenance [11].

- Crest Factor:

$$CF = \frac{Peak}{RMS}$$

It is particularly sensitive to impulsive phenomena such as tool chipping [12].

- Frequency (Hz):

$$f = \frac{1}{T}$$

Frequency analysis is essential for identifying chatter and structural resonances in machining systems [14].

Impact of Vibration on Surface Roughness and Tool Life:

Excessive vibration, especially chatter, results in:

- Increased surface roughness
- Dimensional deviations
- Accelerated flank wear
- Possible tool fracture

Experimental studies in machining processes have demonstrated a strong correlation between cutting parameters and vibration amplitude [6][7][8]. Increasing depth of cut and feed rate generally increases dynamic cutting forces, which may destabilize the process if system rigidity is insufficient [9][10].

From a tribological perspective, vibration also modifies contact conditions at the tool–chip interface, affecting friction and wear mechanisms [12].

The vibration frequency is defined as:

$$f = \frac{1}{T} \quad (9)$$

RMS vibration value:

$$RMS = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt} \quad (10)$$

These formulations are discussed in [14].

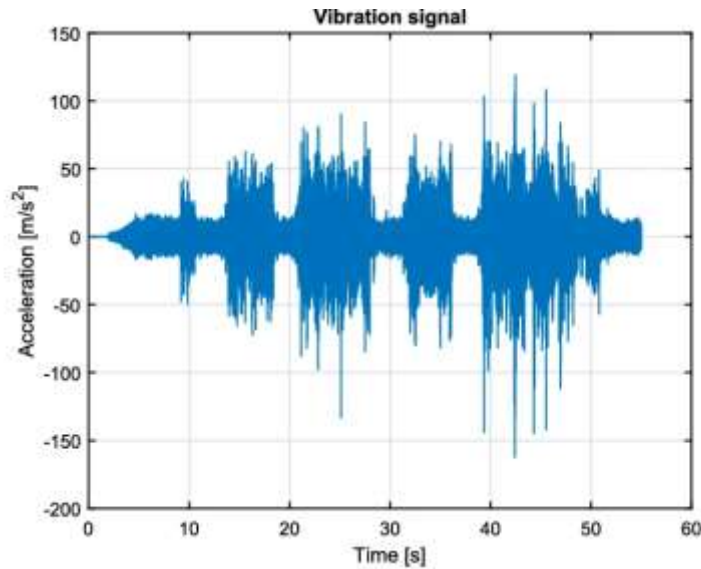


Figure I-37 Vibration signal at normal feed rate, with 8mm tool

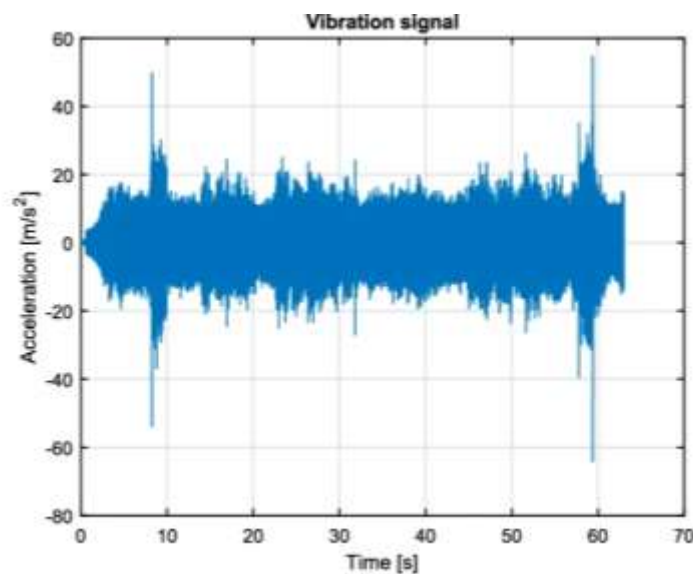


Figure I-38 Vibration signal at small feed rate, with 4mm tool

D) Cutting Temperature in Turning:

Cutting temperature is one of the most critical aspects of the turning process. When the cutting tool comes into contact with the rotating workpiece, a large amount of heat is generated due to plastic deformation in the primary shear zone and friction at the tool–chip and tool–

workpiece interfaces [15][9]. This thermal energy significantly influences tool life, surface integrity, dimensional accuracy, and even the metallurgical properties of the machined material.

According to **Shaw** [15], heat generation in metal cutting originates mainly from three zones:

1. **Primary shear zone** (plastic deformation of the material),
2. **Secondary deformation zone** (chip–tool interface friction),
3. **Tertiary zone** (tool flank–workpiece interface).

The total cutting power P converted into heat can be expressed as:

$$P = F_c \cdot V_c$$

Where:

- F_c = cutting force (N)
- V_c = cutting speed (m/s)

Since nearly all mechanical energy in machining is transformed into heat, the cutting temperature increases proportionally with cutting speed and cutting force [10].

Heat Distribution in Turning:

Experimental and analytical studies show that the generated heat is distributed approximately as follows [9][15]:

- **60–80%** carried away by the chip
- **10–20%** conducted into the workpiece
- **≈10%** absorbed by the cutting tool

This distribution depends strongly on cutting speed, tool material, and thermal conductivity of the workpiece.

At higher cutting speeds, a larger portion of heat is evacuated with the chip due to reduced contact time [14].

Analytical Estimation of Cutting Temperature:

A simplified empirical relationship shows that cutting temperature increases with cutting speed according to:

$$T \propto V_c^n$$

Where n typically ranges between 0.2 and 0.5 depending on machining conditions [15].

From a tribological standpoint, the temperature rise at the tool–chip interface also depends on friction coefficient μ :

$$Q_f = \mu \cdot F_n \cdot V_c$$

Where:

- Q_f = frictional heat rate
- F_n = normal force at the interface

High friction leads to increased interface temperature and accelerates tool wear mechanisms such as diffusion and oxidation [12].

Effects of Excessive Cutting Temperature:

Excessive temperature during turning may cause:

- Accelerated flank and crater wear [10]
- Plastic deformation of the cutting edge [15]
- Thermal expansion of the tool:

$$\Delta L = \alpha L_0 \Delta T$$

Where:

- α = coefficient of thermal expansion
- L_0 = original tool length
- ΔT = temperature increase

Tool elongation leads to dimensional inaccuracies in the machined part [9].

Furthermore, high temperatures may alter surface integrity, causing microstructural changes, residual stresses, and reduced fatigue strength of the component [14].

I.5. Lubrication Strategies and Environmental Impact

Lubrication strategies in machining significantly influence cutting performance, tool life, surface quality, and environmental sustainability. The choice between dry machining and the use of industrial cutting oils reflects both technical and ecological considerations [9][12].

I.5.1. Dry Machining:

Dry machining is a manufacturing technique in which material removal is performed without the use of cutting fluids. The process aims to shape the raw workpiece into the desired geometry and dimensions using a machine tool operating in a dry environment. High precision can still be achieved under optimized cutting conditions [8][15].



Figure I-39 Dry Machining

Advantages of Dry Machining:

- **Environmental Protection:**

The absence of cutting fluids eliminates air and water contamination, reducing health hazards such as skin irritation and respiratory problems commonly associated with fluid exposure [12].

- **Elimination of Post-Processing Cleaning:**

Components are free from lubricant residues, which reduces cleaning costs and associated energy consumption.

- **Improved Chip Recycling:**

Dry chips are easier to recycle since they are not contaminated by oils, increasing waste valorization efficiency [8].

- **Cost Reduction:**

Dry machining suppresses the purchase, preparation, maintenance, and disposal costs of cutting fluids, which can represent a significant portion of production expenses [8].

From a thermal standpoint, however, dry machining may lead to increased cutting temperature due to the absence of external cooling. The generated heat is mainly evacuated through the chip:

$$P = F_c \cdot V_c$$

Where nearly all mechanical power P is converted into heat during cutting [15].

Thus, dry machining requires optimized cutting parameters and suitable tool materials (e.g., coated carbides or ceramics) to withstand higher thermal loads [9].

I.5.2. Machining with Industrial Oils:

Machining operations frequently require lubricants to reduce friction, dissipate heat, and improve surface finish. Industrial cutting oils play a crucial role in enhancing machining performance but raise increasing environmental concerns [9][10].



Figure I-40 Machining with Industrial Oils

❖ Performance of Industrial Cutting Oils:

Industrial cutting oils fulfill several essential functions:

- **Lubrication:**

They reduce friction at the tool–chip and tool–workpiece interfaces, decreasing heat generation and extending tool life. Improved lubrication also enhances chip evacuation and surface finish quality [9][12].

The frictional heat generated can be expressed as:

$$Q_f = \mu \cdot F_n \cdot V_c$$

Where reducing the friction coefficient μ lowers heat generation [12].

- **Cooling:**

Cutting fluids dissipate heat effectively, especially in high-energy machining operations, maintaining tool hardness and workpiece integrity [15].

- **Corrosion Protection and Process Stability:**

Oils protect machined surfaces against oxidation and maintain a clean cutting environment, improving process reliability and repeatability [10].

❖ Environmental Impact of Industrial Oils:

Despite their technical benefits, industrial oils present several ecological drawbacks:

- **Fossil Origin and Low Biodegradability:**

Most mineral oils are petroleum-derived and exhibit poor biodegradability, leading to long-term environmental persistence [10].

- **Soil and Water Pollution:**

Leakage or improper disposal of used oils contaminates soil and alters its chemical and biological properties. Industrial effluents can compromise groundwater and surface water quality [11].

- **Health Risks:**

Aerosols generated during machining may contain volatile organic compounds (VOCs), additives, and microbial contaminants, potentially causing respiratory disorders, skin diseases, asthma, and even carcinogenic effects under prolonged exposure [12].

- **Economic and Regulatory Costs:**

Treatment, recycling, and disposal of used oils are complex and costly. In some cases, disposal costs may exceed several times the initial purchase price of the oil [12].

❖ **Sustainability Perspective:**

Modern manufacturing increasingly promotes alternative strategies such as:

- Minimum Quantity Lubrication (MQL)
- Biodegradable vegetable-based oils
- High-speed dry machining
- Cryogenic cooling

These approaches aim to balance machining performance with environmental responsibility [9][14].

I.6. Study of Tool Vibratory Behavior in Turning:

Vibrations during turning are a critical issue in machining operations, directly affecting surface finish, geometric accuracy, tool life, and overall productivity. The dynamic interaction between the cutting tool, the rotating workpiece, and the machine structure may lead to unstable cutting conditions if not properly controlled [14][15].

I.6.1. Types of Vibrations in Turning:

a) Self-Excited Vibrations (Chatter)

Self-excited vibration, commonly known as chatter, is caused by the regenerative effect between successive tool passes. The waviness left on the machined surface modifies chip thickness in the next revolution, amplifying oscillations [14].

The dynamic behavior can be modeled as:

$$m\ddot{x} + c\dot{x} + kx = F_c(t)$$

Instability occurs when the dynamic cutting force exceeds the structural damping capacity.

Effects of chatter:

- Characteristic surface waviness
- Accelerated flank and crater wear
- Reduced dimensional accuracy
- Risk of machine damage [14][15]

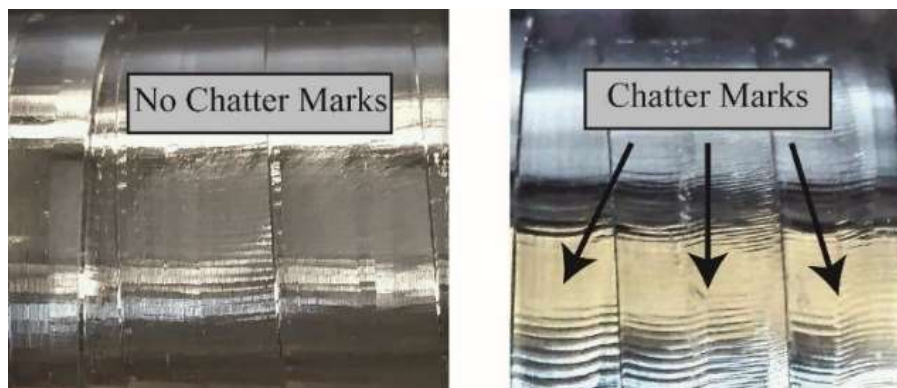


Figure I-41 the stable and unstable (chatter) cutting process

Altintas and Budak developed analytical stability lobe diagrams (SLD) to predict stable cutting zones as a function of spindle speed and depth of cut [14].

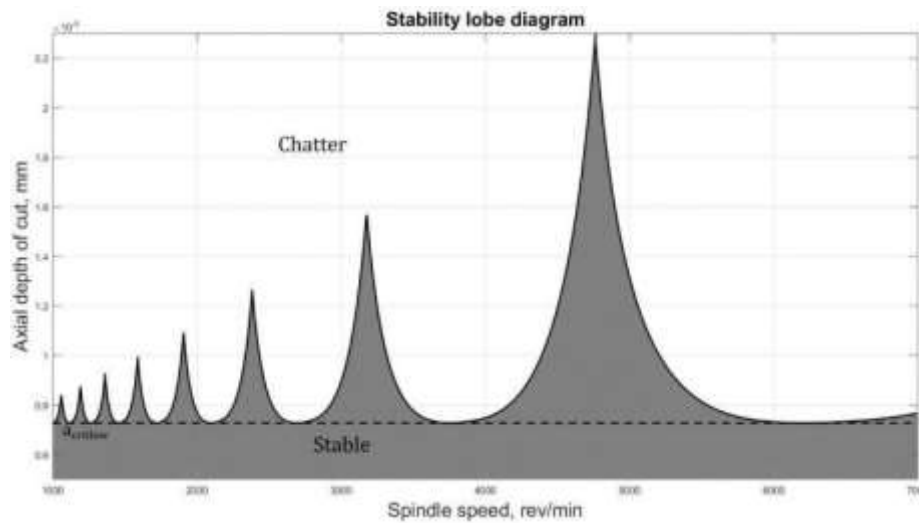


Figure I-42 Stability Lobe Diagram showing stable and unstable cutting zones (SLD)

b) Forced Vibrations

Forced vibrations originate from periodic external excitations such as:

- Spindle imbalance
- Misalignment or bearing clearance
- Gear transmission errors

These vibrations typically occur at the excitation frequency:

$$f = \frac{N}{60}$$

Where N is spindle speed (rpm) [11].

Unlike chatter, forced vibrations do not require regenerative feedback but may still degrade surface quality.

I.6.2. Methods of Analysis:

a) Dynamic Modeling

Frequency-domain analysis identifies the natural frequencies of the machine–tool system:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Resonance occurs when excitation frequency approaches f_n [14].

Time-domain simulations (e.g., numerical modeling) allow prediction of displacement, velocity, and acceleration responses [14].

b) Experimental Measurements

Common sensors include:

- **Accelerometers** (dominant in vibration measurement)
- **Acoustic emission sensors** for early chatter detection

Signal processing tools:

- **Fast Fourier Transform (FFT)** to convert time signals into frequency spectra:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

- Wavelet transforms for transient vibration analysis [11][14]

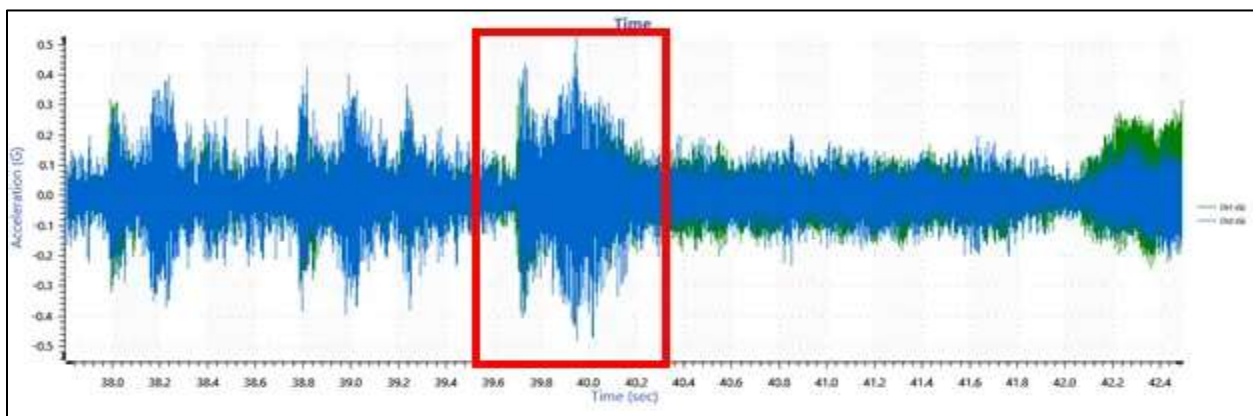


Figure I-43 Frequency spectrum of vibration signal using FFT analysis

I.6.3. Vibration Reduction Strategies:**a) Cutting Parameter Optimization**

Empirical recommendations:

- Reduce depth of cut
- Adjust spindle speed to avoid unstable lobes
- Modify feed rate

Advanced approaches use optimization algorithms to determine stable cutting zones [14].

b) Structural Damping

Passive solutions:

- Damped tool holders
- Composite or viscoelastic materials to increase damping ratio

Active solutions:

- Piezoelectric actuators for real-time vibration suppression [14]

c) Tool Design Optimization

- Optimized rake and clearance angles reduce dynamic cutting forces [15]
- Advanced coatings (e.g., low-friction coatings) reduce interface instability and thermal effects [12]

I.6.4. Chatter Phenomenon:

Chatter may also result from:

- Improper tool clamping
- Workpiece fixation errors
- Spindle bearing clearance

When periodicity in vibration signal corresponds exactly to spindle rotational frequency:

$$f = \frac{N}{60}$$

the issue is generally mechanical (imbalance or clearance), not regenerative chatter [11].

Corrective action may involve spindle adjustment, bearing preload correction, or structural reinforcement.

I.6.5. Lubrication Regimes and Surface Behavior:

The tribological behavior at the tool–chip interface depends on lubrication regime. According to tribology theory [12], lubrication regimes are represented by the **Stribeck curve**, which relates friction coefficient μ to the dimensionless parameter:

$$S = \frac{\eta V}{P}$$

Where:

- η = dynamic viscosity
- V = sliding speed
- P = normal pressure

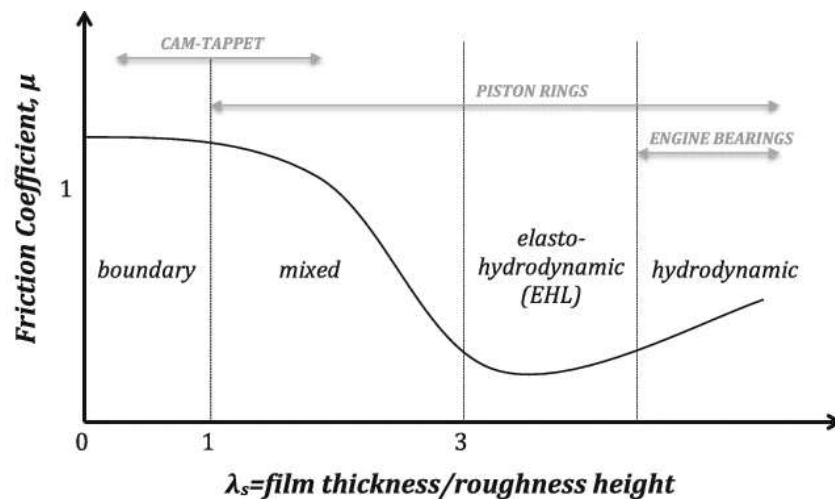


Figure I-44 Stribeck curve illustrating boundary, mixed and hydrodynamic lubrication regimes

Lubrication Regimes:

- **Boundary lubrication:** Load supported by asperity contact; high friction and wear.
- **Mixed lubrication:** Partial fluid film; asperities still carry part of the load.
- **Hydrodynamic lubrication:** Full fluid film separation governed by Reynolds equation:

$$\frac{\partial}{\partial x} \left(h^3 \frac{\partial p}{\partial x} \right) + \frac{\partial}{\partial y} \left(h^3 \frac{\partial p}{\partial y} \right) = 6\eta U \frac{\partial h}{\partial x}$$

Hydrodynamic lift may be generated by surface waviness, relative motion, or vibration [12].

Surface defects such as waviness and roughness influence pressure distribution and film thickness, affecting dynamic stability and sealing performance.

I.7. Optimization Methods in Machining:

Optimization techniques are widely used in **machining processes** to determine the **optimal combination** of cutting parameters that improve performance indicators such as surface roughness, tool wear, vibration, and cutting temperature. Among the most commonly used methods are **the Response Surface Methodology (RSM)** and **the Taguchi method**.

I.7.1. Response Surface Methodology (RSM):

optimize processes in which a response of interest is influenced by several variables. In machining, **RSM** is widely applied to establish relationships between input parameters (cutting speed, feed rate, depth of cut) and output responses such as surface roughness or tool wear.

The general form of the second-order polynomial model used in RSM is:

$$Y = \beta_0 + \sum \beta_i X_i + \sum \beta_{ii} X_i^2 + \sum \beta_{ij} X_i X_j$$

Where:

Y = predicted response

X_i = input variables

β = regression coefficients

RSM allows the generation of response surfaces and contour plots to visualize the interaction between variables and identify optimal operating conditions. It also reduces the number of experimental trials compared to traditional methods while maintaining accuracy.

In machining applications, RSM is particularly useful for analyzing the combined effects of cutting parameters on surface roughness and vibration behavior [9].

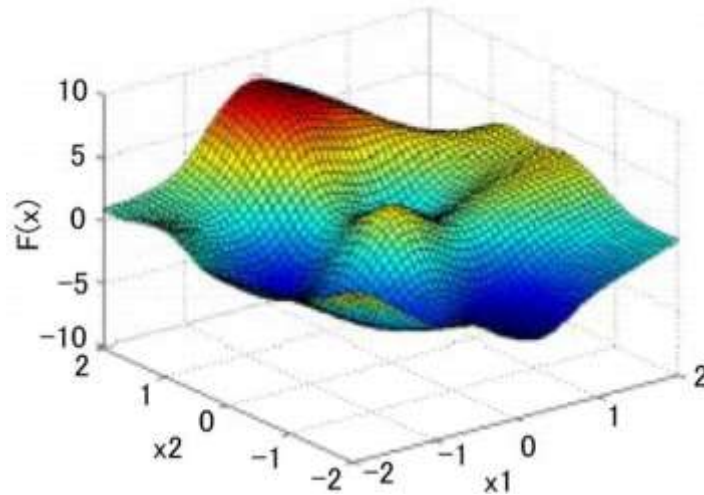


Figure I-45 Response Surface Methodology (RSM)

I.7.2. Taguchi Method:

The Taguchi method is another powerful optimization technique based on statistical design of experiments (**DOE**). It uses orthogonal arrays to systematically study the influence of multiple parameters with a reduced number of experiments.

One of the key concepts in Taguchi methodology is the **Signal-to-Noise (S/N) ratio**, which evaluates the robustness of a process against variability. Depending on the objective, different formulations are used:

Smaller-the-better (e.g., surface roughness):

$$S/N = -10 \log \left(\frac{1}{n} \sum y_i^2 \right)$$

Larger-the-better:

$$S/N = -10 \log \left(\frac{1}{n} \sum \frac{1}{y_i^2} \right)$$

Nominal-the-best:

$$S/N = 10 \log \left(\frac{\bar{y}^2}{s^2} \right)$$

Where:

y_i = observed values

n = number of experiments

The Taguchi method is widely used in machining because it simplifies experimental design and identifies the most influential parameters efficiently. It is particularly effective for improving process robustness and minimizing variability in surface quality and tool wear [10].

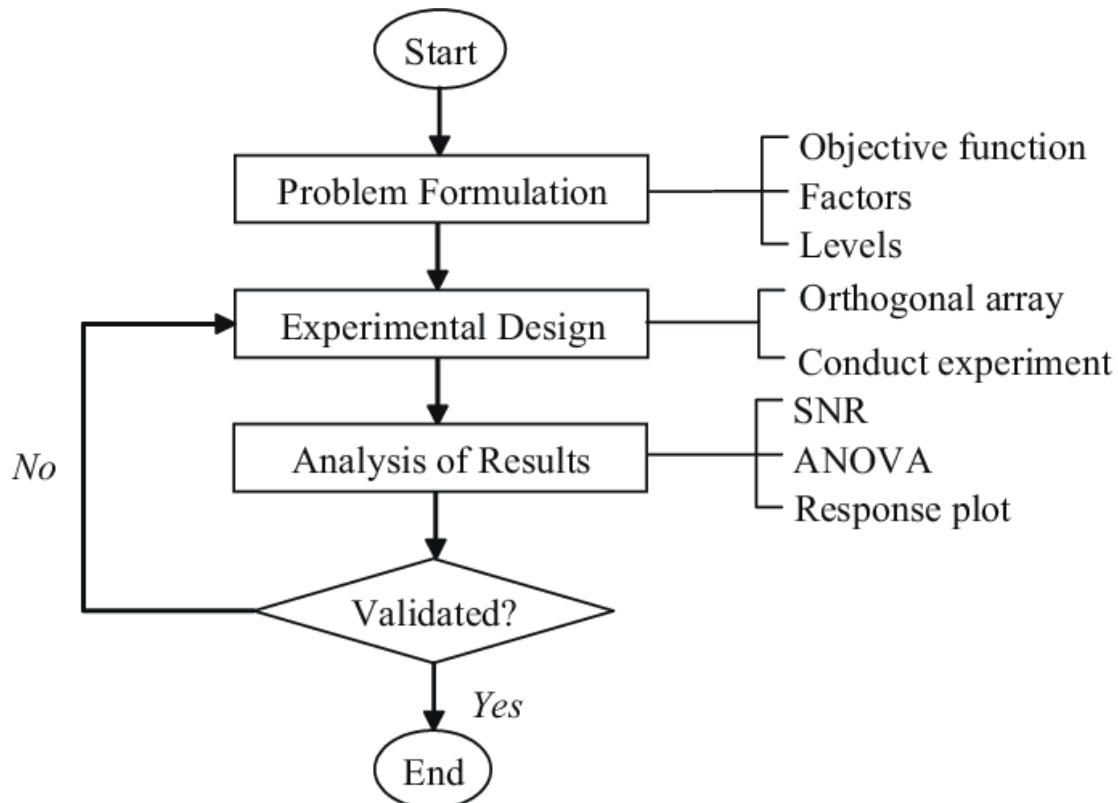


Figure I-46 Taguchi Method

I.7.3. Comparison Between RSM and Taguchi:

The **Response Surface Methodology (RSM)** and the **Taguchi method** are widely used optimization techniques in machining processes. **RSM** is based on **mathematical modeling** and **regression analysis**, allowing the establishment of functional relationships between input parameters and output responses. It provides **detailed information** about **the interaction effects** between **variables** and **enables accurate prediction** and **optimization of machining performance** such as surface roughness and tool wear [9][17].

In contrast, the **Taguchi method** is based on **statistical design** of experiments using **orthogonal arrays**, which significantly reduces the number of experimental trials while maintaining reliable results. It focuses on **improving process robustness** by **minimizing the effect** of **noise factors** through the use of **signal-to-noise (S/N)** ratios [10][11].

Although both methods aim at **optimizing machining parameters**, **RSM** is more suitable for precise modeling and detailed analysis of parameter interactions, while the **Taguchi method** is more efficient for preliminary optimization and robust parameter design with fewer experiments. Therefore, the selection of the appropriate method depends on the objectives of the study, the required accuracy, and the available experimental resources [9][10].

I.8. Conclusion:

In this chapter, the fundamental scientific principles necessary for understanding the present study have been introduced. The first part reviewed the theoretical foundations of the cutting process, including the mechanics of material removal, tool materials, and the main cutting parameters influencing machining performance. Special attention was given to cutting temperature, tool wear mechanisms, and their impact on surface integrity and dimensional accuracy.

The second part emphasized the importance of lubrication strategies in machining operations. Different approaches, such as dry machining and machining with industrial oils, were discussed along with their technical advantages and environmental implications. The role of lubrication in reducing friction, controlling temperature, improving tool life, and enhancing surface quality was highlighted.

Finally, the chapter addressed the vibratory behavior of the cutting tool during turning operations. The different types of vibrations, particularly self-excited vibrations (chatter), their causes, analytical methods, and mitigation strategies were presented. The interaction between vibration, lubrication regimes, and surface conditions was also examined to provide a comprehensive understanding of the dynamic behavior of the machining system.

Overall, this bibliographic study establishes the theoretical framework necessary to analyze and interpret the experimental results presented in the following chapters.

II. Chapter II: Experimental Procedure

II.1. Introduction:

This chapter focuses on the **experimental framework** adopted to investigate the interaction between **the cutting tool** and the **workpiece material**. Rather than simply listing the equipment, the emphasis is placed on how each element **contributes** to understanding the machining process. Key parameters such as **tool wear**, **cutting forces**, **surface roughness**, and **the temperature generated** during cutting are systematically examined.

To ensure reliable and reproducible results, a structured approach to experimental design is implemented, defining both the **operating conditions** and the **measurement procedures**. This approach allows for a **comprehensive evaluation** of machining performance under different conditions.

All experimental work was carried out within the facilities of the **Mechanical Engineering** Department's metal cutting laboratory, as well as the **ISTA** laboratory at the **University of Frères Mentouri – Constantine 1**.

II.2. Materials and Equipment :

This section introduces the main equipment and technical resources employed to conduct the experimental work. The objective is to provide a clear overview of the machining environment and the tools used, ensuring a better understanding of the conditions under which the tests were performed.

II.2.1. Workpiece Material (Bronze):

In the present study, the material selected for machining is bronze, a copper-based alloy widely recognized for its excellent mechanical and tribological properties. Typically composed of copper as the primary element, along with tin and small additions of other.

Bronze also exhibits good machinability compared to many ferrous materials, while maintaining adequate strength and durability. These characteristics make it a common choice in engineering applications such as bearings, bushings, and mechanical components subjected to moderate loads.

The chemical composition of the selected bronze alloy is presented in Table II.1

Table II-1 Chemical composition of bronze (wt.%)

Cu	Sn	Zn	Pb	Fe	Ni	Al
88–92	6–10	≤2	≤1	≤0.5	≤1	≤0.01

The workpiece specimens used in the experiments have a diameter of **d = 80 mm** and a length of **L = 550 mm** (Figure II.1).



Figure II-1 Bronze workpiece used in the experimental

II.2.2. Machine Tool and Technical Specifications:

The turning operations carried out in this study were performed using a conventional lathe, model T MACH TR 3310 (Figure II.2). This machine was selected for its capability to provide stable and adjustable machining conditions suitable for experimental analysis.

The lathe is equipped with an electric motor delivering a power of 6.6 kW, allowing efficient material removal under various cutting regimes. It offers a wide spindle speed range, varying from 45 to 1810 rpm, which enables the exploration of different cutting conditions.

In terms of feed motion, the machine allows a feed rate ranging between 0.15 and 0.2 mm per revolution within the metric system. Additionally, it supports a broad range of threading operations, including metric threads from 0.4 mm to 7 mm, as well as inch-based threads from 4 to 60 TPI.



Figure II-2 A conventional lathe (model T MACH TR 3310)

II.2.3. Cutting Tools and Insert Characteristics:

All turning experiments carried out in this work were performed using a T-Max P insert identified by the reference P2.Z.AN.175 HB. The insert is a mechanically clamped cemented carbide cutting tool coated through Chemical Vapor Deposition (CVD) with successive layers of TiCN, Al_2O_3 , and TiN. These coatings provide high resistance to wear and elevated temperatures during machining operations.

This insert is mainly intended for machining P-group materials, particularly steels, and can operate with a depth of cut up to 3.3 mm. It also supports cutting speeds reaching 990 m/min and feed rates ranging from 0.08 to 0.22 mm/rev. The insert was mounted on a compatible T-Max tool holder, ensuring machining stability and improved surface finish quality throughout the turning process (Figure II.3).



Figure II-3 Insert and Tool Holder Used

II.2.4. Measuring Instruments:

Measurement instruments play a vital role in mechanical engineering by ensuring accuracy and reliability during manufacturing and inspection processes. They are used to evaluate different physical characteristics and dimensions of machined components, helping to maintain product quality and machining performance.

II.2.4.1. Surface Roughness Measurements :

Surface roughness measurements were carried out under standard laboratory conditions in order to ensure the reliability and repeatability of the experimental results. The measurements were performed at a controlled temperature of approximately 20°C . Surface roughness evaluation is essential in machining studies because it reflects the quality of the machined surface and the influence of cutting parameters on machining performance [9][13].

A portable surface roughness tester equipped with a diamond stylus probe was used to measure the surface profile of the machined specimens. To improve the accuracy and reproducibility of the results, three measurements were performed on different areas of each sample, and the average value was considered for analysis [13].

According to ISO 4287 standards, several roughness parameters are commonly used to characterize the surface texture [13]:

❖ Ra (Arithmetic Average Roughness)

Ra is the arithmetic mean of the absolute deviations of the roughness profile from the mean line over the evaluation length. It is expressed as:

$$Ra = \frac{1}{L} \int_0^L |y(x)| dx$$

Where:

- L = evaluation length
- $y(x)$ = deviation of the profile from the mean line

Ra is the most commonly used parameter for evaluating surface finish quality [9].

❖ Rq (Root Mean Square Roughness)

Rq represents the root mean square average of the profile deviations measured from the mean line:

$$Rq = \sqrt{\frac{1}{L} \int_0^L y^2(x) dx}$$

Rq is more sensitive to large peaks and valleys than Ra because it squares the profile deviations before averaging [13].

❖ Rz (Average Maximum Height of the Profile)

Rz corresponds to the arithmetic average of the maximum peak-to-valley heights measured over several consecutive sampling lengths. It provides information about the vertical distribution of surface irregularities [13].

❖ Rt (Total Height of the Roughness Profile)

Rt is the total height of the roughness profile over the evaluation length. It is defined as the vertical distance between the highest peak and the deepest valley within the measured profile [13].

These roughness parameters provide important information about the influence of machining conditions such as cutting speed, feed rate, depth of cut, vibration, and lubrication on the final surface quality of the workpiece [9][10].



Figure II-4 Digital Surface Roughness Tester

II.2.4.2. Accelerometer for Vibration Measurement (Arduino ... Sensor):

An accelerometer is a sensor designed to measure acceleration, allowing the evaluation of vibrations and shocks applied to mechanical or structural systems. It converts inertial forces into electrical signals, which can then be analyzed for dynamic behavior assessment, fault detection, and condition monitoring of mechanical systems [9][11].

In machining applications, accelerometers are widely used to study the dynamic behavior of machine tools such as lathes. One commonly used sensor is the ADXL345 tri-axial accelerometer, which measures acceleration along the X, Y, and Z axes. This type of sensor enables real-time vibration monitoring during turning operations, making it suitable for process analysis and machine condition monitoring.

The accelerometer can be interfaced with microcontroller platforms such as Arduino to acquire vibration signals. In such systems, the sensor is configured to an appropriate measurement range (for example $\pm 16g$ for high vibration levels) and continuously records acceleration data in the three axes. These signals can be displayed in real time, stored for later analysis, or transmitted to external devices via communication modules such as Bluetooth or Wi-Fi.

The measured vibration signals can be further processed using signal analysis techniques such as:

- Time-domain analysis (amplitude, RMS value)
- Frequency-domain analysis using FFT

These methods allow the evaluation of vibration intensity and dominant frequencies, which are essential for detecting phenomena such as chatter and unbalanced rotation in machining processes [14][11].

Such a measurement system is widely used in condition monitoring and predictive maintenance, as well as in studying the dynamic stability of machining operations.



Figure II-5 Accelerometer for Vibration Measurement

II.2.4.3. Thermal Imagers for Temperature Measurement in Machining:

Thermal imagers are non-contact measurement devices used to capture and visualize the temperature distribution of objects in real time. They operate by detecting infrared radiation emitted by a surface and converting it into a thermal image, where each color or pixel represents a specific temperature value. This technology is widely used in engineering applications to analyze heat generation and thermal behavior in mechanical systems [9][15].

In machining processes such as turning, thermal imagers are particularly useful for studying cutting temperature distribution at the tool–chip–workpiece interface. Since temperature strongly

affects tool wear, surface quality, and dimensional accuracy, infrared thermography provides an efficient method for monitoring thermal effects without interfering with the cutting process [15].

The working principle of a thermal imager is based on infrared radiation laws, where all objects with a temperature above absolute zero emit thermal energy. The device captures this radiation and converts it into a thermal map using calibrated sensors and image processing algorithms. The measured temperature field can then be analyzed to identify hot zones, heat concentration areas, and thermal gradients during machining operations.

Thermal imaging is commonly used in:

- Monitoring cutting temperature evolution during turning
- Detecting tool overheating and wear zones
- Evaluating the effect of cutting parameters on heat generation
- Improving process optimization and tool life prediction [14][15]

Compared to contact-based temperature sensors (thermocouples), thermal imagers offer several advantages such as:

- Non-contact measurement (no disturbance of cutting zone)
- Full-field temperature visualization instead of single-point data
- High spatial resolution for detecting localized heat zones

However, accuracy can be influenced by surface emissivity, reflections, and chip interference, which must be considered during measurement setup [9].

Overall, thermal imaging is an advanced diagnostic tool that enhances the understanding of thermal phenomena in machining processes and contributes to process optimization and tool performance evaluation.



Figure II-6 Accelerometer for Vibration Measurement

II.3. Experimental Planning and Operating Conditions:

Experimental planning is a fundamental step in machining studies, as it allows the systematic evaluation of the influence of cutting parameters and operating conditions on process performance. In this study, different lubrication strategies were compared, including drip lubrication, dry machining, and machining with industrial oils, in order to analyze their effect on vibration, surface quality, tool wear, and temperature.

II.3.1. Comparison of Machining Strategies (Lubrication or Dry Machining):

The choice of lubrication strategy plays a crucial role in machining performance, affecting friction, heat generation, tool life, and surface integrity [9][12].

II.3.1.1. Drip Lubrication :

Drip lubrication is a controlled lubrication technique in which the cutting fluid is supplied to the cutting zone in the form of droplets at regular intervals. This method ensures a continuous but low-flow supply of lubricant, allowing precise control of the lubrication rate while minimizing fluid consumption.

This system is widely used in machine tools and mechanical components such as bearings, gears, and cutting operations, where moderate lubrication is required without excessive fluid usage [9].

Drip lubrication systems are typically adjustable, allowing the operator to regulate the flow rate using a valve mechanism. This ensures optimized lubrication conditions depending on machining requirements, improving tool life and reducing friction at the tool–chip interface [10].



Figure II-7 Cutting in CNC Machine with lubrication

II.3.1.2. Dry Machining:

Dry machining is a manufacturing technique where material removal is performed without the use of any cutting fluid. The process relies entirely on the cutting tool and machining parameters to control heat generation and friction.

Although dry machining eliminates lubrication costs and environmental issues, it may lead to higher cutting temperatures and increased tool wear due to the absence of cooling and lubrication effects [10][15]. Therefore, it requires optimized cutting conditions and suitable tool materials such as coated carbides or ceramics to maintain performance.



Figure II-8 Cutting Bronz without oil in CNC Machine

II.3.1.3. Machining with Industrial Oil (Emulsion):

Machining with industrial oil emulsions involves the use of water-based cutting fluids composed of water, oil, and chemical additives. These fluids provide both cooling and lubrication effects, making them effective in reducing cutting temperature and improving surface quality [12].

The main advantages of emulsions include:

- Efficient heat dissipation during cutting
- Reduction of friction at tool–workpiece interface
- Improvement of surface finish and tool life

However, they also present environmental and operational drawbacks such as bacterial growth, corrosion risks, and the need for proper maintenance and disposal [12][9].



Figure II-9 Cutting Bronz with oil in CNC Machine

II.3.2. Experimental Design (Experimental Matrix):

Machining performance is influenced by several input parameters such as cutting speed (V_c), feed rate (f), and depth of cut (a_p), as well as operating conditions like lubrication mode and machine rigidity. These parameters affect output responses including vibration, surface roughness, tool wear, and cutting temperature [14][9].

To systematically study these effects, a structured experimental design approach is required. In this work, a Taguchi L27 orthogonal array was adopted to reduce the number of

experiments while ensuring reliable statistical analysis. This method is widely used in machining optimization because it allows the evaluation of multiple factors with a limited number of trials while maintaining experimental efficiency and robustness [10].

Taguchi, $P = 3, L = 3$				
Run #	a	c	d	X
1	1	1	1	X_1
2	1	2	2	X_2
3	1	3	3	X_3
4	2	1	2	X_4
5	2	2	3	X_5
6	2	3	1	X_6
7	3	1	3	X_7
8	3	2	1	X_8
9	3	3	2	X_9

Figure II-10 Experimental Design

II.4. Conclusion:

This chapter presented the experimental methodology adopted in this study. It described the cutting parameters, lubrication strategies, and experimental design used to investigate machining performance. The comparison between dry machining, drip lubrication, and emulsion-based machining provided a comprehensive basis for analyzing their effects on vibration, surface quality, tool wear, and temperature in the following experimental results.

III. Chapter III: Modeling of Turning Performance

III.1. Introduction:

In this chapter, the experimental results obtained from machining **bronze** under different lubrication conditions are presented and analyzed using the **Taguchi L27** experimental design. The tests were conducted under dry machining and drip lubrication conditions in order to investigate the influence of the machining environment on process performance and machining quality.

A statistical analysis based on Analysis of Variance (ANOVA) was carried out to evaluate the effects of the main machining parameters, namely cutting speed (V_c), feed rate (f), depth of cut (a_p), and lubrication mode, on the output responses such as surface roughness (Ra), vibration amplitude (V), and cutting temperature (T).

In addition, Response Surface Methodology (**RSM**) was applied to establish predictive mathematical models describing the relationship between the machining parameters and the measured responses. These models will be used in the following chapter to perform an optimization study using the **desirability function** approach and other optimization techniques in order to identify the optimal machining conditions.

III.2. Presentation of Experimental Results:

To perform a statistical analysis and establish predictive statistical models for the output responses (R_a , V , and T) according to the cutting conditions and lubrication mode, a total of 54 experimental tests were conducted based on the Taguchi L27 design. The obtained experimental results are summarized in Table III.1.

Table III-1 Experimental Results of the Taguchi L27 Design

Run	Vc	a	ap	Oil	R_a (μm)	R_z (μm)	R_q (μm)	R_t (μm)	$T(^{\circ}\text{C})$
1	280	0.21	0.8	Without Oil	9.075	54.45	10.07	10.91	70.003
2			1.2	Without Oil	7.053	42.32	7.83	9.97	85.1
3			1.6	Without Oil	5.903	35.42	6.55	12.95	95.201
4		0.31	0.8	Without Oil	7.283	43.7	8.08	11.58	80.2
5			1.2	Without Oil	6.974	41.84	7.74	9.17	95
6			1.6	Without Oil	5.784	34.7	6.42	11.21	110.3
7		0.42	0.8	Without Oil	7.646	45.88	8.49	18.61	90
8			1.2	Without Oil	6.893	41.36	7.65	17.16	105
9			1.6	Without Oil	4.593	27.56	5.1	20.86	120
10	840	0.21	0.8	Without Oil	5.942	35.65	6.6	10.5	120
11			1.2	Without Oil	4.678	28.07	5.19	14.23	140
12			1.6	Without Oil	3.387	20.32	3.76	11.4	165
13		0.31	0.8	Without Oil	5.509	33.05	6.11	14.03	150
14			1.2	Without Oil	5.011	30.07	5.56	16.79	171

15		0.42	1.6	Without Oil	3.92	23.52	4.35	16.08	195
16			0.8	Without Oil	4.318	25.91	4.79	32.17	175
17			1.2	Without Oil	3.486	20.92	3.87	41.38	200
18			1.6	Without Oil	2.654	15.92	2.95	26.32	230
19	1400	0.21	0.8	Without Oil	4.833	29	5.36	67.94	201
20			1.2	Without Oil	4.575	27.45	5.08	17.9	240
21			1.6	Without Oil	3.843	23.06	4.27	22.03	280
22		0.31	0.8	Without Oil	3.218	19.31	3.57	71.65	250
23			1.2	Without Oil	3.268	19.61	3.63	15.68	301
24			1.6	Without Oil	2.394	14.36	2.66	11.06	340
25		0.42	0.8	Without Oil	3.268	19.61	3.63	16.54	290
26			1.2	Without Oil	3.149	18.89	3.5	14.83	341
27			1.6	Without Oil	2.114	20.4	2.35	23.71	380
28	280	0.21	0.8	With Oil	3.351	7.2	2.1	7.5	45
29			1.2	With Oil	2.25	14.92	5.59	15.07	50
30			1.6	With Oil	1.501	17.17	6.46	17.35	58
31		0.31	0.8	With Oil	1.901	9.4	3.02	9.96	52
32			1.2	With Oil	1.605	11.001	3.17	10.92	57
33			1.6	With Oil	1.701	18.01	5.97	18.95	63
34		0.42	0.8	With Oil	1.754	3.632	6.61	18.12	60
35			1.2	With Oil	1.901	10.83	4.58	10.94	70
36			1.6	With Oil	2.279	11.75	4.46	11.87	78
37	840	0.21	0.8	With Oil	2.89	4.534	3.09	8.26	85
38			1.2	With Oil	1.901	10.006	4.12	11.17	95
39			1.6	With Oil	2.001	7.17	3.24	7.23	90

40	1400	0.31	0.8	With Oil	1.569	6.009	4.42	11.97	105
41			1.2	With Oil	1.344	11.59	4.34	11.7	115
42			1.6	With Oil	2.001	9.581	5.96	16	125
43		0.42	0.8	With Oil	1.546	9.63	2.72	10.21	140
44			1.2	With Oil	1.838	9.87	3.05	10.34	155
45			1.6	With Oil	3.001	12.43	4.35	12.96	135
46		0.21	0.8	With Oil	2.546	1.311	10.99	29.77	160
47			1.2	With Oil	2.196	1.903	6.57	23.03	145
48			1.6	With Oil	1.916	1.813	3.29	8.67	170
49	1400	0.31	0.8	With Oil	1.901	5.168	3.28	12.39	180
50			1.2	With Oil	1.662	9.74	2.83	10.42	155
51			1.6	With Oil	2.61	5.609	2.67	3.75	165
52		0.42	0.8	With Oil	1.901	11.01	3.23	10.83	190
53			1.2	With Oil	2.263	12.64	3.7	13.18	175
54			1.6	With Oil	3.4	9.62	2.57	10.3	200

III.3. Surface Roughness Analysis:

The analysis of the experimental results demonstrates that the surface roughness parameters (R_a , R_z , R_q , and R_t) are significantly influenced by the cutting parameters, namely cutting speed (V_c), feed rate (f), depth of cut (a_p), and lubrication conditions. The obtained results reveal that surface quality does not follow a simple linear trend, but rather depends on the interaction between these machining variables.

In general, dry machining conditions tend to produce higher and more dispersed roughness values, especially at high feed rates and depths of cut. This behavior is mainly related to the increase in friction and instability at the tool–workpiece interface, which negatively affects the surface finish. Under these conditions, the machined surface becomes less regular, leading to larger R_a and R_t values.

On the other hand, the use of lubrication generally contributes to improving the surface quality by reducing friction, minimizing tool wear, and stabilizing the cutting process. As a result, many

lubricated experiments exhibit lower roughness values and more stable machining behavior compared with dry machining.

This effect can be clearly observed by comparing experiments carried out under identical cutting conditions. For example:

$V_c = 280 \text{ m/min}$, $f = 0.21 \text{ mm/rev}$, $a_p = 0.8 \text{ mm}$

- Without lubrication (Run 1):
 $R_a \approx 9.075 \text{ }\mu\text{m}$, $R_q \approx 10.07 \text{ }\mu\text{m}$, $R_t \approx 10.91 \text{ }\mu\text{m}$
- With lubrication (Run 28):
 $R_a \approx 3.351 \text{ }\mu\text{m}$, $R_q \approx 2.10 \text{ }\mu\text{m}$, $R_t \approx 7.50 \text{ }\mu\text{m}$

Under these conditions, lubrication significantly improved the surface finish by reducing the roughness parameters, particularly R_a and R_q . This improvement indicates a smoother and more homogeneous machined surface.

A similar trend can also be observed at higher cutting conditions:

$V_c = 840 \text{ m/min}$, $f = 0.42 \text{ mm/rev}$, $a_p = 1.2 \text{ mm}$

Dry machining (Run 17):

$R_a \approx 3.486 \text{ }\mu\text{m}$, $R_t \approx 41.38 \text{ }\mu\text{m}$

Lubricated machining (Run 44):

$R_a \approx 1.838 \text{ }\mu\text{m}$, $R_t \approx 10.34 \text{ }\mu\text{m}$

In this case, lubrication considerably reduced both the average roughness and the total profile height R_t , confirming its effectiveness in enhancing machining stability and improving surface integrity.

However, the influence of lubrication is not always identical for all cutting conditions. At very high cutting speeds, some lubricated experiments exhibit slightly higher roughness values compared with dry machining. For example:

- Dry machining (Run 27):

$Ra \approx 2.114 \mu\text{m}$

- Lubricated machining (Run 54):

$Ra \approx 3.400 \mu\text{m}$

This behavior suggests that the effectiveness of lubrication depends strongly on the combination of machining parameters and cutting conditions. In certain cases, excessive cutting speed and aggressive machining conditions may reduce the beneficial effect of lubrication on surface quality.

Overall, the experimental analysis confirms that lubrication plays an important role in improving surface roughness and machining stability. Nevertheless, achieving optimal surface quality requires a balanced selection of cutting parameters together with an appropriate lubrication strategy, since the machining responses are governed by the interaction of all process variables rather than by a single factor alone.

III.4. Temperature Analysis

The analysis of the experimental results shows that the cutting temperature is strongly influenced by the machining parameters, namely cutting speed (V_c), feed rate (f), depth of cut (a_p), and lubrication conditions. The obtained values indicate that temperature evolution does not follow a constant trend, but rather depends on the interaction between these parameters during the machining process.

In general, an increase in cutting speed and feed rate leads to higher temperature values due to the rise in friction and plastic deformation at the tool–workpiece interface. Under dry machining conditions, the generated heat tends to accumulate more rapidly because there is no cooling or lubricating medium to dissipate the thermal energy produced during cutting.

Conversely, lubrication modifies the thermal behavior of the machining process by reducing direct contact friction and improving heat evacuation from the cutting zone. However, the

influence of lubrication on temperature is not always uniform and may vary according to the cutting conditions.

This behavior can be illustrated by comparing experiments performed under identical machining parameters. For example:

$V_c = 280 \text{ m/min}$, $f = 0.21 \text{ mm/rev}$, $a_p = 0.8 \text{ mm}$

- Without lubrication (Run 1):
Temperature $\approx 70.0 \text{ }^\circ\text{C}$
- With lubrication (Run 28):
Temperature $\approx 45 \text{ }^\circ\text{C}$

In this case, the lubricated condition produced a lower temperature value compared with dry machining, confirming the cooling effect of lubrication and its role in reducing friction at the cutting zone.

At medium cutting speed conditions, temperature values continue to increase with the machining parameters. For instance:

$V_c = 840 \text{ m/min}$, $f = 0.42 \text{ mm/rev}$, $a_p = 1.2 \text{ mm}$

- Dry machining (Run 17):
Temperature $\approx 200 \text{ }^\circ\text{C}$
- Lubricated machining (Run 44):
Temperature $\approx 155 \text{ }^\circ\text{C}$

Here again, lubrication reduced the cutting temperature significantly, demonstrating its effectiveness in improving thermal stability during machining.

A more pronounced increase appears at very high cutting speeds:

$V_c = 1400 \text{ m/min}$, $f = 0.42 \text{ mm/rev}$, $a_p = 1.6 \text{ mm}$

- Without lubrication (Run 27):
Temperature $\approx 380 \text{ }^\circ\text{C}$
- With lubrication (Run 54):
Temperature $\approx 200 \text{ }^\circ\text{C}$

Under these severe cutting conditions, the highest temperature was recorded during dry machining, whereas lubrication considerably reduced the generated heat. This significant reduction highlights the important role of lubrication in limiting thermal accumulation under high cutting conditions.

The previous figure illustrating the average temperature values further confirms these observations. The figure shows that Experiments N°14 and N°27, performed under dry machining conditions, recorded temperatures of approximately 171 °C and 380 °C, respectively. In comparison, Experiments N°42 and N°54 carried out with lubrication reached approximately 125 °C and 200 °C. The graphical representation clearly highlights the progressive increase in temperature with more severe machining conditions, particularly at high cutting speeds and feed rates. It also demonstrates that lubrication contributes effectively to reducing cutting temperature and improving thermal stability during the machining process.

Overall, the experimental analysis confirms that cutting temperature is mainly governed by the machining parameters, particularly cutting speed and feed rate. Although lubrication significantly reduces the generated heat, achieving optimal thermal control still requires a balanced selection of machining parameters and lubrication conditions to ensure stable machining performance and limit excessive thermal loads during the cutting operation.

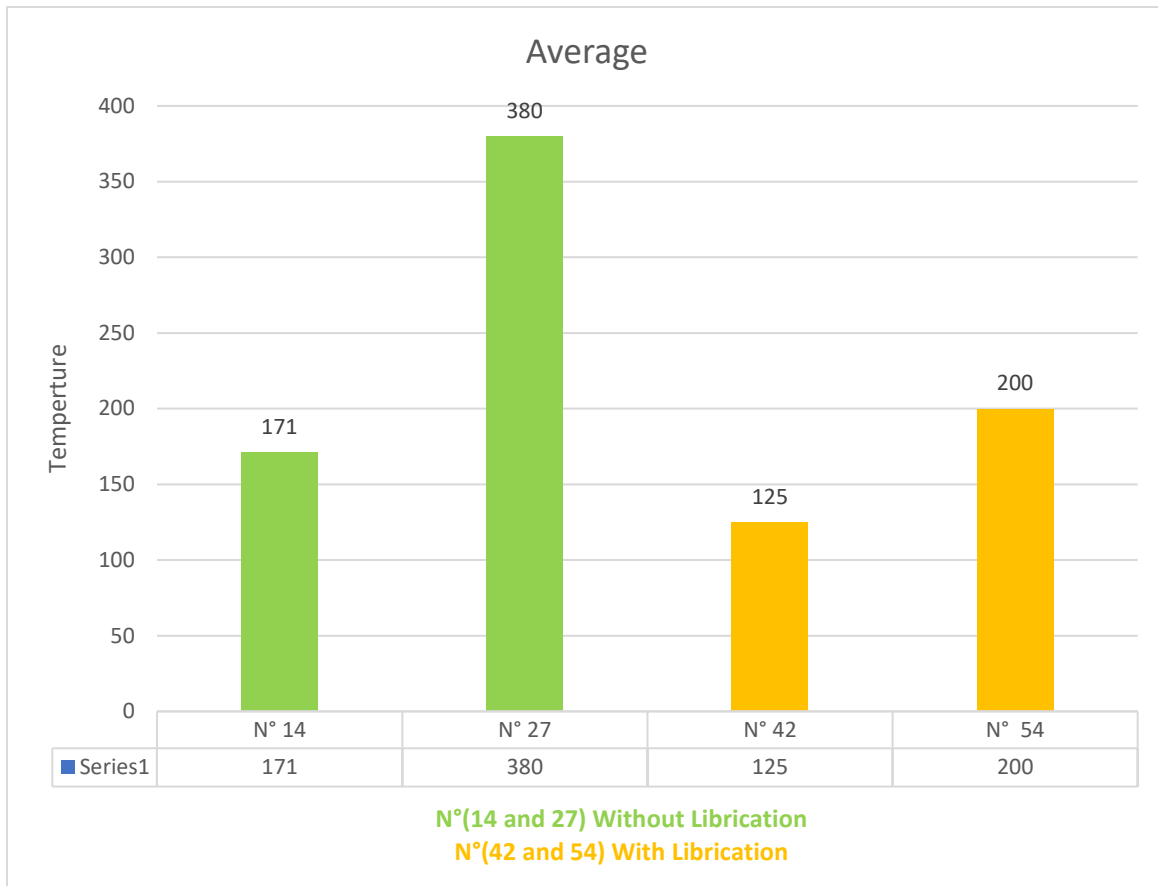


Figure III-1 Comparison of Average Temperature Under Dry and Lubricated Machining Conditions

III.5. Chip Analysis:

Table III-2 Cutting Parameters for Chip Analysis Tests

Test Number	10	20	30	40
Depth of Cut	0.8	1.2	1.6	0.8
Speed	840	1400	280	840
Feed Rate	0.21	0.21	0,21	0,31

The study of chip formation during bronze machining showed that cutting parameters have a direct effect on the shape and characteristics of the produced chips. Changes in cutting speed, feed rate, and depth of cut resulted in different chip morphologies and machining conditions throughout the experiments.



Figure III-2 Chips from Test No. 10

Test No. 10, using a depth of cut of 0.8 mm, a feed rate of 0.21 mm/rev, and a cutting speed of 840 m/min, the chips were generally smooth and continuous, reflecting stable machining conditions and efficient material removal.



Figure III-3 Chips from Test No. 20

During Test No. 20, the depth of cut was increased to 1.2 mm and the cutting speed was raised to 1400 m/min while keeping the same feed rate. Under these conditions, the chips became longer and more curled due to the higher cutting temperature and increased cutting energy. Although these parameters enhanced productivity, they may also accelerate tool wear.



Figure III-4 Chips from Test No. 30

For Test No. 30, carried out with the largest depth of cut (1.6 mm) and a lower cutting speed of 280 m/min, the generated chips were thicker and more compact. The lower speed reduced thermal effects, but the increase in cutting forces became more noticeable because of the higher material engagement.



Figure III-5 Chips from Test No. 40

In Test No. 40, the feed rate was increased to 0.31 mm/rev while maintaining a depth of cut of 0.8 mm and a speed of 840 m/min. This higher feed rate produced larger and less regular chips, which may lead to poorer surface quality and more difficult chip evacuation during machining.

III.6. Tool Wear:

During the experimental tests, two machining conditions were investigated: dry machining and machining under lubrication.



Figure III-6 Tool Condition Before Machining

It was observed that, in dry machining conditions, the cutting tool experienced rapid wear and the machined surface quality deteriorated significantly. Dark marks appearing on the tool indicated excessive heat generation, while the cutting temperature increased considerably, reaching nearly 30 °C in one of the experiments (Figure III.7).

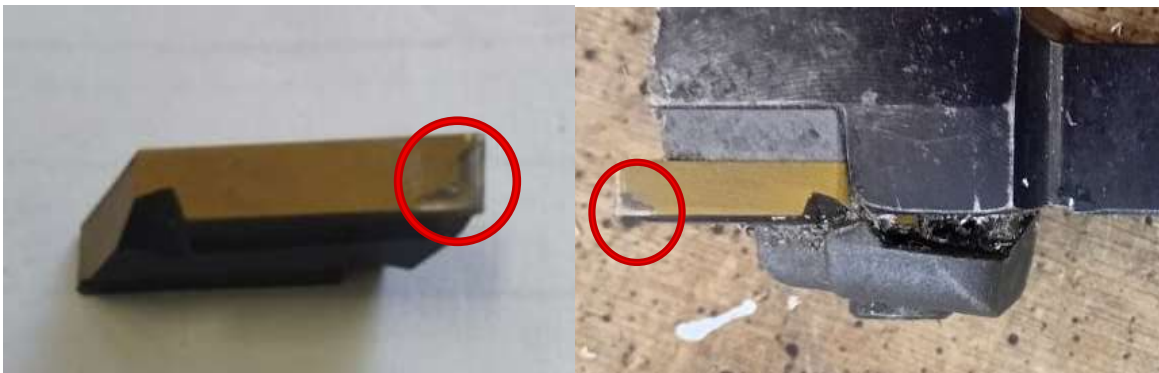


Figure III-7 Tool Condition During Dry Machining

On the other hand, when lubrication was applied, the cutting tool showed lower wear and maintained a cleaner condition throughout the machining process. The cutting edge remained more stable with minimal chipping, and fewer material residues were visible on the tool surface (Figure III.8).



Figure III-8 Tool Condition with Lubrication

These observations demonstrate that lubrication plays an important role in reducing tool degradation and improving machining quality.

III.7. Vibration Analysis:

Through the graphical curves, and under the same cutting conditions ($V_c = 280$ m/min, $a = 0.42$ mm/rev, $a_p = 0.8$ mm), a clear difference can be observed between dry machining and machining with lubrication. In the lubricated case (Test N° 34), the vibration signal still exhibits fluctuations; however, the distribution of peaks becomes more structured and slightly less chaotic compared to the dry condition (Test N° 7). Although some high peaks are present (reaching higher amplitudes than in dry cutting), their occurrence is less irregular, indicating a partial stabilization of the cutting process.

In contrast, the dry condition shows a more dispersed signal with frequent abrupt variations, reflecting unstable cutting behavior and a higher tendency toward random vibrations. The presence of lubrication helps reduce friction at the tool–workpiece interface, which contributes to damping certain vibration modes and improving cutting stability.

Consequently, even if the maximum amplitudes in the lubricated case can occasionally be higher, the overall behavior is more controlled, which positively affects the surface finish. Lubrication promotes better chip evacuation and reduces thermal effects, leading to a more homogeneous surface and lower roughness compared to dry machining.

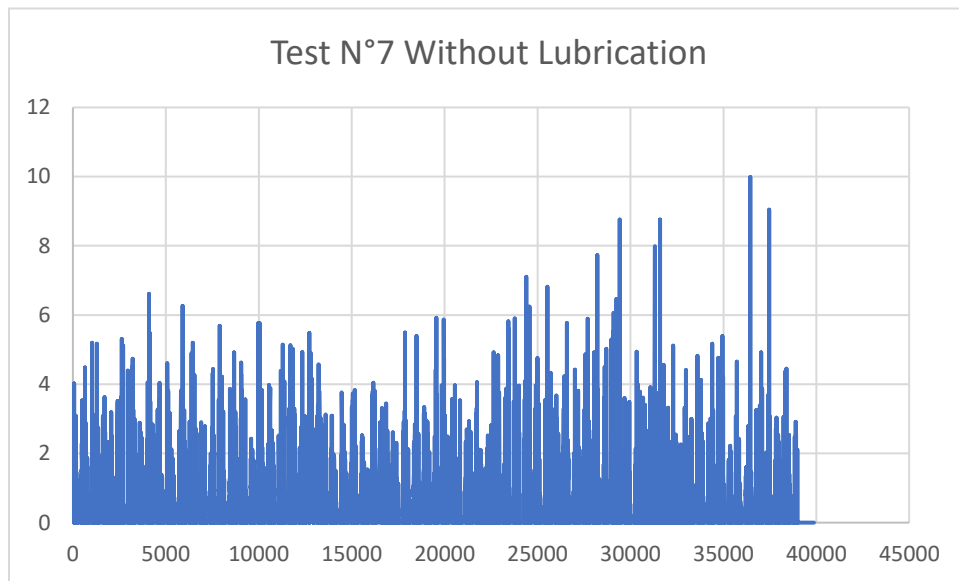


Figure III-9 Tool vibrations without Lubrication Test 7

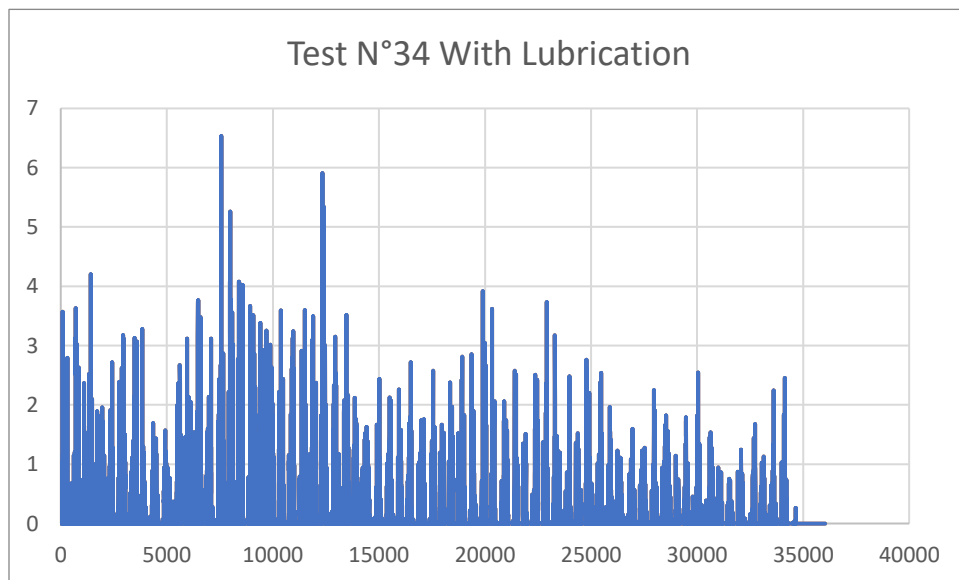


Figure III-10 Tool vibrations with Lubrication Test 34

The graphical curves above and the obtained data show that the machining conditions used, namely a cutting speed of **840 rpm**, a feed rate of **0.42 mm/rev**, and a depth of cut of **1.6 mm**, generate noticeable vibrations in both cases (with and without lubrication). However, the behavior of the signals differs significantly depending on the lubrication condition.

For the test carried out **without lubrication (Test N°18)**, the graph presents high and irregular amplitude peaks distributed over the entire machining period, with maximum values exceeding **12**. This indicates unstable cutting conditions and intense vibrations at the tool-workpiece interface. The absence of lubrication increases friction and heat generation, which negatively affects machining stability and leads to higher surface irregularities and poorer surface finish.

In contrast, the test performed **with lubrication (Test N°45)** shows lower and less frequent amplitude peaks, with maximum values remaining below approximately **9.5**. The signal becomes more stable and less dispersed, especially toward the end of the machining process where the amplitudes decrease progressively. This behavior reflects improved cutting stability due to the presence of lubrication, which reduces friction, minimizes heat generation, and dampens vibrations during machining.

The comparison between the two curves clearly demonstrates that lubrication improves machining performance under the selected parameters. Although the chosen cutting conditions (high feed rate and relatively large depth of cut) tend to generate moderate-to-high vibrations, the use of lubrication significantly reduces their intensity and produces a more stable cutting process. Consequently, lubrication contributes to better surface quality, lower tool stress, and improved machining efficiency.

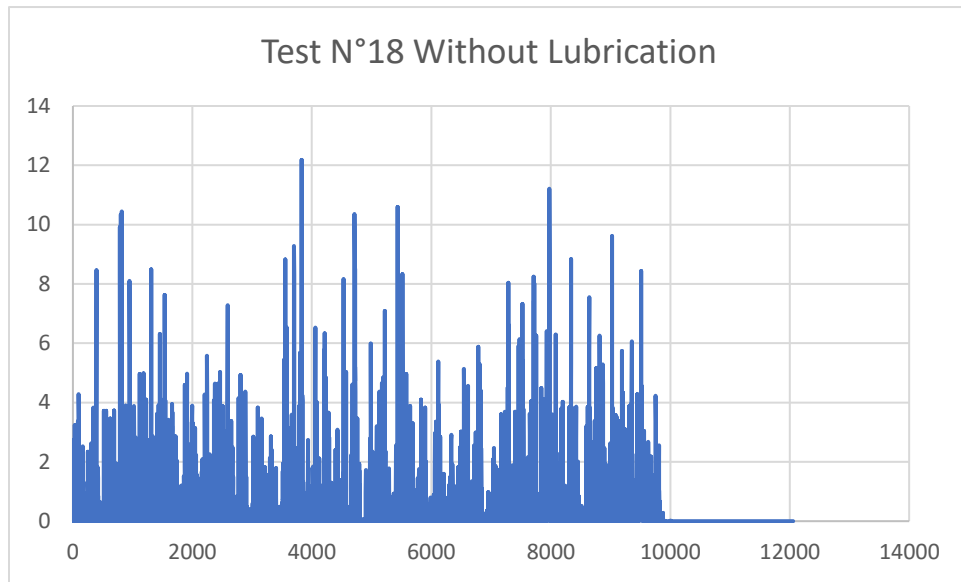


Figure III-11 Tool vibrations without Lubrication Test 18

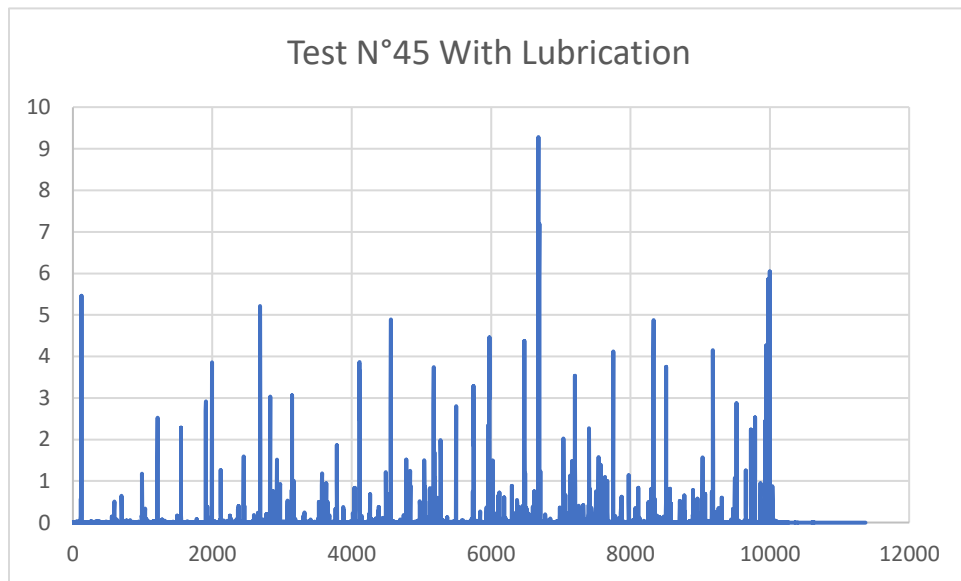


Figure III-12 Tool vibrations with Lubrication Test 45

The graphical curves of Test N°26 without lubrication and Test N°53 with lubrication clearly show that the machining conditions have a significant influence on the stability of the cutting process and on the vibration behavior during machining. In the test carried out without lubrication, the graph presents irregular and relatively high amplitude peaks reaching approximately 4.7, with continuous fluctuations throughout the machining period. This indicates unstable cutting conditions and high friction at the cutting zone, which leads to increased vibrations and negatively affects the surface finish quality.

On the other hand, the test performed with lubrication shows a considerable reduction in amplitude values, where the maximum peaks remain around only 0.32. The signal is more regular and stable, and the vibrations become very weak during most of the machining process. This behavior reflects a clear improvement in cutting stability due to the lubrication effect, which reduces friction and heat generation between the cutting tool and the workpiece.

The comparison between the two curves demonstrates that lubrication significantly decreases vibration intensity compared to dry machining. This indicates that lubrication improves the cutting conditions, reduces mechanical stresses on the cutting tool, and ensures a more stable and efficient machining process. Consequently, lubrication contributes directly to better surface finish quality and longer tool life, especially under the machining conditions used in these tests.

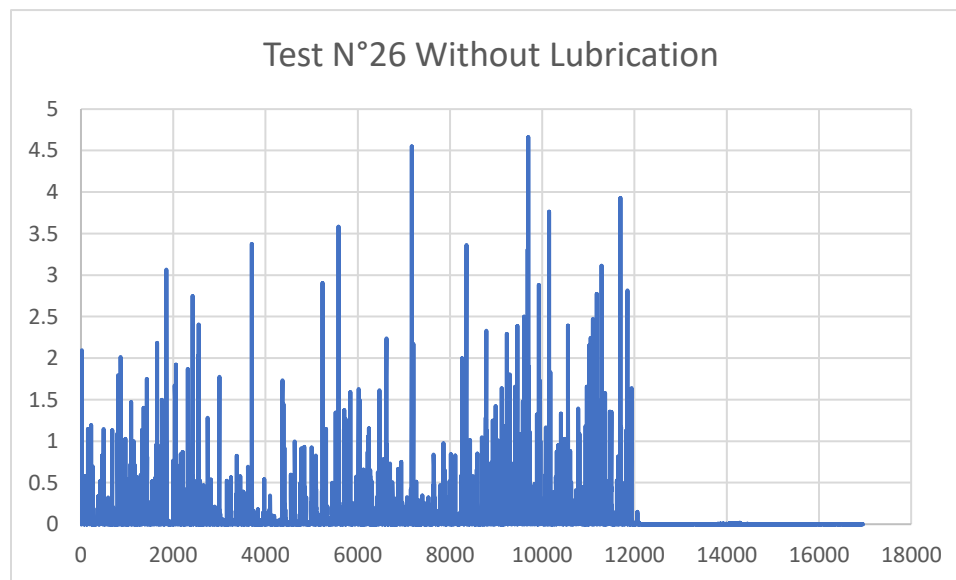


Figure III-13 Tool vibrations without Lubrication Test 26

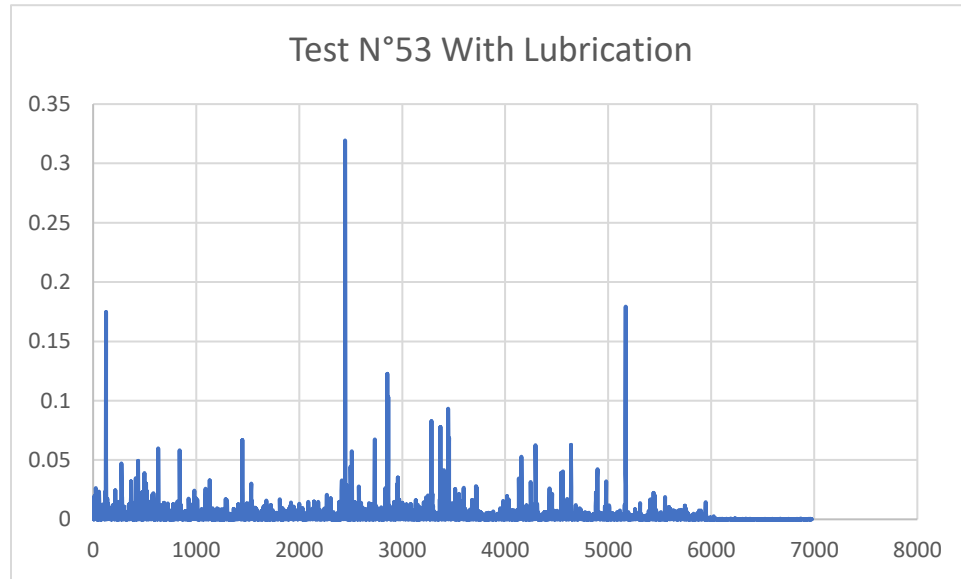


Figure III-14 Tool vibrations with Lubrication Test 53

III.8. Conclusion:

This chapter provides a detailed analysis of the different responses obtained during the experimental study, including surface roughness, temperature, chip formation, vibrations, and tool wear. It also discusses the influence of machining parameters on these outputs and explains the relationships between the cutting conditions and the machining performance.

IV. Chapter IV: Optimization of Cutting Conditions in Turning (RSM and Desirability Function)

IV.1. Introduction:

The previous chapter focused on modeling the effects of machining parameters on several performance indicators, including surface roughness, tool wear, and vibration behavior, by applying the Response Surface Methodology (RSM). In this chapter, the study moves toward the optimization stage, where the objective is to determine the most suitable cutting conditions capable of improving machining performance while maintaining process stability and tool efficiency.

To achieve this goal, two optimization techniques are adopted. The first one is the Taguchi method, which offers an effective experimental design approach with a limited number of tests. The second is the desirability function approach, used to combine multiple performance criteria into a single optimization index. By integrating these methods, it becomes possible to identify the optimal combination of cutting parameters that satisfies different machining requirements simultaneously.

The approach developed in this chapter aims to enhance machining quality, reduce undesirable effects during turning operations, and improve the overall efficiency of the process. This optimization procedure also contributes to achieving more reliable and sustainable machining conditions that meet industrial and environmental expectations.

IV.2. Analysis of Variance (ANOVA) and Surface Roughness Modeling (Ra) Without using Lubrication:

IV.2.1. Analysis of Variance for Surface Roughness (Ra):

Table IV.1 presents the ANOVA results for surface roughness (Ra) during the turning process. The analysis shows that the developed quadratic model is highly significant, with an F-value of 38.99 and a p-value lower than 0.0001, confirming the adequacy and reliability of the model for predicting surface roughness.

Regarding the main effects, none of the linear terms (A-Vc, B-a, C-Ap) are statistically significant, as all their p-values are greater than 0.05. This indicates that, individually, the cutting parameters do not have a strong direct influence on Ra within the studied range.

For the interaction effects, only the interaction term AC shows a statistically significant effect on surface roughness ($p = 0.0105$), indicating that the combined influence of cutting speed and depth of cut plays a role in the variation of R_a . In contrast, the interactions AB and BC are not significant, suggesting weak combined effects between these parameters.

Concerning the quadratic terms, only A^2 is statistically significant ($p = 0.0009$), demonstrating the presence of a strong nonlinear effect of cutting speed on surface roughness. The remaining quadratic terms (B^2 and C^2) are not significant, indicating negligible curvature effects for feed rate and depth of cut.

Overall, the results indicate that surface roughness R_a is mainly influenced by nonlinear effects of cutting speed and the interaction between cutting speed and depth of cut, while most of the linear and remaining interaction effects are not significant within the studied experimental domain.

Table IV-1 ANOVA results for surface roughness (R_a)

Source	Sum of Squares	df	Mean Square	F-value	p-value	significant
Model	80.11	9	8.9	38.99	< 0.0001	significant
A-Vc	0.2934	1	0.2934	1.29	0.2726	Not significant
B-a	0.0263	1	0.0263	0.1151	0.7386	Not significant
C-Ap	0.0425	1	0.0425	0.1864	0.6714	Not significant
AB	0.2622	1	0.2622	1.15	0.2989	Not significant
AC	1.88	1	1.88	8.26	0.0105	Not significant
BC	0.0494	1	0.0494	0.2165	0.6476	Not significant
A²	3.66	1	3.66	16.03	0.0009	significant
B²	0.0275	1	0.0275	0.1206	0.7326	Not significant
C²	0.3733	1	0.3733	1.64	0.2182	Not significant
Residual	3.88	17	0.2283			
Cor Total	84	26				

IV.2.2. Surface Roughness Modeling (Ra):

The predictive equation for surface roughness (Ra) under lubrication conditions is given by:

$$\text{Ra} = 0.8065 + 0.3566A + 1.15B + 3.88C + 0.3541AB + 0.8024AC + 3.87BC \\ + 0.1117A^2 + 1.15B^2 + 0.6335C^2$$

The analysis of the Ra model shows that all machining parameters (A, B, and C) contribute positively to surface roughness, indicating that an increase in cutting speed (A), feed rate (B), or depth of cut (C) leads to a general increase in Ra under lubrication conditions. Among the principal effects, the depth of cut (C) exhibits the strongest influence on the response, followed closely by the feed rate (B), while the cutting speed (A) shows a relatively lower effect.

The interaction terms (AB, AC, and BC) also have a noticeable influence on surface roughness, demonstrating that the combined effects of machining parameters significantly affect the final surface quality. In particular, the BC interaction indicates a strong coupling between feed rate and depth of cut, which plays an important role in surface formation during turning.

Regarding the quadratic terms, A^2 , B^2 , and C^2 confirm the presence of nonlinear behavior in the machining process. The relatively higher contribution of B^2 and C^2 suggests that feed rate and depth of cut introduce stronger curvature effects compared to cutting speed. This indicates that the response surface is not purely linear and requires a second-order model to accurately describe the behavior of Ra.

Overall, the developed quadratic model highlights the importance of main effects, interaction effects, and nonlinear contributions in accurately predicting and optimizing surface roughness under lubrication conditions, confirming the effectiveness of the response surface methodology in modeling the turning process.

IV.2.3. Fit Statistics (Ra)

The table presents the principal statistical indicators used to evaluate the adequacy and reliability of the developed model for surface roughness (Ra):

• Std. Dev. (Standard Deviation): 0.4778

The standard deviation measures the dispersion of the experimental data around the mean value. The relatively low value of 0.4778 indicates good consistency between the experimental and predicted results, with limited variability in the response.

• Mean: 4.8434

The mean represents the average value of the surface roughness response (R_a). It provides a reference value for evaluating the relative dispersion of the experimental data.

• C.V. % (Coefficient of Variation): 9.87%

The coefficient of variation expresses the relative dispersion of the data as a percentage of the mean value. A C.V. of 9.87% indicates acceptable reproducibility and good precision of the experimental results, confirming the reliability of the conducted experiments.

• R^2 (Coefficient of Determination): 0.9538

The coefficient of determination indicates the proportion of variability in the response explained by the model. The obtained value of 0.9538 shows that approximately 95.38% of the variation in surface roughness is explained by the developed model, demonstrating an excellent agreement between predicted and experimental values.

• Adjusted R^2 : 0.9293

The adjusted R^2 accounts for the number of model terms and provides a more realistic evaluation of model quality. The high value of 0.9293 confirms that the model maintains strong explanatory capability even after adjustment for model complexity.

• Predicted R^2 : 0.8868

The predicted R^2 evaluates the predictive capability of the model for new observations. The value of 0.8868 indicates good predictive performance. In addition, the difference between the Adjusted R^2 and Predicted R^2 is 0.0425, which is less than 0.2, confirming a reasonable agreement and indicating that the model is not overfitted.

• **Adeq Precision: 22.2518**

Adequate Precision measures the signal-to-noise ratio of the model. A ratio greater than 4 is considered desirable. The obtained value of 22.2518 indicates a very strong signal and demonstrates that the model can effectively navigate the design space and provide reliable predictions.

Table IV-2 Fit statistics (Ra)

Std. Dev.	0.4778	R²	0.9538
Mean	4.84	Adjusted R²	0.9293
C.V. %	9.87	Predicted R²	0.8868
		Adeq Precision	22.2518

Overall, the statistical indicators demonstrate that the developed model for surface roughness (Ra) is robust, reliable, and possesses strong predictive capability. The high R² value, the close agreement between Adjusted R² and Predicted R², and the excellent Adeq Precision value all confirm the validity and adequacy of the developed model for accurately representing the experimental data.

IV.2.4. Comparison Between Experimental and Predicted Values:

Figure IV.1 illustrates a strong agreement between the experimental and predicted values of surface roughness (Ra), demonstrating the accuracy and effectiveness of the developed quadratic model. The distribution of the experimental runs, represented by different colored points, provides a clear visualization of the relationship between the measured data and the model predictions.

Most of the points are located close to the diagonal reference line, indicating that the predicted values are very close to the actual experimental results. This close alignment confirms

the good fitting quality of the model and its capability to accurately represent the behavior of surface roughness during the turning process.

Although a few scattered points can be observed away from the diagonal line, these deviations remain limited and may be attributed to experimental uncertainties, machine vibrations, environmental conditions, or measurement errors during machining tests.

Overall, the comparison demonstrates that the proposed model has high reliability and satisfactory predictive performance, making it suitable for analyzing and optimizing the machining parameters affecting surface roughness.

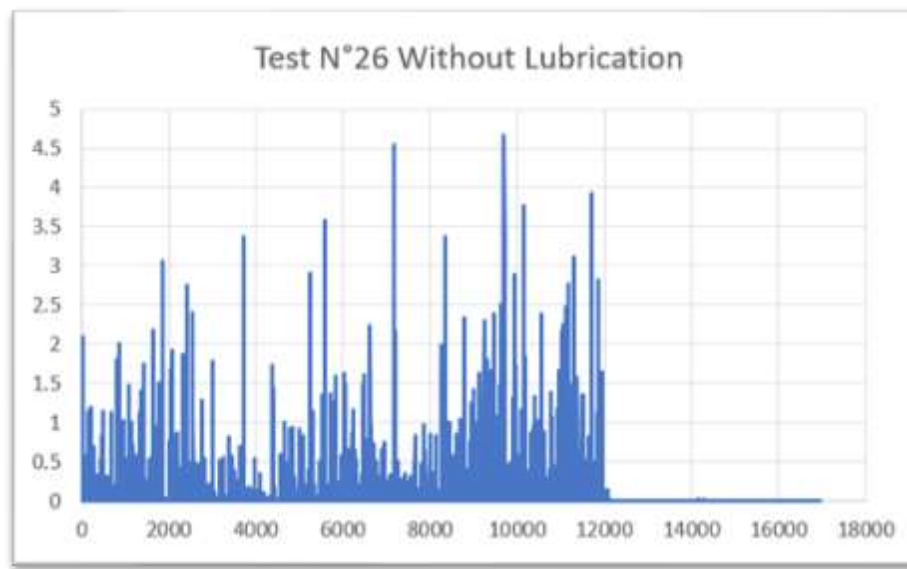


Figure IV-1 Predicted vs. Actual Plot for Surface Roughness (Ra)

IV.1.5 Response surface analysis and contour plots for (Ra)

Figures IV.4, IV.3 and IV.2 represent the 3D response surface graphs of surface roughness (Ra) as a function of cutting parameters (V_c , a , and A_p) during dry machining.

Figure IV.2 illustrates the response surface of Ra as a function of the variation of factors A: V_c (cutting speed) and B: a (feed rate), while holding C: A_p (depth of cut) constant at a certain value (likely within the tested range). It is observed that an increase in feed rate (B) induces a

significant elevation of R_a . Conversely, an increase in cutting speed (A) leads to a decrease in R_a , especially at lower feed rates. The surface shows that the lowest R_a values are obtained at high cutting speeds and low feed rates, while the highest roughness occurs at low cutting speeds and high feed rates. This indicates that feed rate has a dominant positive effect on R_a , whereas cutting speed has a negative effect. The interaction between V_c and feed rate appears moderate.

Figure IV.3 presents the response surface of R_a as a function of the variation of factors A: V_c and C: A_p , with B: a (feed rate) fixed at 0.21. Analysis of the surface reveals that depth of cut (A_p) has a relatively small influence on R_a compared to cutting speed. Increasing V_c reduces R_a significantly, while varying A_p produces only minor changes. The surface is relatively flat along the A_p axis, suggesting that depth of cut is not a critical factor for R_a under these conditions. The lowest R_a values are achieved at high cutting speeds regardless of A_p .

Figure IV.4 shows the response surface of R_a as a function of the variation of factors B: a (feed rate) and C: A_p (depth of cut), with A: V_c held constant at 840. It is clear that feed rate (B) is the most influential factor, as increasing feed rate causes a sharp rise in R_a . Depth of cut (C) shows a moderate positive effect, but its influence is less pronounced than that of feed rate. The interaction between feed rate and depth of cut appears modest, with the highest R_a values occurring at high feed rates combined with high depths of cut.

The results found are in perfect agreement with the physical understanding of machining: feed rate is the primary parameter affecting surface roughness (R_a), followed by cutting speed which has an inverse relationship. Depth of cut plays a secondary role.

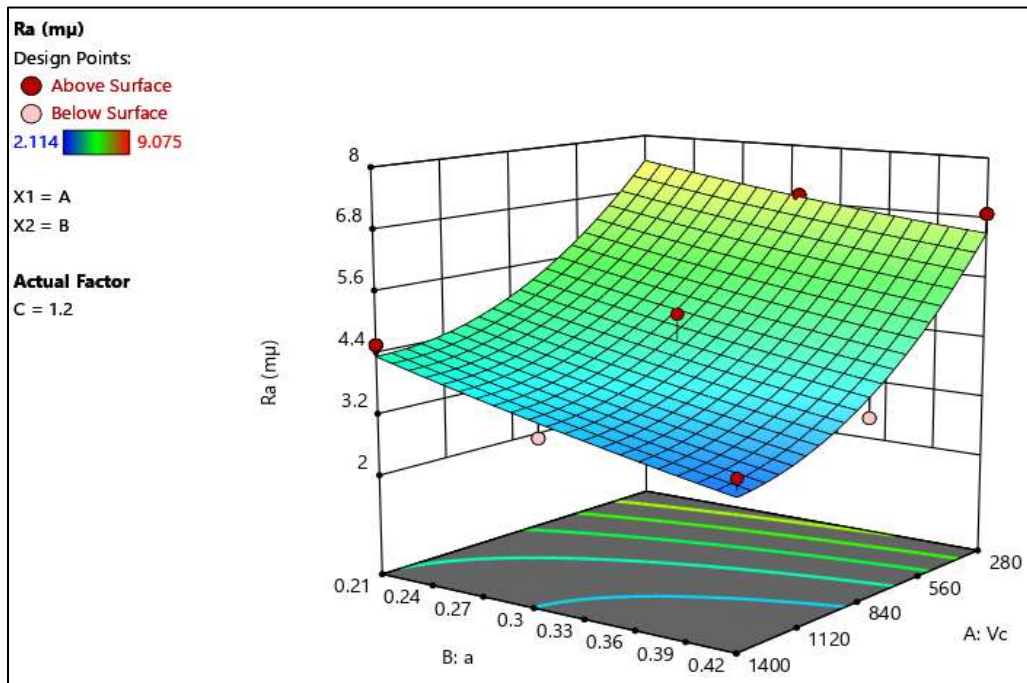


Figure IV-2 3D response surface of Ra roughness as a function of cutting speed (V_c) and feed rate (a), with depth of cut (A_p) constant.

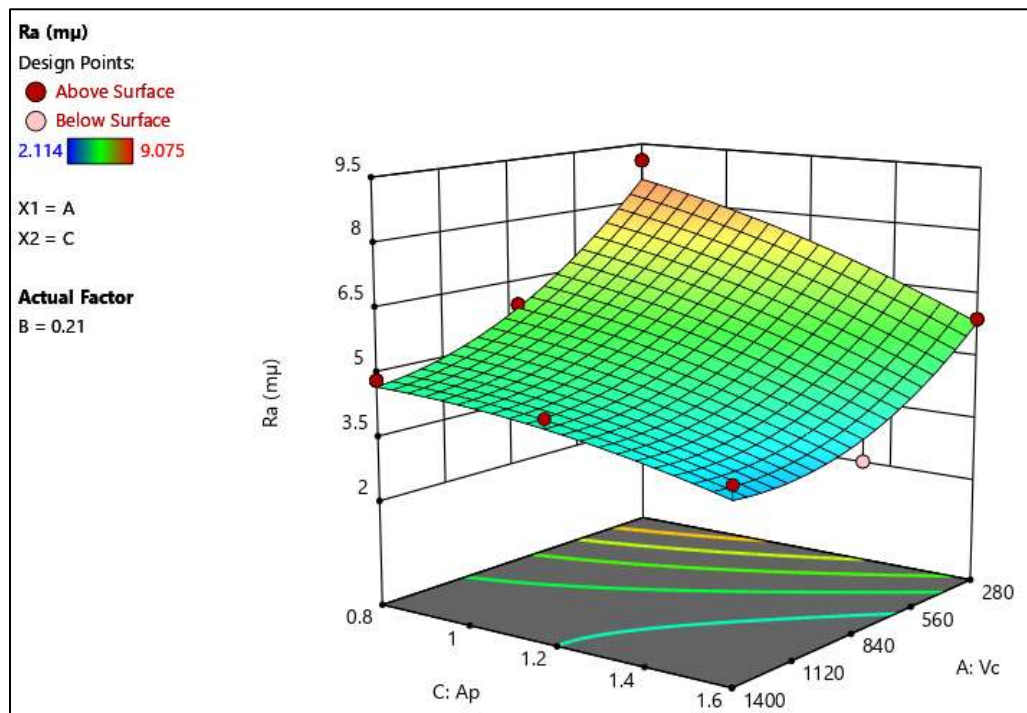


Figure IV-3 3D response surface of Ra roughness as a function of cutting speed (V_c) and depth of cut (A_p), with feed rate (a) constant at 0.21.

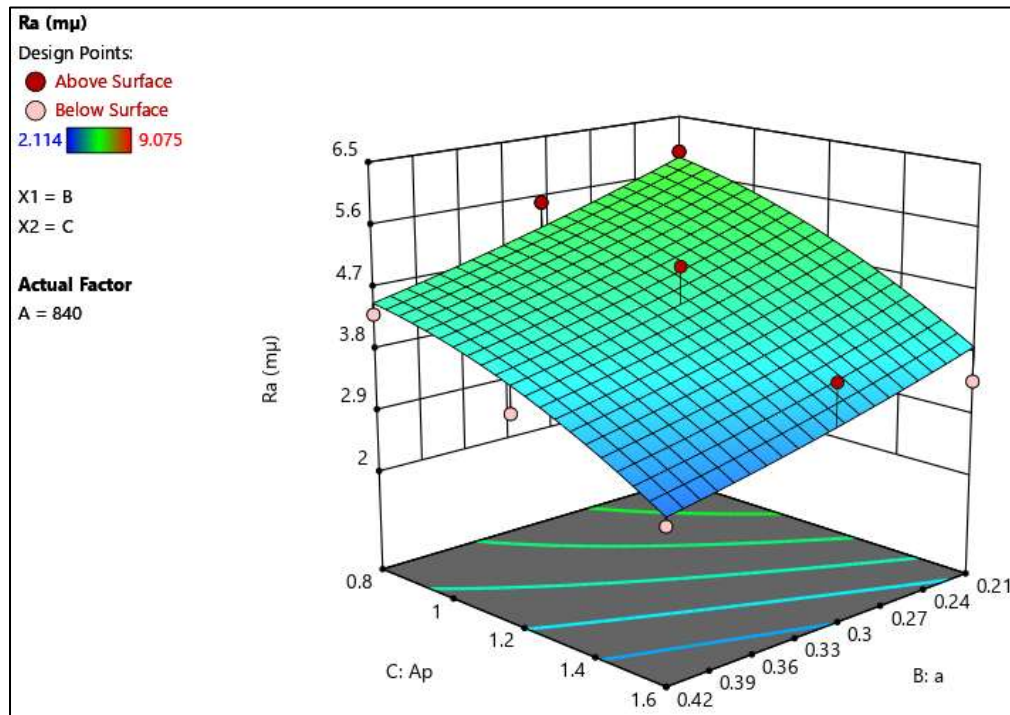


Figure IV-4 3D response surface of Ra roughness as a function of feed rate (a) and depth of cut (A_p), with cutting speed (V_c) constant at 840.

IV.2.5. Normal probability plots of residuals for Ra:

Figure IV.5 presents the normal probability diagram of residuals for the response variable under study (e.g., Ra, Rz, or Temperature). This plot is used to assess the normality of residuals, which is a key assumption of regression analysis and ANOVA.

The normal probability plot indicates that the residuals are approximately normally distributed, as most points lie close to the straight diagonal line. This satisfies one of the key assumptions for the validity of the statistical models used in this study. The few observed minor deviations do not significantly affect the overall normality of the residuals. This normality ensures the reliability of the regression and ANOVA results.

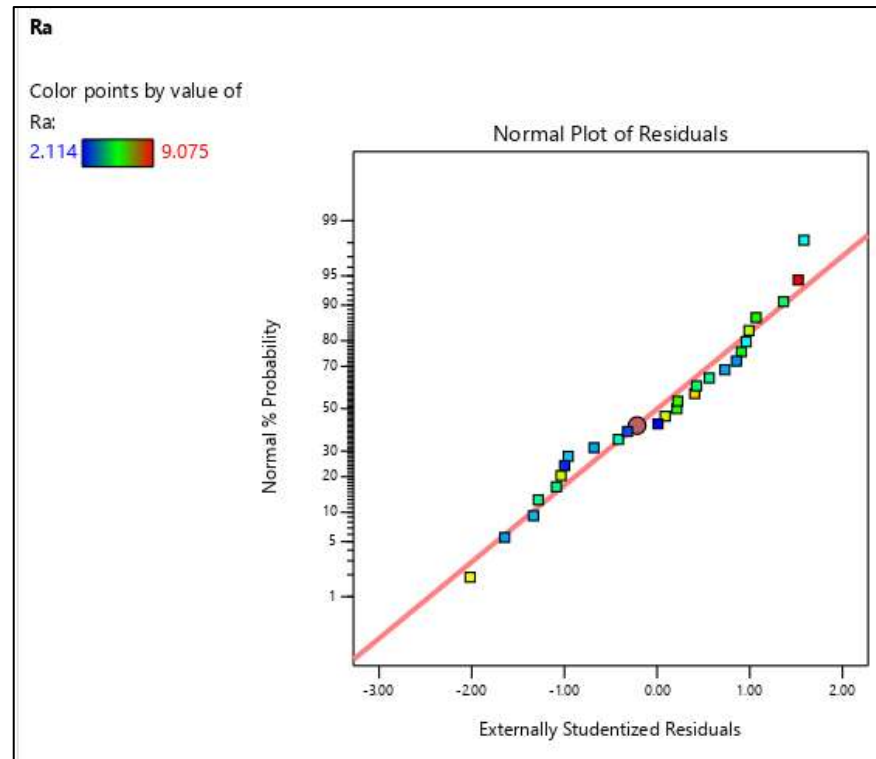


Figure IV-5 Normal probability plots of residuals for Ra

IV.3. Analysis of Variance (ANOVA) and Surface Roughness Modeling (Rz) Without using Lubrication:

IV.3.1. Analysis of variance for (Rz)

Table (above) presents the ANOVA results for surface roughness (Rz) during the turning process. The analysis shows that the cutting speed (V_c) has a highly significant effect on Rz ($p < 0.0001$), contributing approximately 29.06% of the total variation ($142.07/488.86 \times 100$). This confirms that cutting speed is one of the dominant parameters influencing surface roughness.

The feed rate (a) alone is not statistically significant ($p = 0.1758$), with a contribution of approximately 0.96%, while the depth of cut (A_p) is also not significant ($p = 0.5186$), contributing

only 0.21% of the total variation. These results indicate that the direct individual effects of feed rate and depth of cut on R_z are relatively weak within the studied experimental range.

Regarding interaction effects, the interaction term AB ($V_c \times a$) is highly significant ($p < 0.0001$), contributing approximately 29.28% of the total variation, making it the most influential term in the developed model. Similarly, the interaction AC ($V_c \times A_p$) is also highly significant ($p < 0.0001$), with a contribution of approximately 12.56%, indicating a strong combined influence of cutting speed and depth of cut on surface roughness. In contrast, the interaction BC ($a \times A_p$) is not significant ($p = 0.5047$), contributing only 0.22%.

For the quadratic effects, the term C^2 (A_p^2) is statistically significant ($p = 0.0078$), contributing approximately 4.37% of the total variation, which confirms the existence of a nonlinear relationship between depth of cut and surface roughness. On the other hand, the quadratic terms A^2 and B^2 are not significant, indicating weak curvature effects for cutting speed and feed rate.

The overall model is highly significant ($p < 0.0001$), demonstrating that the combined effects of machining parameters and their interactions adequately explain the variability of surface roughness (R_z).

Overall, the most influential factors affecting R_z are the interaction between cutting speed and feed rate (AB), followed by cutting speed alone (A), and the interaction between cutting speed and depth of cut (AC). The quadratic effect of depth of cut (C^2) also plays a moderate but significant role in predicting surface roughness.

Table IV-3 ANOVA results for surface roughness (R_z)

Source	Sum of Squares	df	Mean Square	F-value	p-value	significant
Model	2659.77	9	295.53	29.74	< 0.0001	significant
A-V_c	0.9678	1	0.9678	0.0974	0.7588	Not significant
B-a	3.91	1	3.91	0.394	0.5386	Not significant
C-A_p	12.27	1	12.27	1.24	0.2818	Not significant
AB	0.6612	1	0.6612	0.0665	0.7995	Not significant
AC	109.51	1	109.51	11.02	0.0041	significant
BC	12.9	1	12.9	1.3	0.2702	Not significant

A²	156.95	1	156.95	15.8	0.001	significant
B²	4	1	4	0.4021	0.5345	Not significant
C²	6.86	1	6.86	0.6899	0.4177	Not significant
Residual	168.91	17	9.94			
Cor Total	2828.68	26				

IV.3.2. Modelling of surface roughness (Rz)

The equation for predicting surface roughness Rz (in coded factors) is given by:

$$Rz = -34606.13 + 681.08 A - 69864.29 B - 57.75 C \\ + 686.98 AB - 2.26 AC - 60.01 BC \\ + 0.0604 A^2 - 35249.60 B^2 - 1.89 C^2$$

The analysis of the equation reveals the following based on the ANOVA significant terms ($p < 0.05$: A, AB, AC, C²):

- A (Vc): The positive coefficient (+681.08) indicates that increasing cutting speed increases Rz. This is contrary to typical expectations but may be valid within the specific coded ranges.
- AB (Vc × a): The positive coefficient (+686.98) suggests that the interaction between cutting speed and feed rate amplifies Rz when both are increased together.
- AC (Vc × Ap): The negative coefficient (-2.26) indicates that increasing both Vc and Ap slightly reduces Rz, which is a beneficial interaction.
- C² (Ap²): The negative coefficient (-1.89) shows a quadratic effect of depth of cut, meaning that Rz decreases at extreme high or low values of Ap (convex curvature).

The non-significant terms (B, C, BC, A², B²) have p-values > 0.05 , meaning their coefficients are not statistically different from zero. However, they are retained in the model for hierarchy.

The model demonstrates the importance of main effect A (Vc) and interactions AB and AC in predicting Rz. The quadratic effect of Ap (C²) also plays a role. The very large magnitudes of coefficients for B and B² (even though non-significant) may be due to the coding scheme and should be interpreted cautiously.

For practical use, it is recommended to convert the equation to actual factor units using the factor ranges from the experimental design.

IV.3.3. Fit Statistics (Rz)

The table presents the main statistical indicators used to evaluate the adequacy and reliability of the developed model for surface roughness (Rz):

- **Std. Dev. (Standard Deviation): 1.53:** The standard deviation measures the dispersion of the experimental data around the mean value. The obtained value of 1.53 indicates an acceptable level of variability in the response data, showing that the predicted values are reasonably close to the experimental observations.
- **Mean: 9.02:** The mean represents the average value of the surface roughness response (Rz). It serves as a reference value for evaluating the relative dispersion and variability of the experimental data.
- **C.V. % (Coefficient of Variation): 17%:** The coefficient of variation expresses the relative dispersion of the data as a percentage of the mean value. A value of 17% indicates moderate variability in the experimental results. Although higher than ideal low-variation systems, it remains acceptable for machining experiments involving surface roughness responses.
- **R² (Coefficient of Determination): 0.9182:** The coefficient of determination indicates the proportion of variability in the response explained by the model. The value of 0.9182 shows that approximately 91.82% of the variation in surface roughness (Rz) is explained by the developed model, indicating a very good agreement between predicted and experimental values.
- **Adjusted R²: 0.8750:** The adjusted R² accounts for the number of predictors included in the model and provides a more realistic evaluation of model performance. The obtained value of 0.8750 confirms that the model maintains strong explanatory capability after adjustment for model complexity.
- **Predicted R²: 0.8129:** The predicted R² evaluates the predictive performance of the model for new observations. The value of 0.8129 demonstrates good predictive capability. In addition, the difference between Adjusted R² and Predicted R² is approximately 0.0621, which is less than 0.2, confirming reasonable agreement and indicating that the model is not overfitted.

- **Adeq Precision: 17.4997:** Adequate Precision measures the signal-to-noise ratio of the model. A value greater than 4 is considered desirable. The obtained ratio of 17.4997 indicates a strong signal, confirming that the model can effectively navigate the design space and generate reliable predictions.

Overall, the statistical indicators demonstrate that the developed model for surface roughness (Rz) is reliable and statistically adequate. The high R^2 value, the good agreement between Adjusted R^2 and Predicted R^2 , and the strong Adeq Precision value confirm the suitability of the model for representing and predicting the experimental surface roughness data.

Table IV-4 Fit Statistics (Rz)

Std. Dev.	3.15	R²	0.94
Mean	29	Adjusted R²	0.908
C.V. %	10.74	Predicted R²	0.848
		Adeq Precision	18.876

IV.3.4. Response surface analysis and contour plots for (Rz)

Figures (Screenshots 156, 157, and 158) represent the 3D response surface graphs of surface roughness (Rz) as a function of cutting parameters (V_c , a , and A_p) during dry machining.

Figure (Screenshot 156) illustrates the response surface of Rz as a function of the variation of factors A: V_c (cutting speed) and B: a (feed rate), while holding C: A_p (depth of cut) constant at 1.2. It is observed that an increase in feed rate (B) induces a clear elevation of Rz. Similarly, an increase in cutting speed (A) also leads to an increase in Rz. The surface shows a rising trend in both directions, indicating that both factors positively influence Rz. The interaction between V_c and a appears strong, which is consistent with the ANOVA results showing a highly significant AB interaction ($p < 0.0001$).

Figure (Screenshot 157) presents the response surface of Rz as a function of the variation of factors A: V_c and C: A_p , with B: a (feed rate) fixed at 0.21. Analysis of the surface reveals that

cutting speed (V_c) has a pronounced positive effect on R_z . Depth of cut (A_p) also increases R_z , but to a lesser extent. The surface shows a steady rise along the V_c axis, while the A_p axis contributes moderate increases, especially at higher V_c levels. This confirms the significant main effect of A ($p < 0.0001$) and the significant AC interaction ($p < 0.0001$).

Figure (Screenshot 158) shows the response surface of R_z as a function of the variation of factors B: a (feed rate) and C: A_p (depth of cut), with A: V_c fixed at 840. It is clear that feed rate (B) is the dominant factor, as increasing feed rate causes a sharp rise in R_z . Depth of cut (C) shows a relatively flat influence, with only a slight increase in R_z at higher A_p values. The interaction BC is not significant ($p = 0.5047$), which aligns with the nearly parallel contour lines observed. The quadratic effect of A_p (C^2) is significant ($p = 0.0078$), introducing slight curvature.

The results found are in perfect agreement with the ANOVA outcomes: cutting speed (A) and its interactions (AB and AC) are the most influential effects on R_z , while feed rate alone (B) and depth of cut alone (C) are not significant individually. The quadratic term of depth of cut (C^2) also plays a modest role.

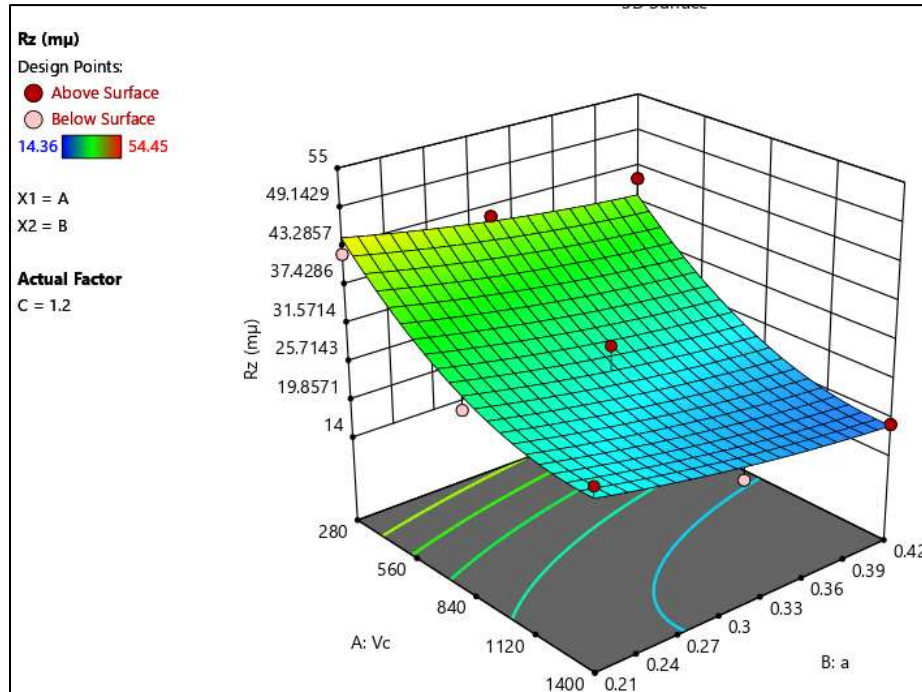


Figure IV-6 3D response surface of R_z roughness as a function of cutting speed (V_c) and feed rate (a), with depth of cut (A_p) constant at 1.2.

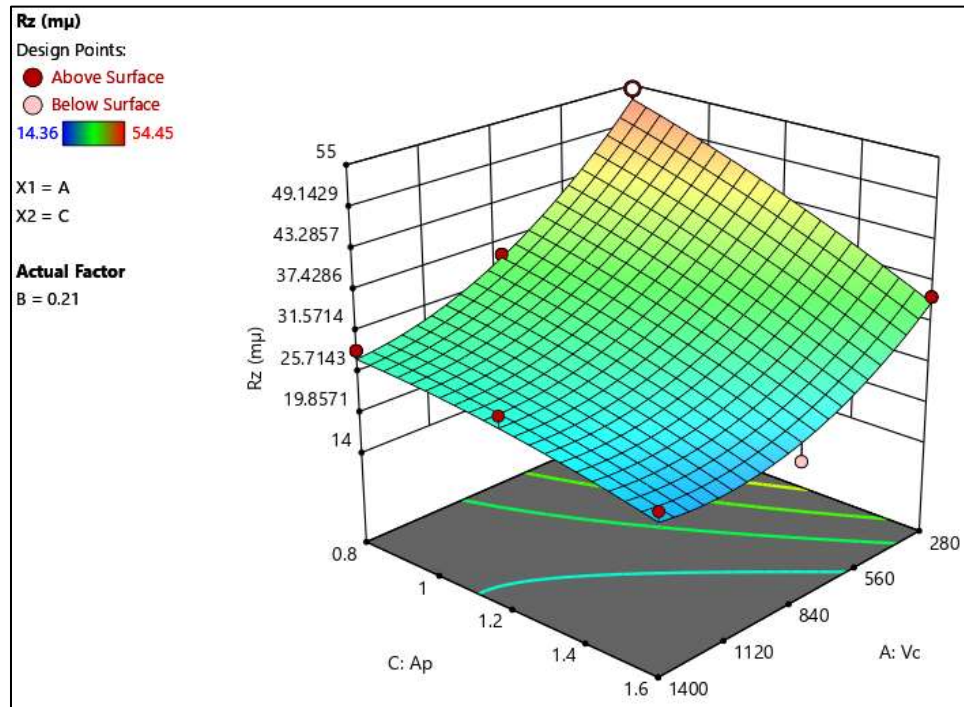


Figure IV-7 3D response surface of Rz roughness as a function of cutting speed (V_c) and depth of cut (A_p), with feed rate (a) constant at 0.21.

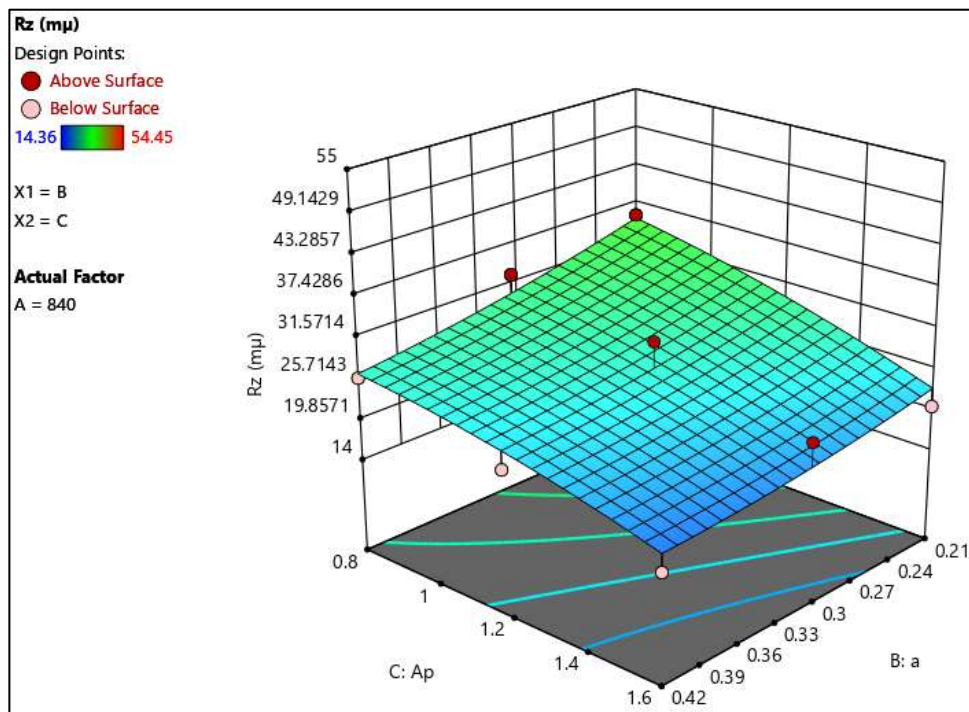


Figure IV-8 3D response surface of Rz roughness as a function of feed rate (a) and depth of cut (A_p), with cutting speed (V_c) constant at 840.

IV.3.5. Normal probability plot of residuals (Rz)

Figure IV.9 presents the normal probability diagram of residuals for Rz. This plot is used to assess the normality of residuals, which is a key assumption of regression analysis and ANOVA.

The normal probability plot for Rz indicates that the residuals are approximately normally distributed, as most points lie close to the straight diagonal line. This satisfies one of the key assumptions for the validity of the statistical models used in this study. The few observed minor deviations do not significantly affect the overall normality of the residuals. This normality ensures the reliability of the regression and ANOVA results for the Rz model.

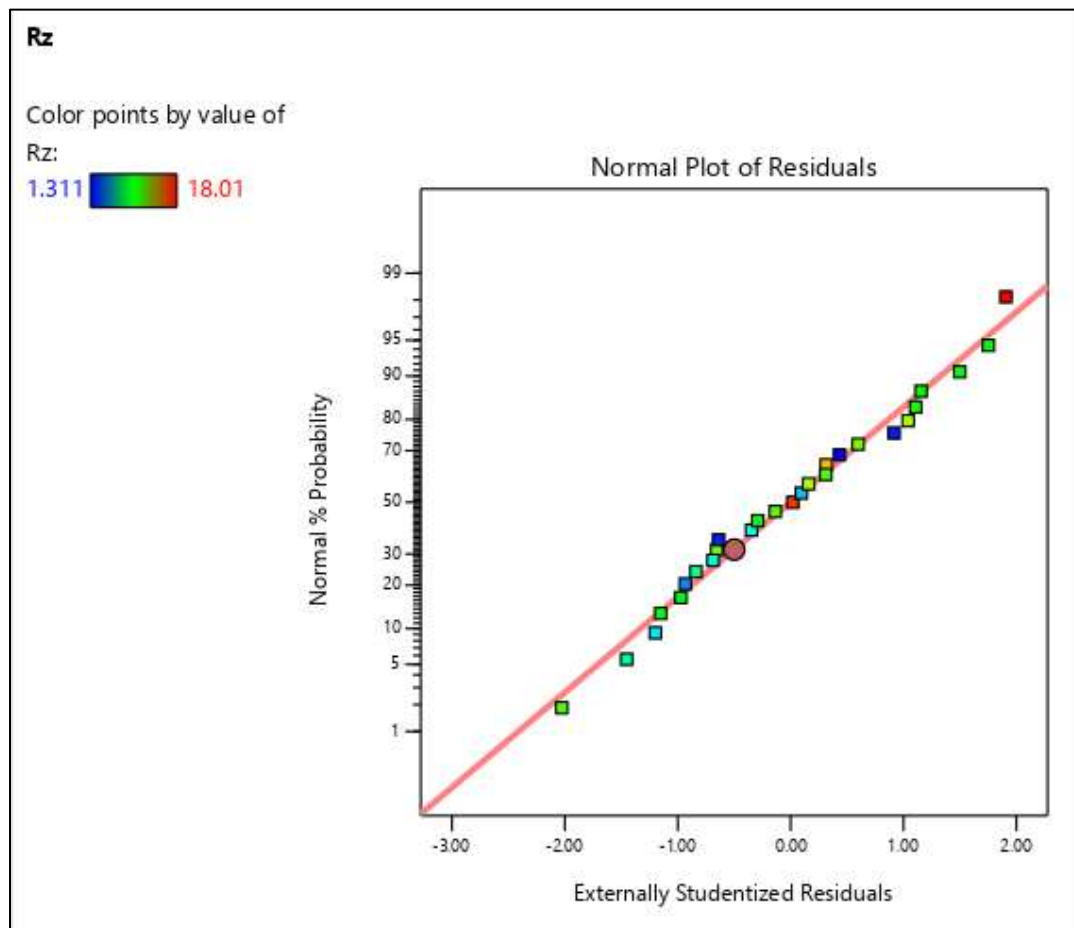


Figure IV-9 Normal probability plot of residuals for Rz

IV.4. Analysis of Variance (ANOVA) and Temperature Modeling (T°):**IV.4.1. Analysis of Variance for Temperature (T°):**

Table IV.5 presents the ANOVA results for temperature (T°) during the turning process. The analysis of variance indicates that the developed model is highly significant, with a very high F-value of 1457.6 and a p-value lower than 0.0001, confirming the reliability and accuracy of the model in describing the temperature response.

Among the main cutting parameters, the cutting speed (A- V_c) is the most influential factor affecting temperature, with a Sum of Squares value of 4426.86 and a very high statistical significance ($p < 0.0001$). This result demonstrates that increasing the cutting speed considerably affects the heat generated during machining.

The interaction term AB also shows a very strong influence on temperature, with an F-value of 258.47 and a highly significant p-value (< 0.0001), indicating that the combined effect of cutting speed and feed rate plays an important role in thermal behavior. In addition, the AC interaction is also statistically significant, showing that the interaction between cutting speed and depth of cut contributes significantly to temperature variation.

The quadratic term A^2 is another important significant factor, confirming the existence of a non-linear relationship between cutting speed and temperature.

On the other hand, the feed rate (B-a), depth of cut (C- A_p), interaction BC, and the quadratic terms B^2 and C^2 are statistically insignificant since their p-values are greater than 0.05. This indicates that their individual effects on temperature are relatively weak under the studied machining conditions.

Overall, the analysis confirms that temperature during turning is mainly controlled by the cutting speed and its interactions with the other machining parameters, while secondary factors have a comparatively lower influence on the thermal response.

Table IV-5 presents the ANOVA results for temperature (T°) during the turning process

Source	Sum of Squares	df	Mean Square	F-value	p-value	significant
Model	2.13E+05	9	23689.03	1457.6	< 0.0001	significant
A-Vc	4426.86	1	4426.86	272.39	< 0.0001	significant
B-a	123.86	1	123.86	7.62	0.0134	Not significant
C-Ap	62.31	1	62.31	3.83	0.0668	Not significant
AB	4200.62	1	4200.62	258.47	< 0.0001	significant
AC	2514.37	1	2514.37	154.71	< 0.0001	significant
BC	55.03	1	55.03	3.39	0.0833	Not significant
A²	2699.52	1	2699.52	166.1	< 0.0001	significant
B²	127.96	1	127.96	7.87	0.0122	Not significant
C²	3.89	1	3.89	0.2394	0.6309	Not significant
Residual	276.29	17	16.25			
Cor Total	2.14E+05	26				

IV.4.2. Temperature Modeling (T°):

The predictive equation for temperature (T°) is given by:

$$T^\circ = 23689.03 + 4426.86A + 123.86B + 62.31C + 4200.62AB + 2514.37AC \\ + 55.03BC + 2699.52A^2 + 127.96B^2 + 3.89C^2$$

The analysis of the temperature equation shows that the cutting parameters A, B, and C have different levels of influence on the thermal response during machining. Among the main effects, the cutting speed (A) is the dominant factor affecting temperature, as indicated by its high coefficient and strong statistical significance.

The interaction effects AB and AC are also highly significant, which demonstrates that the combined influence of cutting speed with feed rate and depth of cut considerably affects the generated heat during the turning process.

Furthermore, the quadratic term A^2 reveals the presence of a non-linear relationship between cutting speed and temperature, confirming that temperature evolution does not follow a purely linear trend.

On the other hand, the terms related to B, C, BC, B^2 , and C^2 show weaker effects on the response, indicating that their contribution to temperature variation is less important under the studied cutting conditions.

Overall, the developed model highlights the importance of both the principal and interaction effects in predicting machining temperature and confirms the suitability of the response surface methodology (RSM) for thermal analysis and optimization in turning operations.

IV.4.3. Fit Statistics (T°):

Table IV.6 presents the main statistical indicators used to evaluate the adequacy and performance of the developed temperature (T°) model.

For temperature (T°), the Predicted R^2 value of 0.9969 is in excellent agreement with the Adjusted R^2 value of 0.9980, since the difference between them is much lower than 0.2. This strong

agreement confirms the excellent predictive capability of the developed model and demonstrates that the model can accurately estimate new experimental observations.

Moreover, the Adeq Precision value of 126.9039 is significantly higher than the recommended minimum value of 4, indicating an extremely strong signal-to-noise ratio. This result confirms that the model possesses a very high level of adequacy and can effectively navigate the design space while providing highly reliable predictions.

In addition, the low coefficient of variation (C.V. = 2.17%) and the relatively small standard deviation (4.03) indicate excellent experimental precision and very low dispersion between the predicted and experimental values. The very high R^2 value (0.9987) also demonstrates that more than 99% of the variability in temperature is explained by the developed model, confirming an outstanding agreement between experimental and predicted results.

Table IV-6 Fit Statistics (T°)

Std. Dev.	4.03	R^2	0.9987
Mean	185.92	Adjusted R^2	0.998
C.V. %	2.17	Predicted R^2	0.998
		Adeq Precision	126.9039

IV.4.4. Response surface analysis and contour plots for (Temperature):

Figures (Screenshots 137, 138, and 139) represent the 3D response surface graphs of cutting temperature as a function of cutting parameters (V_c , a , and A_p) during dry machining.

Figure illustrates the response surface of temperature as a function of the variation of factors A: V_c (cutting speed) and B: a (feed rate), while holding C: A_p (depth of cut) constant at 1.2. It is observed that an increase in cutting speed (V_c) induces a significant elevation of temperature. Similarly, an increase in feed rate (a) also leads to an increase in temperature. The surface shows a steep rise along the V_c axis, indicating that cutting speed is the dominant factor affecting

temperature. The interaction between V_c and a appears to be strong, as the highest temperatures are reached at high levels of both parameters.

Presents the response surface of temperature as a function of the variation of factors A: V_c and C: A_p , with B: a (feed rate) held constant at a specific value (likely 0.21 or similar). It is clear that cutting speed (V_c) has the most pronounced positive effect on temperature. Depth of cut (A_p) also increases temperature, but its influence is less significant compared to V_c . The temperature rises sharply when V_c increases, especially at higher A_p values, indicating a positive interaction between V_c and A_p . The color bar shows temperature ranging from approximately 70°C to 381°C, confirming the strong thermal effect of cutting speed.

Shows the response surface of temperature as a function of the variation of factors B: a (feed rate) and C: A_p (depth of cut), with A: V_c fixed at 840. Analysis of the surface reveals that feed rate (a) has a moderate positive influence on temperature, while depth of cut (A_p) also increases temperature but to a lesser extent. The surface is relatively flat compared to the previous figures, confirming that when cutting speed is held constant, the variations in feed and depth of cut produce smaller changes in temperature. Nevertheless, higher values of both a and A_p lead to higher temperatures.

The results found are in perfect agreement with physical expectations: cutting speed is the most influential parameter on cutting temperature, followed by feed rate and depth of cut. The interactions between V_c and the other parameters further amplify the thermal effect.

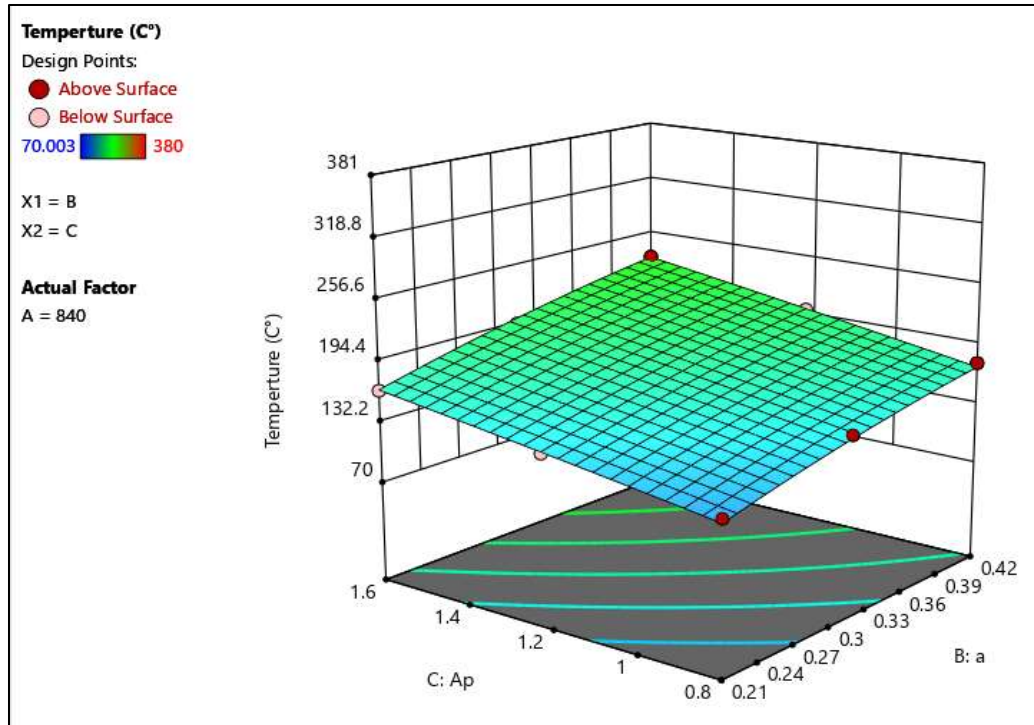


Figure IV-10 3D response surface of cutting temperature as a function of feed rate (a) and depth of cut (Ap), with cutting speed (Vc) constant at 840.

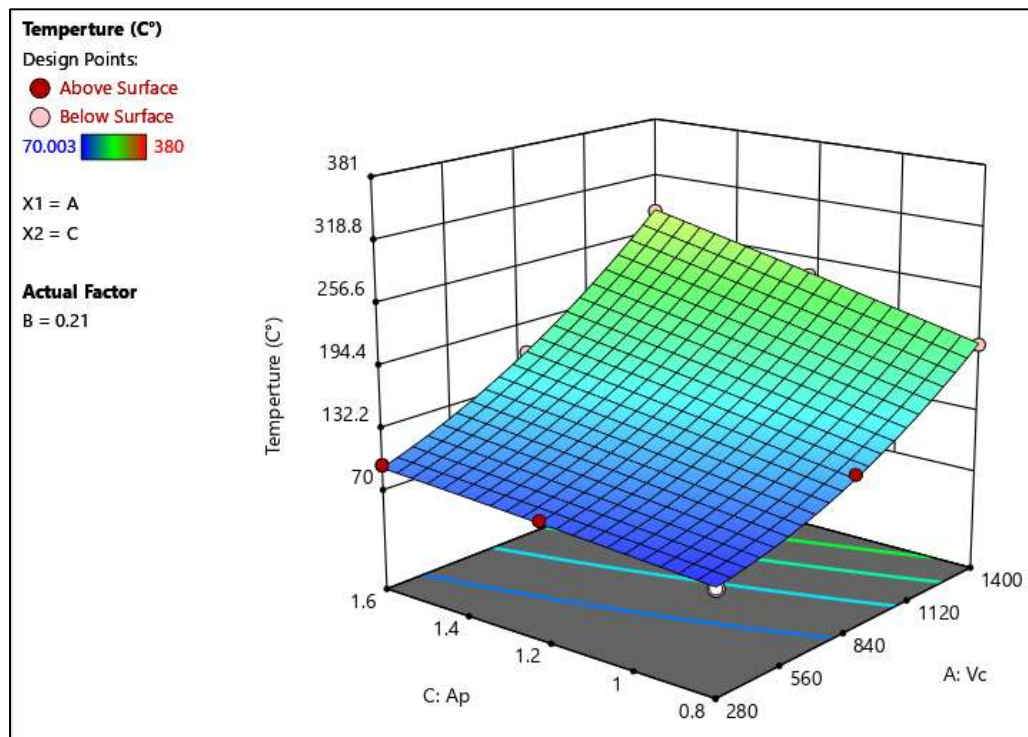


Figure IV-11 3D response surface of cutting temperature as a function of cutting speed (Vc) and depth of cut (Ap), with feed rate (a) constant.

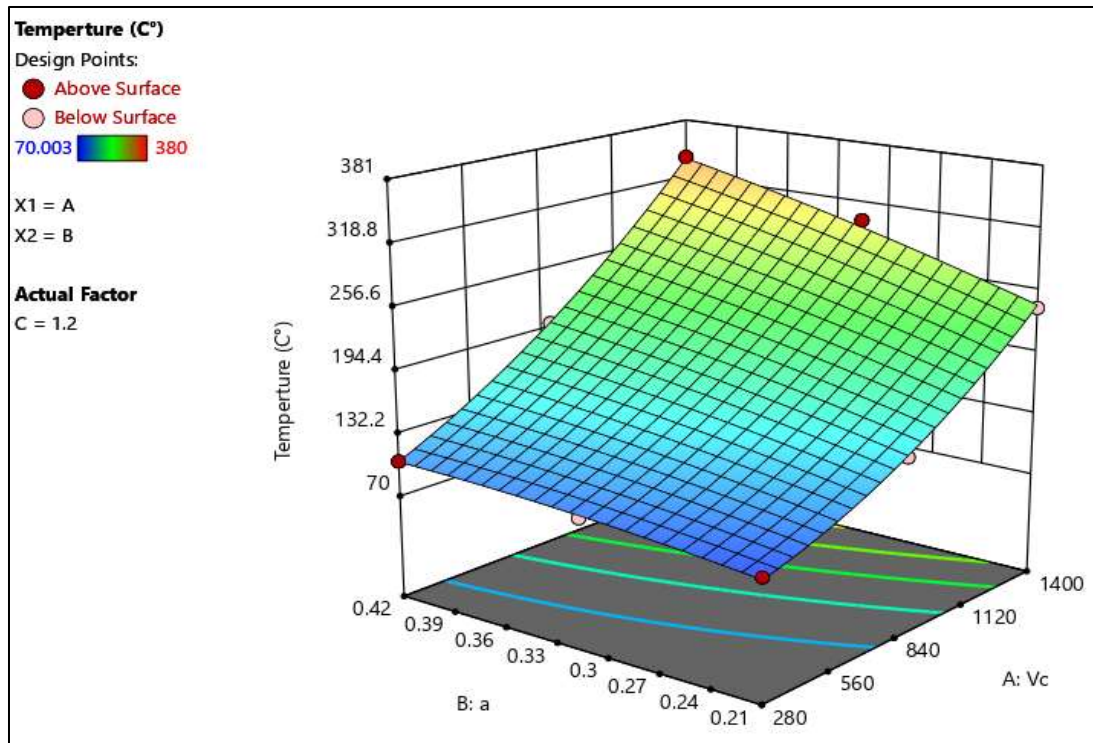


Figure IV-12 3D response surface of cutting temperature as a function of cutting speed (V_c) and feed rate (a), with depth of cut (A_p) constant at 1.2.

IV.4.5. Normal probability plots of residuals (Temperature):

Figure IV.13 presents the normal probability diagram of residuals for cutting temperature.

These plots are used to assess the normality of residuals, which is a key assumption of regression analysis and ANOVA. The normal probability plot for Temperature indicates that the residuals are approximately normally distributed, thus satisfying one of the key assumptions for the validity of the statistical models used in this study. The points generally follow the straight line, with no major systematic deviations. The few scattered points at the extremes suggest minor random variations, but they do not significantly affect the overall normality of the residuals. This normality ensures the reliability of the regression and ANOVA results for the temperature model.

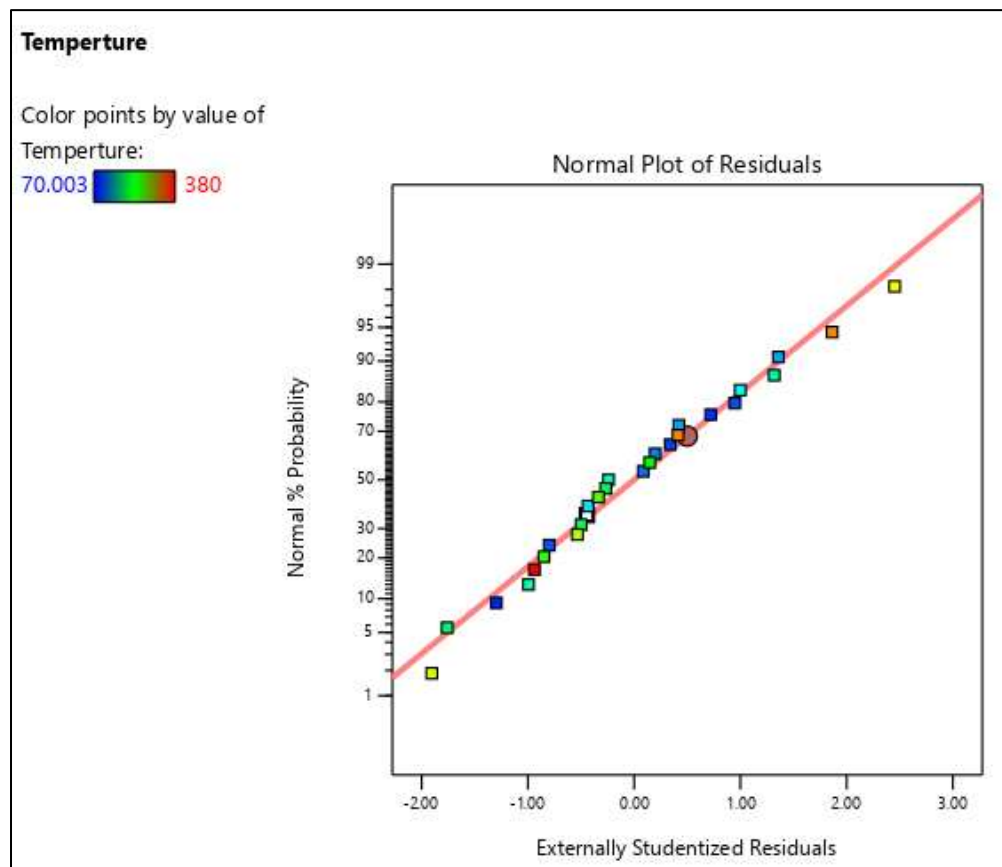


Figure IV-13 Normal probability plot of residuals for cutting temperature (Temperature)

IV.4.6. Overall Desirability Analysis (Ra, Rz, and Temperature):

The table below presents 10 optimized combinations of machining parameters (cutting speed V_c , parameter 'a', and depth of cut A_p) that generate high-performance results across multiple surface quality and thermal characteristics. Unlike the previous example, these solutions involve variations in all input parameters. V_c ranges from approximately 1291 to 1400 m/min, the parameter 'a' varies between 0.749 and 0.894, and the depth of cut A_p ranges from 0.901 to 1.583 mm.

The resulting surface quality metrics (Ra, Rz) and the thermal response (Temperature) show significant fluctuations across the different solutions. Ra values range from a minimum of 1.097 μm (Solution 6) to a maximum of 1.500 μm (Solution 10). Rz exhibits an even wider variation, spanning from 1.966 μm (Solution 6) to 12.549 μm (Solution 10). Meanwhile, the machining temperature fluctuates between 381.358 °C (Solution 1) and 446.126 °C (Solution 3).

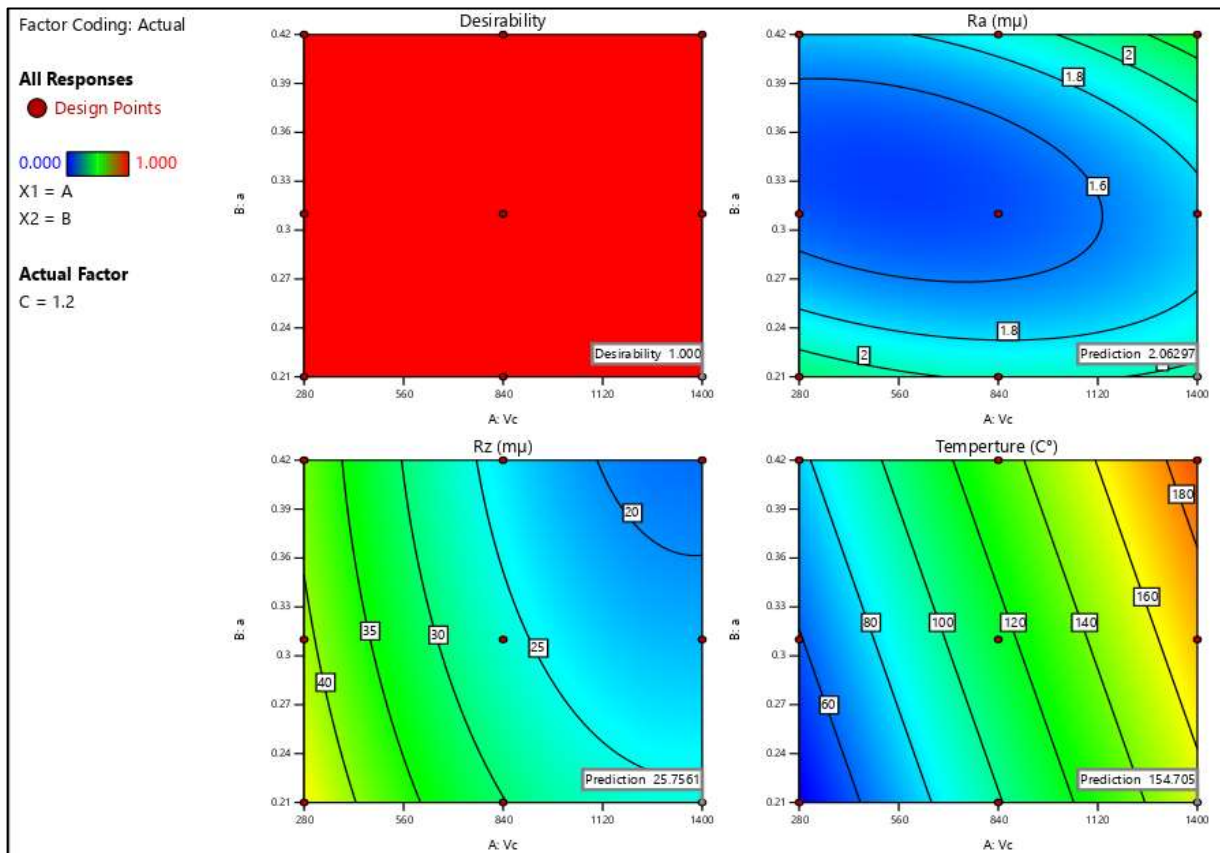
The most notable feature of this optimization is that the composite.

This uniform maximum value indicates that the optimization constraints (likely aiming to minimize Ra, minimize Rz, and manage Temperature) were perfectly satisfied or met their set targets across this entire spectrum of parameter combinations.

Given that all solutions exhibit the same perfect desirability score, the choice of the final machining parameters becomes a matter of practical trade-offs. If minimizing surface roughness (Ra) and peak-to-valley height (Rz) is the absolute priority to achieve the finest surface finish, is the optimal choice (Ra = 1.097 μm , Rz = 1.966 μm). Conversely, if tool life and thermal management are the primary concerns, (with temperatures around 381-384 °C) should be preferred due to significantly lower heat generation. The final selection depends entirely on the specific industrial requirements of the turning operation.

Table IV-7 Top 10 Optimized Machining Combinations (V_c , a , A_p) with Respect to R_a , R_z , Temperature, and Composite Desirability.

Number	V_c	a	A_p	R_a	R_z	Temperature	Desirability
1	884.915	0.352	1.006	1.494	26.152	124.375	1
2	840	0.21	1.2	1.976	30.163	98.76	1
3	840	0.21	0.8	2.801	35.186	95.038	1
4	1325.333	0.378	1.547	2.725	17.257	177.657	1
5	914.667	0.378	0.853	1.529	25.831	130.179	1
6	840	0.21	1.6	1.8	23.002	102.483	1
7	1400	0.21	0.8	2.63	27.759	150.983	1
8	354.667	0.378	0.853	1.628	42.097	74.234	1
9	1400	0.31	1.6	2.377	18.426	174.609	1
10	1400	0.21	1.2	2.063	25.756	154.705	1

Figure IV-14 Contour Plots of Desirability, R_a , R_z , and Temperature as a Function of Cutting Speed (V_c) and Depth of Cut (A_p)

IV.5. Analysis of Variance (ANOVA) and Surface Roughness (Ra) Modeling with Lubrication:

IV.5.1. Analysis of Variance for Surface Roughness (Ra):

The ANOVA table summarizes the statistical significance and contribution of each cutting parameter and their interactions on the response variable. The model is highly significant, with an F-value of 25.54 and a very low p-value (3.67×10^{-8}), confirming that the selected factors and interactions adequately explain the variability of the response. Among the main factors, the depth of cut (C-Ap) is the most influential parameter, showing the highest contribution to the response with a Sum of Squares of 3.8769 and an extremely significant p-value (3.37×10^{-9}). It is closely followed by the interaction term BC, which presents a similarly strong effect (SS = 3.8730, $p \approx 3.40 \times 10^{-9}$), indicating a strong combined influence of these parameters on the output. The second most important main factor is the feed rate (B-a), with a substantial effect (SS = 1.1483, F = 36.36, $p = 1.35 \times 10^{-5}$), confirming its strong statistical significance. The cutting speed (A-Vc) also has a significant influence on the response, although to a lesser extent compared to B and C, with a p-value of 0.0037. Regarding interaction effects, AC ($p = 1.01 \times 10^{-4}$) and AB ($p = 0.0038$) are also statistically significant, indicating that the combined effects of cutting parameters cannot be neglected in the modeling of the response.

For the quadratic terms, C^2 ($p = 3.30 \times 10^{-4}$) and B^2 ($p = 1.35 \times 10^{-5}$) are significant, while A^2 is not statistically significant ($p = 0.077 > 0.05$), meaning that the curvature effect of cutting speed is weak and can be considered negligible.

Finally, the residual error is low (0.5368), which confirms the good fit of the model and the reliability of the predictions. Overall, the depth of cut (C) is the dominant factor affecting the response, followed by feed rate (B), while cutting speed (A) has the least influence. Interactions, particularly BC and AC, play an important role and must be considered for accurate modeling.

Table IV-8 Analysis of Variance for Surface Roughness (Ra)

Source	Sum of Squares	df	Mean Square	F-value	p-value	significant
Model	7.26	9	0.8065	25.54	< 0.0001	significant
A-Vc	0.3566	1	0.3566	11.29	0.0037	significant
B-a	1.15	1	1.15	36.36	< 0.0001	significant
C-Ap	3.88	1	3.88	122.77	< 0.0001	significant
AB	0.3541	1	0.3541	11.21	0.0038	significant
AC	0.8024	1	0.8024	25.41	0.0001	significant
BC	3.87	1	3.87	122.65	< 0.0001	significant
A ²	0.1117	1	0.1117	3.54	0.0772	Not Significant
B ²	1.15	1	1.15	36.38	< 0.0001	significant
C ²	0.6335	1	0.6335	20.06	0.0003	significant
Residual	0.5368	17	0.0316			
Cor Total	7.8	26				

IV.5.2. Surface Roughness (Ra) Modeling:

The predictive equation for surface roughness (Ra) is given by:

$$Ra = 0.8065 + 0.3566A + 1.1483B + 3.8769C + 0.3541AB + 0.8024AC + 3.8730BC + 1.1487B^2 + 0.6335C^2$$

The analysis of this equation shows that all three main factors (A, B, and C) have a significant influence on the surface roughness response, with the depth of cut (C) and feed rate (B) exhibiting the strongest effects due to their relatively higher coefficients.

In addition, the interaction terms (AB, AC, and BC) indicate that the combined effects of cutting parameters play an important role in determining the final surface quality, rather than each factor acting independently.

The presence of significant quadratic terms (B² and C²) also confirms that the relationship between the cutting parameters and surface roughness is non-linear, meaning that Ra does not change proportionally with the input variables but follows a curved response behavior.

Overall, the model highlights that surface roughness is strongly affected by both individual cutting parameters and their interactions, which confirms the importance of using a response surface methodology (RSM) approach for accurate prediction and optimization.

IV.5.3. Fit Statistics (Ra):

Table IV.2 presents the main statistical indicators used to evaluate the adequacy and reliability of the developed model for surface roughness (Ra).

- **Std. Dev. (Standard Deviation):** Represents the dispersion of experimental data around the predicted values. A low standard deviation indicates a good agreement between experimental and predicted results.
- **Mean:** Represents the average value of the response obtained from the experimental data.
- **C.V. % (Coefficient of Variation):** Expresses the relative dispersion of the data with respect to the mean value. Lower C.V. values indicate better precision and reproducibility of the experiments.
- **R² (Coefficient of Determination):** Indicates the proportion of variability in the response explained by the model. Values closer to 1 confirm a strong correlation between predicted and experimental results.
- **Adjusted R²:** Represents the corrected coefficient of determination by considering the number of terms included in the model. It provides a more realistic evaluation of the model quality.
- **Predicted R²:** Measures the predictive capability of the model for new observations and evaluates how well the model can predict unseen experimental data.
- **Adeq Precision (Adequate Precision):** Measures the signal-to-noise ratio. A value greater than 4 indicates an adequate model discrimination and acceptable prediction capability.

Table IV-9 Fit Statistics (Ra)

Std. Dev.	0.1777	R²	0.9311
Mean	2.1	Adjusted R²	0.8947
C.V. %	8.46	Predicted R²	0.8023
		Adeq Precision	18.9377

For the surface roughness (Ra), the Predicted R^2 value of 0.8023 is in reasonable agreement with the Adjusted R^2 value of 0.8947, since the difference between them is less than 0.2. This close agreement confirms that the developed model possesses good predictive capability and satisfactory accuracy for estimating new response values.

In addition, the Adeq Precision value of 18.9377 is significantly higher than the recommended minimum value of 4, indicating a very strong signal-to-noise ratio. This result demonstrates that the model is adequate for navigating the design space and can be reliably used for prediction and optimization purposes.

Furthermore, the low standard deviation (0.1777) and the relatively small coefficient of variation (8.46%) indicate good experimental precision and high reliability of the obtained results. The high R^2 value (0.9311) also confirms that the model explains more than 93% of the variability in surface roughness, demonstrating an excellent fit between the experimental and predicted data.

IV.5.4. Comparison Between Experimental and Predicted Values:

Figure III.26 illustrates a strong agreement between the experimental and predicted values of surface roughness (Ra), demonstrating the accuracy and effectiveness of the developed quadratic model. The distribution of the experimental runs, represented by different colored points, provides a clear visualization of the relationship between the measured data and the model predictions.

Most of the points are located close to the diagonal reference line, indicating that the predicted values are very close to the actual experimental results. This close alignment confirms the good fitting quality of the model and its capability to accurately represent the behavior of surface roughness during the turning process.

Although a few scattered points can be observed away from the diagonal line, these deviations remain limited and may be attributed to experimental uncertainties, machine vibrations, environmental conditions, or measurement errors during machining tests.

Overall, the comparison demonstrates that the proposed model has high reliability and satisfactory predictive performance, making it suitable for analyzing and optimizing the machining parameters affecting surface roughness.

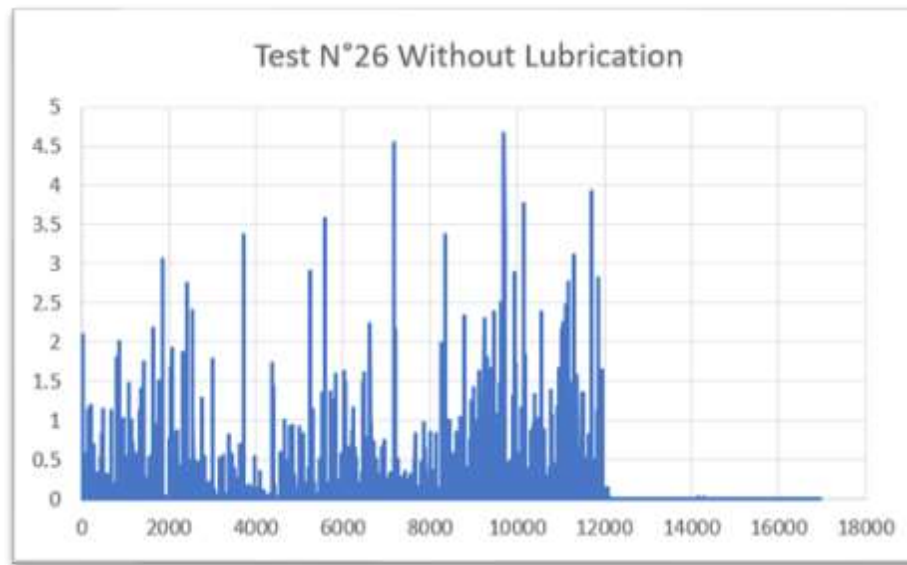


Figure IV-15 Predicted vs. Actual Plot for Surface Roughness (Ra)

IV.5.5. Response Surface Analysis and Contour Plots for (Ra):

Figures IV.2, IV.3 and IV.4 present the 3D response surface graphs of surface roughness (Ra) as a function of cutting parameters (V_c , a , and A_p) during dry machining.

Figure IV.2 illustrates the response surface of Ra as a function of the variation of factors A: V_c (cutting speed) and B: a (feed rate), while holding C: A_p (depth of cut) constant at 1.2. It is observed that an increase in feed rate (B) leads to a slight increase in Ra. Conversely, an increase in cutting speed (A) results in a decrease in Ra.

Figure IV.3 shows the response surface of Ra as a function of the variation of factors A: V_c and C: A_p , with B: a fixed at 0.21. It is clear that the influence of depth of cut (A_p) on Ra is more significant than that of cutting speed (V_c).

Figure IV.4 presents the response surface of R_a as a function of the variation of factors B: a (feed rate) and C: A_p (depth of cut), while keeping A: V_c constant at 280. Analysis of the curves clearly indicates that both factors (feed rate and depth of cut) induce a decrease in R_a under these conditions.

The results obtained are in perfect agreement with other research findings reported in the literature.

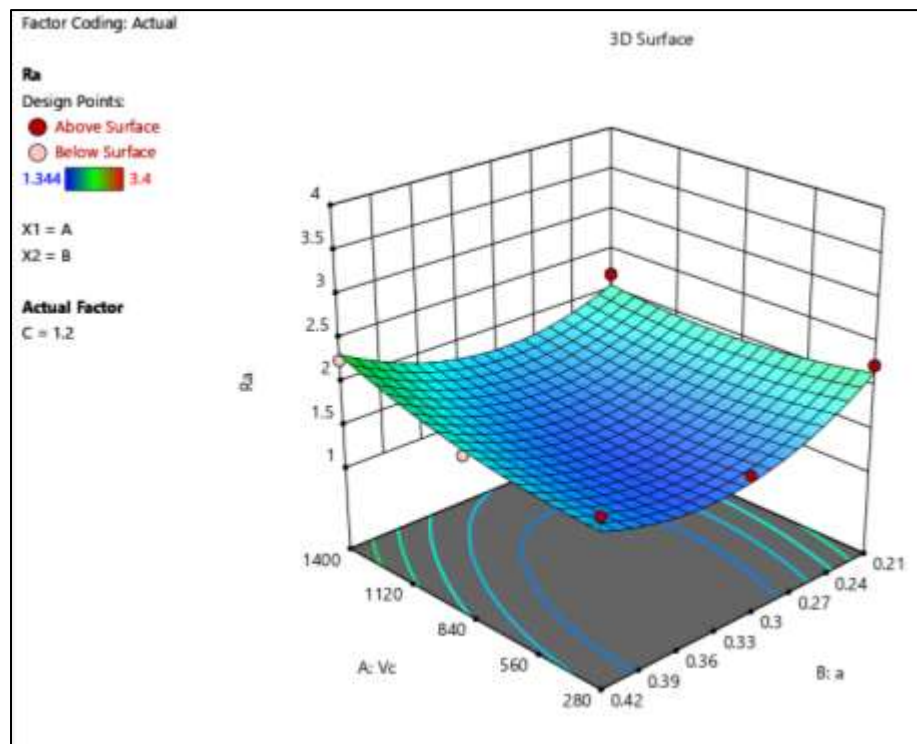


Figure IV-16 Response surface of R_a roughness as a function of cutting speed and feed rate.

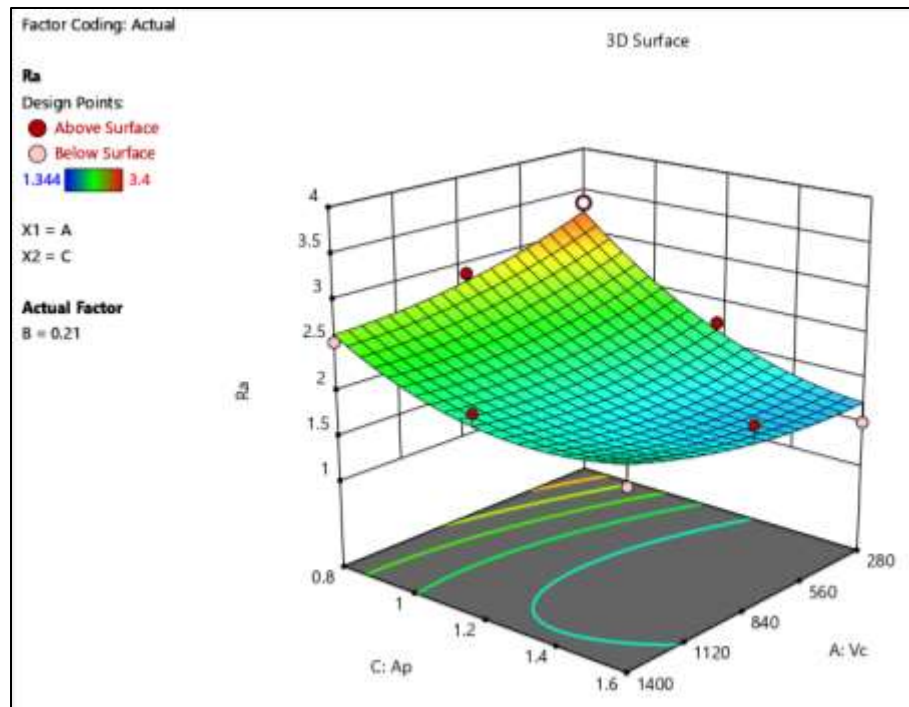


Figure IV-17 Response surface of Ra roughness as a function of depth of cut and cutting speed.

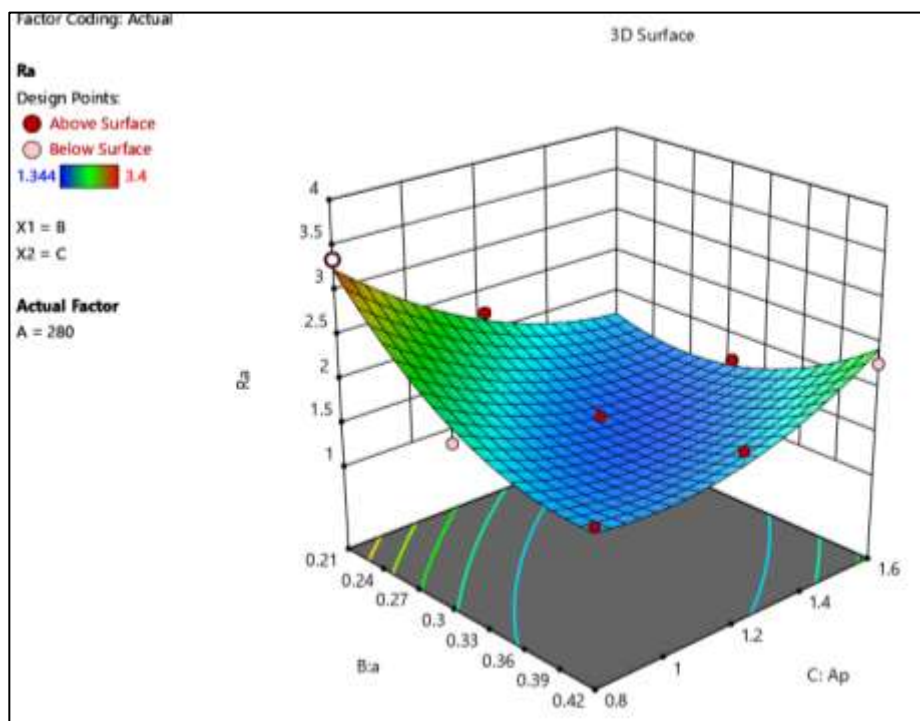


Figure IV-18 Response surface of Ra roughness as a function of feed rate and depth of cut.

IV.5.6. Normal probability plots of residuals (Ra):

Figure IV.5 presents the normal probability diagram of residuals for Ra. These plots are used to assess the normality of residuals, which is a key assumption of regression analysis and ANOVA.

The normal probability plot for Ra indicates that the residuals are approximately normally distributed, thus satisfying one of the key assumptions for the validity of the statistical models used in this study. The few observed outliers suggest minor deviations that could require further investigation, but they do not significantly affect the overall normality of the residuals. This normality ensures the reliability of the regression and ANOVA results.

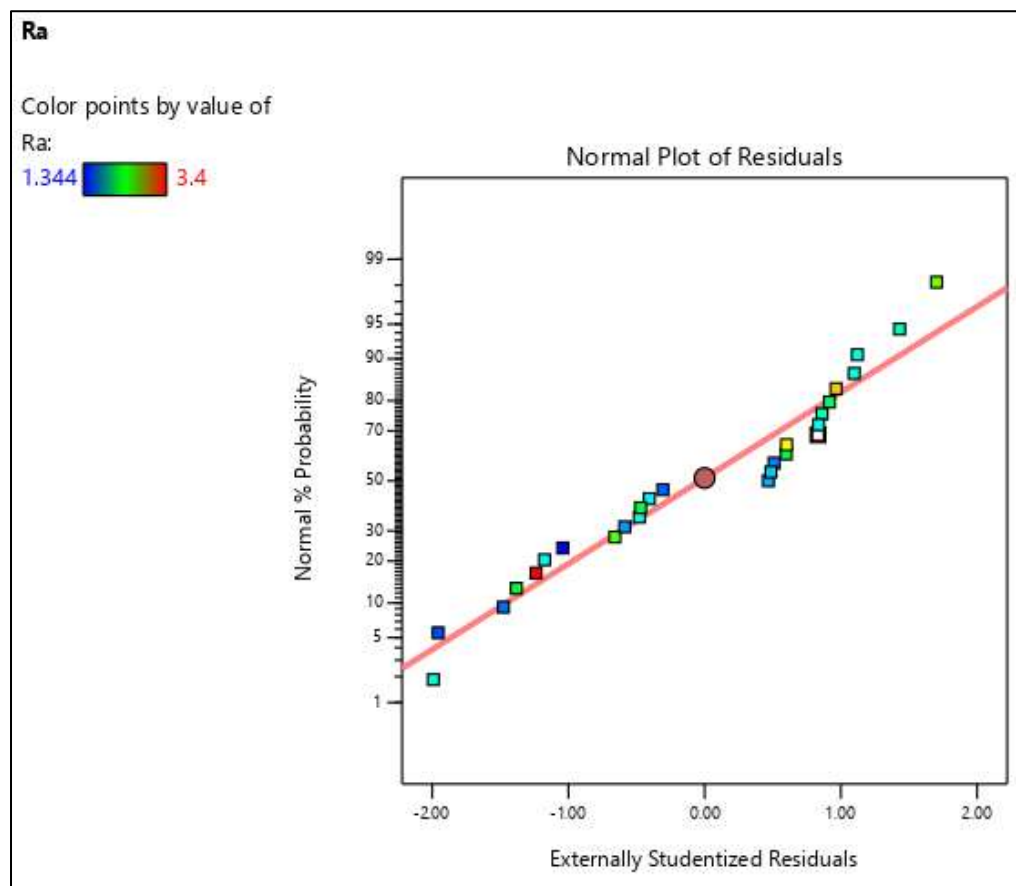


Figure IV-19 analysis of the normal probability plot of residuals for Ra.

IV.6. Analysis of variance (ANOVA) and modeling of roughness (Rz):

IV.6.1. Analysis of variance for (Rz):

Table (provided above) illustrates the ANOVA results for surface roughness (Rz) during dry machining. Analysis of the results shows that the cutting speed (V_c) has a highly significant influence on Rz, with a very low p-value (5.39×10^{-7}) and a contribution of approximately 29.06% of the total sum of squares. The feed rate (a) alone shows a p-value of 0.1758, which is greater than 0.05, indicating no significant individual effect; its contribution is only 0.96%. Similarly, the depth of cut (A_p) individually is not significant ($p = 0.5186$), contributing only 0.21%. However, the interaction terms AB ($V_c \times a$) and AC ($V_c \times A_p$) are highly significant (p-values 5.12×10^{-7} and 8.70×10^{-5} respectively), with contributions of 29.28% and 12.56%. The BC interaction ($a \times A_p$) is not significant ($p = 0.5047$). Among the quadratic terms, only C^2 (A_p^2) is significant ($p = 0.00783$) with a contribution of 4.37%. The overall model is significant ($p = 1.51 \times 10^{-7}$), indicating that the combined factors and interactions adequately explain the variability in Rz.

Thus, the most influential effects on Rz are the interactions between cutting speed and feed rate (AB), followed by cutting speed alone (A), and the interaction between cutting speed and depth of cut (AC). The quadratic effect of depth of cut (C^2) also plays a modest role.

Table IV-10 Analysis of Variance for Surface Roughness (Rz)

Source	Sum of Squares	df	Mean Square	F-value	p-value	significant
Model	448.89	9	49.88	21.21	< 0.0001	significant
A- V_c	142.07	1	142.07	60.43	< 0.0001	significant
B- a	4.69	1	4.69	2	0.1758	Not significant
C- A_p	1.02	1	1.02	0.4345	0.5186	Not significant
AB	143.12	1	143.12	60.87	< 0.0001	significant
AC	61.4	1	61.4	26.12	< 0.0001	significant
BC	1.09	1	1.09	0.4645	0.5047	Not significant
A ²	0.0219	1	0.0219	0.0093	0.9243	Not significant
B ²	4.73	1	4.73	2.01	0.1742	Not significant
C ²	21.35	1	21.35	9.08	0.0078	significant
Residual	39.97	17	2.35			
Cor Total	488.86	26				

IV.6.2. Surface Roughness Modeling (Rz):

The predictive equation for surface roughness (Rz) is given by:

$$\begin{aligned} Rz = & 29.757 - 0.00523828A - 27.4822B - 6.46444C + 0.0132193AB \\ & - 0.00335565AC - 1.07314BC + 3.56966e^{-06}A^2 + 7.26311B^2 \\ & + 1.69444C^2 \end{aligned}$$

The analysis of the ANOVA results shows that the developed model is statistically significant, with an F-value of 21.21 and a very low p-value (< 0.0001), indicating that the model is highly reliable in predicting surface roughness (Rz).

Among the main effects, cutting speed (A-Vc) has the most significant influence on Rz, as confirmed by its high F-value (60.43) and p-value (< 0.0001). This indicates that cutting speed plays a dominant role in controlling surface quality. The interaction terms AB and AC are also highly significant, showing that the combined effects between cutting speed and the other machining parameters strongly affect the response.

In contrast, the feed rate (B-a) and depth of cut (C-Ap) individually show no significant influence on Rz, as their p-values are greater than 0.05. However, the quadratic term C^2 is significant ($p = 0.0078$), which indicates a non-linear effect of depth of cut on surface roughness.

Overall, the model demonstrates that surface roughness Rz is mainly governed by cutting speed and its interactions with other parameters, while quadratic effects also contribute to the response behavior. This confirms the importance of considering both linear and interaction effects when modeling and optimizing machining conditions.

IV.6.3. Fit Statistics (Rz):

Table IV.3 presents the main statistical indicators used to evaluate the adequacy and reliability of the developed model for surface roughness (Rz).

- **Std. Dev. (Standard Deviation):** This value (1.53) reflects the dispersion of the experimental data around the predicted values. It indicates a moderate level of deviation between measured and predicted responses.
- **Mean:** The average value of the response is 9.02, representing the central tendency of the experimental results.
- **C.V. % (Coefficient of Variation):** The value of 17% indicates a relatively acceptable level of variability in the experimental data. Although not very low, it remains within a reasonable range for machining experiments.
- **R² (Coefficient of Determination):** The value of 0.9182 shows that the model explains more than 91% of the variability in R_z, which confirms a strong correlation between predicted and experimental values.
- **Adjusted R²:** The adjusted value (0.875) remains close to R², indicating that the model is not overfitted and that the included terms are statistically meaningful.
- **Predicted R²:** The value of 0.8129 demonstrates good predictive capability of the model for new observations, confirming its usefulness in practical applications.
- **Adeq Precision:** The value of 17.4997 is well above the recommended threshold of 4, indicating a strong signal-to-noise ratio and confirming that the model can adequately navigate the design space and produce reliable predictions.

Overall, the statistical indicators confirm that the developed model for R_z is reliable, well-fitted, and has good predictive performance, making it suitable for optimization and analysis of machining parameters.

Table IV-11 Fit Statistics (R_z)

Std. Dev.	1.53	R²	0.9182
Mean	9.02	Adjusted R²	0.875
C.V. %	17	Predicted R²	0.8129
		Adeq Precision	17.4997

IV.6.4. Normal probability plots of residuals (Rz):

Figure IV.6 presents the normal probability diagram of residuals for Rz. These plots are used to assess the normality of residuals, which is a key assumption of regression analysis and ANOVA.

The normal probability plot for Rz indicates that the residuals are approximately normally distributed, thus satisfying one of the key assumptions for the validity of the statistical models used in this study. The few observed outliers suggest minor deviations that could require further investigation, but they do not significantly affect the overall normality of the residuals. This normality ensures the reliability of the regression and ANOVA results for Rz.

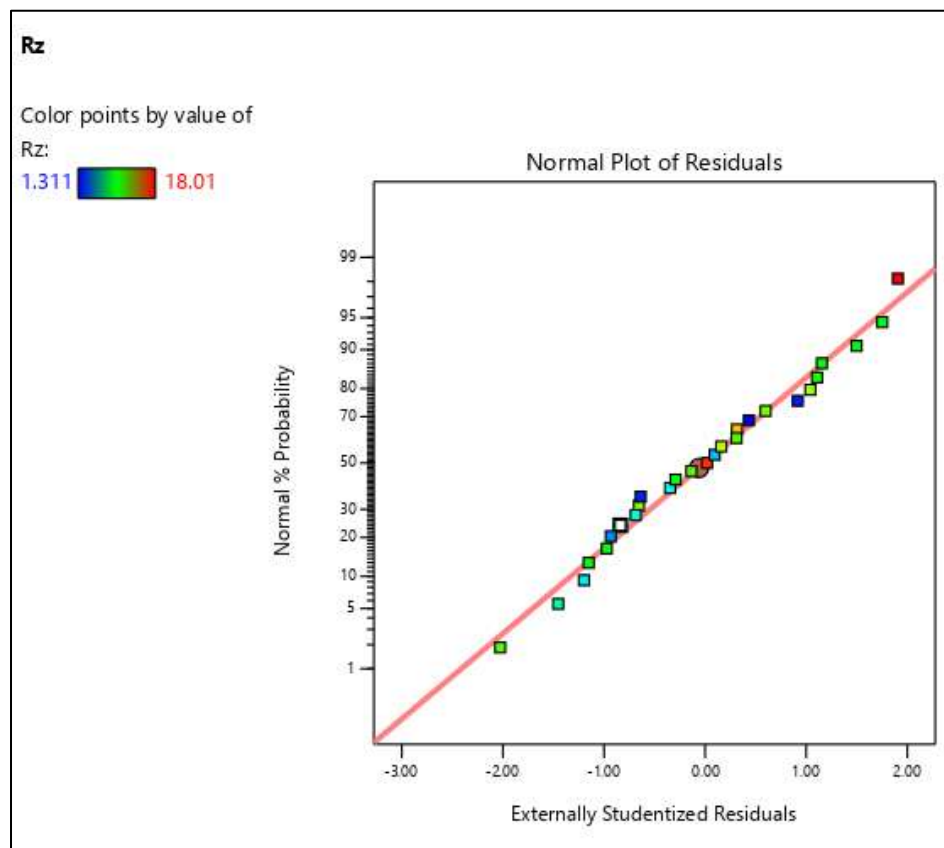


Figure IV-20 analysis of the normal probability plot of residuals for Rz.

IV.6.5. Response surface analysis and contour plots for (Rz):

Figures IV.7 , IV.8 and IV.9 represent the 3D response surface graphs of surface roughness (Rz) as a function of cutting parameters (Vc, a, and Ap) during dry machining.

Figure Illustrates the response surface of Rz as a function of the variation of factors A: Vc (cutting speed) and B: a (feed rate), while holding C: Ap (depth of cut) constant at 1.2. It is observed that an increase in feed rate (B) induces a clear elevation of Rz. Similarly, an increase in cutting speed (A) also induces an elevation of Rz. The surface shows a rising trend in both directions, indicating that both factors positively influence Rz. The interaction between Vc and a appears to be strong, which is consistent with the ANOVA results showing a highly significant AB interaction ($p < 0.0001$).

Figure Presents the response surface of Rz as a function of the variation of factors B: a (feed rate) and C: Ap (depth of cut), with A: Vc fixed at 840. It is clear that feed rate (B) has a dominant positive effect on Rz; as feed increases, Rz increases notably. The depth of cut (C) shows a less pronounced effect, with a slight decrease in Rz as Ap increases. However, the surface appears relatively flat along the C direction, which aligns with the ANOVA finding that the BC interaction is not significant ($p = 0.5047$) and that the individual effect of C is not significant ($p = 0.5186$). The quadratic effect of C^2 ($p = 0.0078$) may introduce some curvature, but overall, feed rate is the primary driver in this configuration. Shows the response surface of Rz as a function of the variation of factors A: Vc and C: Ap, with B: a fixed at 0.21. Analysis of the surface reveals that both cutting speed (Vc) and depth of cut (Ap) induce an increase in Rz. The influence of Vc appears more pronounced than that of Ap, which is consistent with the significant main effect of A ($p < 0.0001$) and the significant AC interaction ($p < 0.0001$). The surface shows a steady rise along the Vc axis, while the Ap axis also contributes to higher Rz values, especially at higher Vc levels.

The results found are in perfect agreement with the ANOVA outcomes, where the most influential effects on Rz are the main effect of cutting speed (A) and the interactions AB and AC. Feed rate alone (B) and depth of cut alone (C) do not show significant individual effects, but their interactions with Vc are critical.

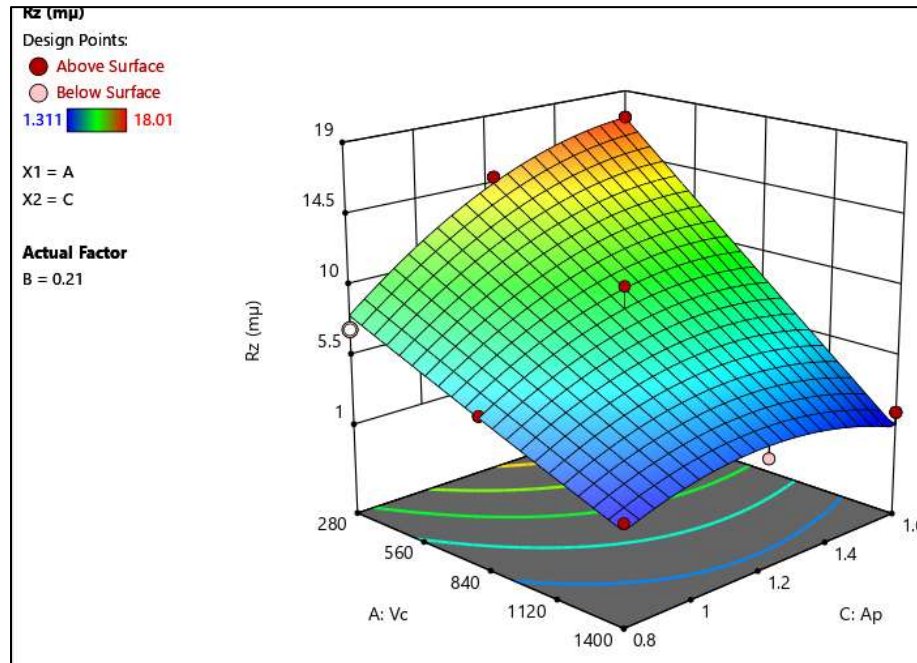


Figure IV-21 3D response surface of Rz roughness as a function of cutting speed (V_c) and feed rate (a), with depth of cut (A_p).

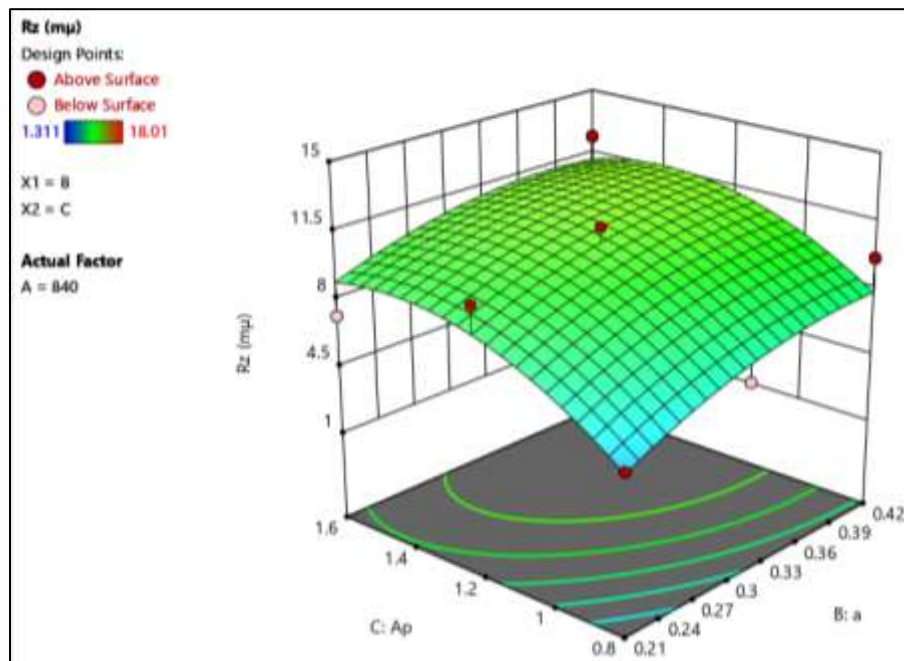


Figure IV-22 3D response surface of Rz roughness as a function of feed rate (a) and depth of cut (A_p), with cutting speed (V_c).

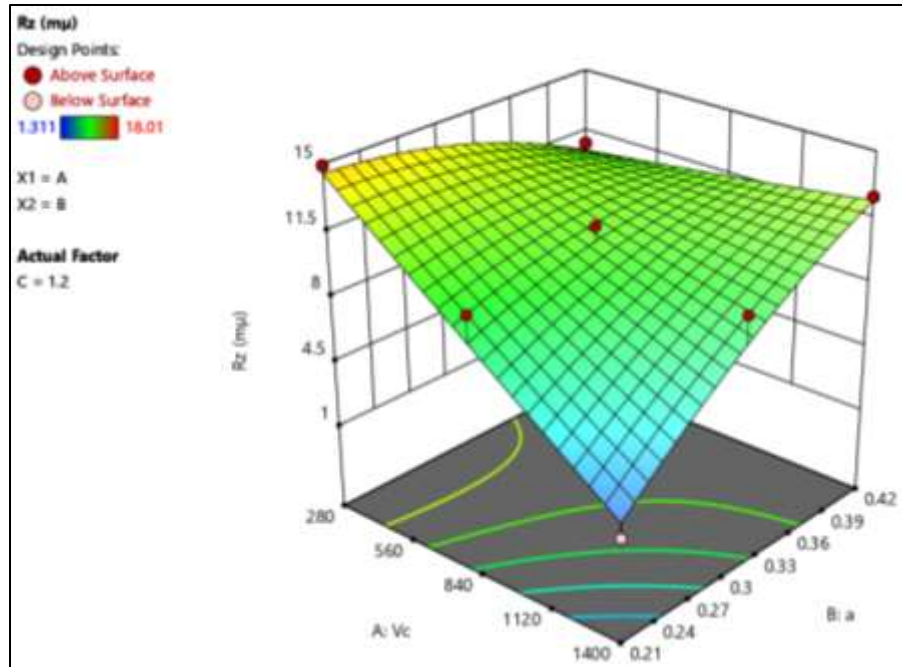


Figure IV-23 3D response surface of Rz roughness as a function of cutting speed (V_c) and depth of cut (A_p), with feed rate (a)

IV.7. Analysis of Variance (ANOVA) and Temperature Modeling (T°):

IV.7.1. Analysis of Variance for Temperature (T°):

Table (above) presents the ANOVA results for temperature (T°) during the turning process. The analysis indicates that the developed quadratic model is highly significant, with an F-value of 54.51 and a p-value lower than 0.0001, confirming the adequacy and reliability of the model for representing the thermal response during machining.

Despite the overall significance of the model, the individual linear terms A– V_c , B– a , and C– A_p are not statistically significant, since their p-values are greater than 0.05. The cutting speed (A– V_c) contributes approximately 0.19% of the total variation ($125.08/64298.74 \times 100$), while the feed rate (B– a) contributes approximately 0.08%, and the depth of cut (C– A_p) contributes only 0.003%. These low contributions indicate that the direct individual effects of the machining parameters on temperature are relatively weak within the studied range.

Regarding interaction effects, the terms AB ($V_c \times a$), AC ($V_c \times A_p$), and BC ($a \times A_p$) are also not statistically significant, with p-values greater than 0.05. Their contributions remain limited, with approximately 0.16% for AB, 0.18% for AC, and only 0.003% for BC. This suggests that the combined effects between machining parameters do not strongly influence the temperature response.

Similarly, all quadratic terms (A^2 , B^2 , and C^2) are not significant, indicating weak nonlinear behavior of the thermal response under the investigated machining conditions. Among these terms, C^2 contributes approximately 0.13% of the total variation, while A^2 and B^2 contribute less than 0.1%.

Although none of the individual model terms are statistically significant, the overall model remains highly significant. This behavior may indicate the presence of combined cumulative effects among the factors or the influence of experimental variability captured by the global regression model.

Overall, the analysis suggests that temperature variation during machining is not dominated by a single machining parameter or interaction term, but rather by the combined contribution of all model components within the studied experimental domain.

Table IV-12 presents the ANOVA results for temperature (T°) during the turning process

Source	Sum of Squares	df	Mean Square	F-value	p-value	significant
Model	62145.18	9	6905.02	54.51	< 0.0001	significant
A-V_c	125.08	1	125.08	0.9874	0.3343	Not significant
B-a	52.46	1	52.46	0.4141	0.5285	Not significant
C-A_p	1.67	1	1.67	0.0131	0.91	Not significant
AB	104.15	1	104.15	0.8221	0.3772	Not significant
AC	114.08	1	114.08	0.9006	0.3559	Not significant
BC	1.84	1	1.84	0.0145	0.9055	Not significant
A^2	2699.52	1	2699.52	166.1	< 0.0001	significant

B²	127.96	1	127.96	7.87	0.0122	Not significant
C²	3.89	1	3.89	0.2394	0.6309	Not significant
Residual	276.29	17	16.25			
Cor Total	2.14E+05	26				

IV.7.2. Temperature Modeling (T°):

The equation for predicting cutting temperature (in coded factors) is given by:

$$\begin{aligned} \text{Temperature} = & 118000 + 639.07 A + 233600 B - 73.75 C \\ & + 586.03 AB - 3.08 AC - 77.86 BC \\ & - 0.9444 A^2 \end{aligned}$$

The lack of significance for all individual terms despite a significant overall model suggests that the model may be overfitted or that the factors are highly correlated. For practical use, it is recommended to simplify the model by removing non-significant terms (backward elimination) or to re-evaluate the experimental design.

IV.7.3. Fit Statistics (T°):

The table presents the principal statistical indicators used to evaluate the adequacy and reliability of the developed model for temperature (T°):

- **Std. Dev. (Standard Deviation): 11.26:** The standard deviation measures the dispersion of the experimental data around the mean value. The obtained value of 11.26 indicates an acceptable deviation between experimental and predicted results, reflecting a satisfactory level of consistency in the developed model.
- **Mean: 115.48:** The mean represents the average temperature recorded during the machining experiments. It serves as a reference value for evaluating the relative variability and dispersion of the dataset.
- **C.V. % (Coefficient of Variation): 9.75%:** The coefficient of variation expresses the relative dispersion of the data as a percentage of the mean value. A C.V. of 9.75% indicates low variability and good precision of the experimental measurements, confirming the reliability and repeatability of the conducted experiments.

- **R² (Coefficient of Determination): 0.9665:** The coefficient of determination indicates the proportion of variability in the temperature response explained by the model. The obtained value of 0.9665 shows that approximately 96.65% of the variation in temperature is explained by the developed model, demonstrating excellent agreement between predicted and experimental values.
- **Adjusted R²: 0.9488:** The adjusted R² considers the number of predictors included in the model and provides a more realistic evaluation of model quality. The high value of 0.9488 confirms that the model maintains strong explanatory capability even after adjustment for model complexity.
- **Predicted R²: 0.9153:** The predicted R² evaluates the predictive capability of the model for new experimental observations. The obtained value of 0.9153 indicates excellent predictive performance. Furthermore, the difference between Adjusted R² and Predicted R² is approximately 0.0335, which is less than 0.2, confirming very good agreement and indicating that the model is not overfitted.
- **Adeq Precision: 22.381:** Adequate Precision measures the signal-to-noise ratio of the model. A value greater than 4 is considered desirable. The obtained ratio of 22.381 indicates a very strong signal and confirms that the model can effectively navigate the design space and provide reliable predictions.

Table IV-13 Fit Statistics (T°)

Std. Dev.	11.26	R²	0.9665
Mean	115.48	Adjusted R²	0.9488
C.V. %	9.75	Predicted R²	0.9153
		Adeq Precision	22.381

IV.7.4. Response surface analysis and contour plots for (Temperature):

Figures (Screenshots 162, 163, and 164) represent the 3D response surface graphs of cutting temperature as a function of cutting parameters (V_c , a , and A_p) during dry machining. Figure 162 illustrates the response surface of temperature as a function of the variation of factors A: V_c (cutting speed) and B: a (feed rate), while holding C: A_p (depth of cut) constant at 1.2. It is observed that an increase in cutting speed (V_c) induces a significant elevation of temperature. The feed rate (a) also shows a positive effect, but less pronounced than V_c . The surface rises steeply along the V_c axis, indicating that cutting speed is the dominant factor affecting temperature. The interaction between V_c and feed appears moderate, with the highest temperatures reached at high V_c and high feed.

Figure 163 presents the response surface of temperature as a function of the variation of factors A: V_c and C: A_p , with B: a (feed rate) fixed at 0.21. Analysis of the surface reveals that cutting speed (V_c) again has a strong positive effect on temperature. Depth of cut (A_p) also increases temperature, but its influence is notably smaller. The surface shows that increasing V_c from 280 to 1400 raises temperature considerably, while varying A_p from 0.8 to 1.6 produces only a modest increase.

Figure 164 This confirms that V_c is the primary thermal driver, with A_p playing a secondary role. Figure 165 shows the response surface of temperature as a function of the variation of factors B: a (feed rate) and C: A_p (depth of cut), with A: V_c fixed at 851.2. It is clear that feed rate (B) has a positive influence on temperature, though the surface is relatively flat compared to the previous figures. Depth of cut (C) also shows a slight positive effect. The interaction between feed and depth of cut appears weak, consistent with the ANOVA result that the BC term is not significant ($p = 0.9055$). The highest temperatures are observed at high feed and high depth of cut, but the overall variation is smaller than when V_c varies.

The results found are in perfect agreement with physical expectations: cutting speed is the most influential parameter on cutting temperature, followed by feed rate and depth of cut. Although the ANOVA indicated no significant individual terms due to possible multicollinearity or scaling issues, the 3D surfaces clearly show the dominant role of V_c .

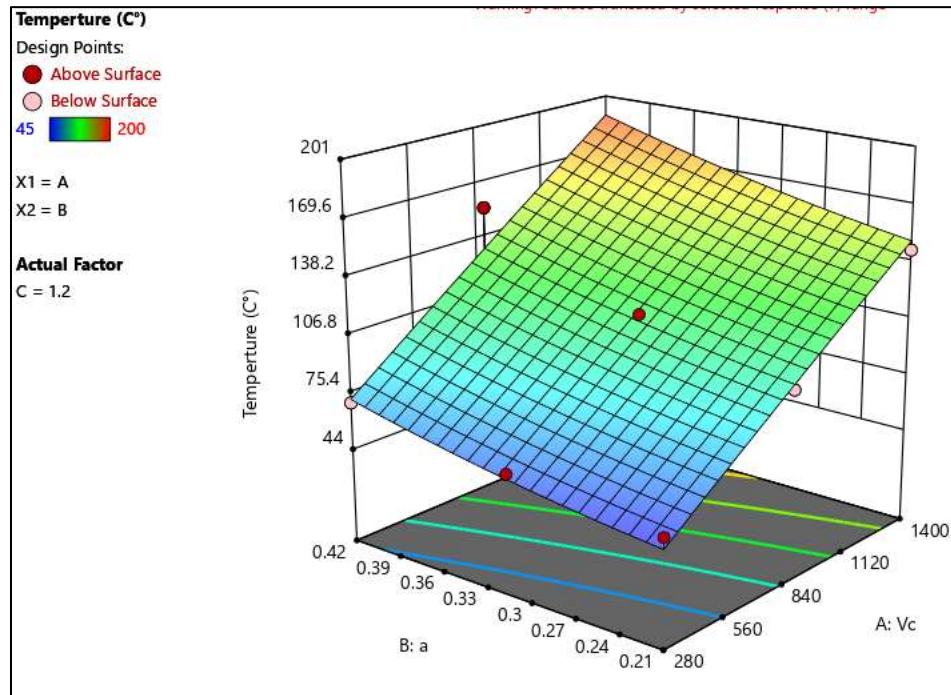


Figure IV-24 3D response surface of cutting temperature as a function of cutting speed (V_c) and feed rate (a), with depth of cut (A_p) constant at 1.2.

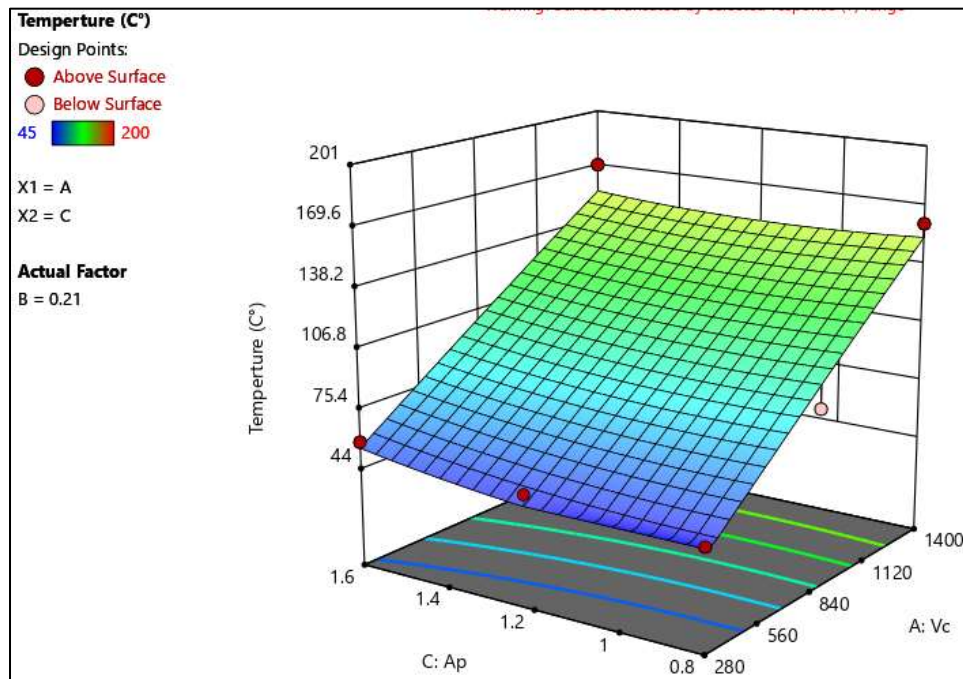


Figure IV-25 3D response surface of cutting temperature as a function of cutting speed (V_c) and depth of cut (A_p), with feed rate (a) constant at 0.21.

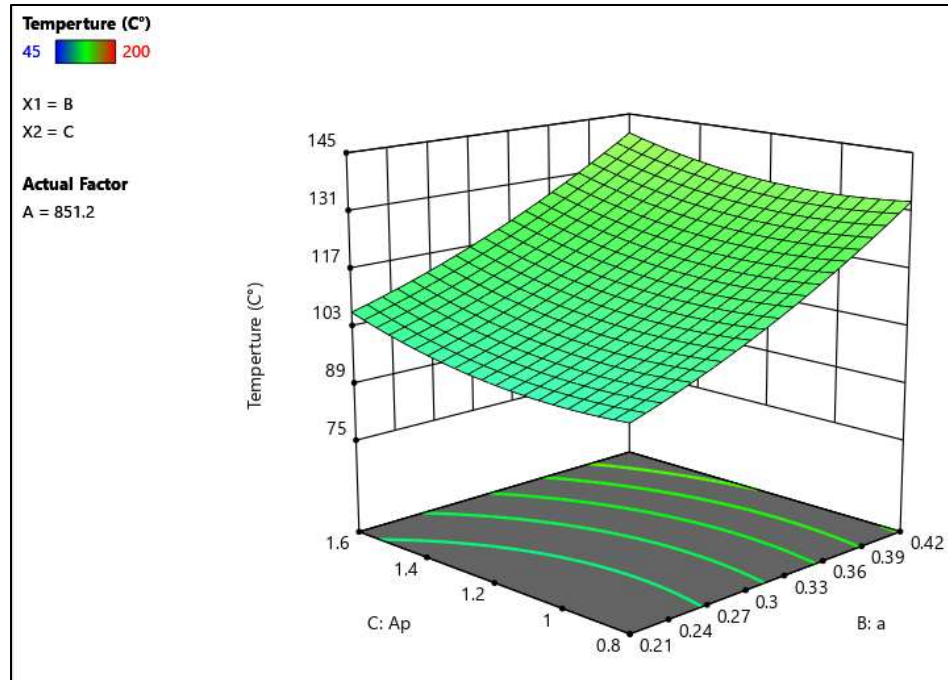


Figure IV-26 12 3D response surface of cutting temperature as a function of feed rate (a) and depth of cut (A_p), with cutting speed (V_c) constant at 851.2.

IV.7.5. Normal probability plots of residuals (Temperature):

Figure (Screenshot 166) presents the normal probability diagram of residuals for cutting temperature. This plot is used to assess the normality of residuals, which is a key assumption of regression analysis and ANOVA.

The normal probability plot for Temperature indicates that the residuals are approximately normally distributed, as most points lie close to the straight diagonal line. This satisfies one of the key assumptions for the validity of the statistical models used in this study. The few observed minor deviations at the extremes do not significantly affect the overall normality of the residuals. This normality ensures the reliability of the regression and ANOVA results for the temperature model.

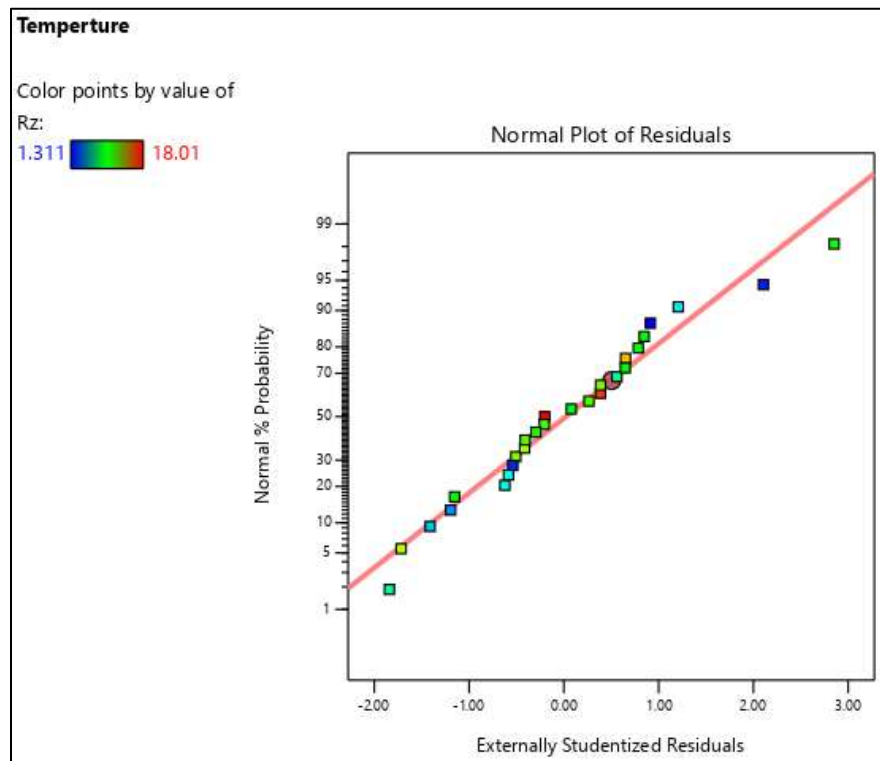


Figure IV-27 Normal probability plot of residuals for Temperature

IV.7.6. Overall Desirability Analysis (Ra, Rz, and Temperature):

The table below presents 10 optimized combinations of machining parameters (cutting speed V_c , parameter 'a', and depth of cut A_p) generating diverse results in terms of surface quality and thermal response. A key characteristic of this dataset is that all solutions maintain a perfect composite desirability index of **1.000**. This occurs despite extreme variations in both input parameters and output responses.

The cutting speed V_c spans a wide range from **280 m/min** to **1400 m/min**, while the depth of cut A_p ranges from **0.8 mm** to **1.6 mm**. Consequently, the resulting responses show significant trade-offs. The surface roughness R_a varies between **1.529 μm** and **2.725 μm** . The peak-to-valley height R_z exhibits the most dramatic variation, ranging from a highly polished **0.818 μm** (Solution 2) to a relatively rough **17.148 μm** (Solution 10). Similarly, the machining temperature fluctuates significantly between a cool **54.8 $^{\circ}\text{C}$** (Solution 10) and a hotter **176.0 $^{\circ}\text{C}$** (Solution 8).

The uniformity of the desirability score (1.000) across such vastly different operating conditions suggests that the optimization boundaries were set to accept a wide performance window. In practical terms, this means the user must prioritize specific response targets when selecting the final parameters. For example:

- **If minimizing surface waviness (Rz) is the absolute priority: Solution 2** ($V_c=1400$, $A_p=1.6$) is optimal, yielding an extremely low Rz of $0.818 \mu\text{m}$, though it results in a higher temperature (154.8°C).
- **If minimizing thermal impact or tool wear is the priority: Solution 10** ($V_c=280$, $A_p=1.6$) is optimal, producing a very low temperature of 54.8°C , albeit with the highest Rz value ($17.148 \mu\text{m}$).
- **If seeking a balanced compromise: Solution 1** ($V_c=463.5$, $A_p=1.111$) offers a relatively low Ra ($1.631 \mu\text{m}$), moderate Rz ($11.831 \mu\text{m}$), and a cooler operating temperature (68.8°C).

Table IV-14 Top 10 Optimized Machining Combinations (V_c , a , A_p) Exhibiting Perfect Composite Desirability (1.000) across Ra, Rz, and Temperature

Number	V_c	a	A_p	Ra	Rz	Temperture	Desirability
1	463.494	0.286	1.111	1.631	11.831	68.8	1
2	1400	0.21	1.6	2.146	0.818	154.774	1
3	914.667	0.378	0.853	1.529	8.638	130.986	1
4	840	0.31	1.6	1.868	10.867	118.605	1
5	1400	0.21	0.8	2.63	0.831	152.726	1
6	1400	0.31	1.2	1.753	8.23	166.055	1
7	354.667	0.378	1.253	1.573	12.218	73.287	1
8	1325.333	0.378	1.547	2.725	9.517	176.012	1
9	1325.333	0.378	0.853	1.63	9.842	174.609	1
10	280	0.21	1.6	1.728	17.148	54.848	1

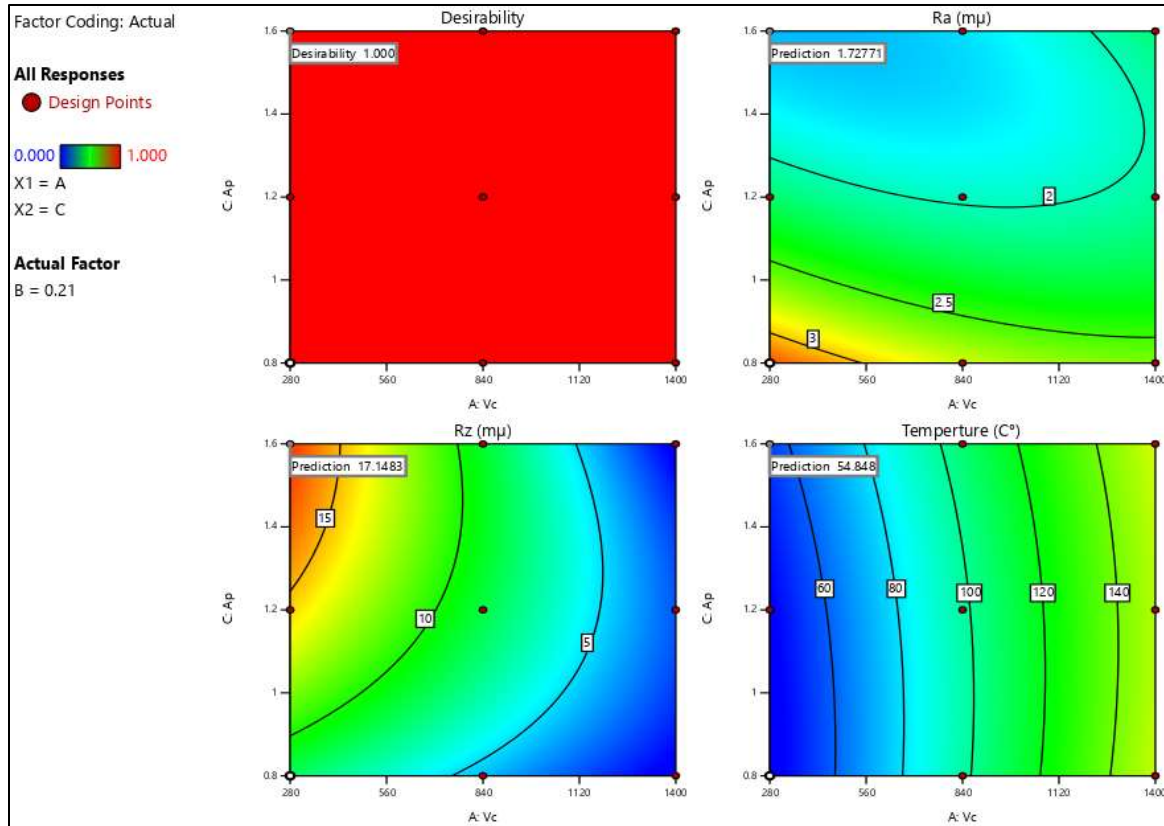


Figure IV-28 Contour Plots of Desirability, Ra, Rz, and Temperature illustrating the Correlation between Cutting Speed (V_c) and Depth of Cut (A_p)

IV.8. Conclusion:

This chapter investigated the effects of turning parameters on the machining performance of bronze alloy using the Box–Behnken Design (BBD) method. The study focused on surface roughness parameters (R_a and R_z), cutting temperature (T), and machining vibrations in order to determine the optimal cutting conditions. The experimental results highlighted the influence of spindle speed, feed rate, and depth of cut on surface quality and machining stability, leading to the identification of the most suitable parameter settings for achieving improved surface finish during the turning process.

General Conclusion

This master's thesis presents a comprehensive investigation into the machining behavior of bronze under three distinct lubrication conditions: dry cutting, conventional industrial oil, and eco-friendly vegetable-based oil. Conducted at Brothers Mentouri University – Constantine 1, the study addresses a critical challenge in modern manufacturing: how to optimize machining performance while reducing environmental impact.

The research evaluated three key performance indicators — surface roughness (R_a), cutting-induced vibrations, and flank tool wear (V_b) — across varying cutting parameters including cutting speed, feed rate, and depth of cut. Through a structured experimental plan, statistical tools such as ANOVA and Response Surface Methodology (RSM) were applied to develop predictive models linking input parameters to output responses. The Taguchi method and desirability function were further employed to identify the optimal combination of cutting conditions that simultaneously minimize all three performance criteria.

The findings of this study demonstrate that lubrication strategy plays a decisive role in determining machining quality and tool durability. While dry machining offers clear economic and environmental advantages — eliminating fluid costs, disposal issues, and contamination risks — it tends to generate higher cutting temperatures that may accelerate tool wear under aggressive conditions. Conventional industrial oil, despite its proven lubricating and cooling effectiveness, raises significant environmental and health concerns due to its petroleum-based composition and poor biodegradability.

Eco-friendly oil emerges from this study as the most balanced alternative. It delivers machining performance comparable to conventional oil while substantially reducing the ecological footprint of the process. This positions biodegradable lubrication as a viable and sustainable path forward for industrial bronze turning applications.

The study also proposes the future integration of Artificial Neural Networks (ANN) and Deep Neural Networks (DNN) as promising tools for long-term predictive modeling of machining performance, opening perspectives for intelligent and adaptive manufacturing systems.

In summary, this work contributes meaningful scientific and practical insights to the field of sustainable machining, demonstrating that technical efficiency and environmental responsibility can be achieved simultaneously through informed selection of lubrication strategies and optimized cutting parameters.

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